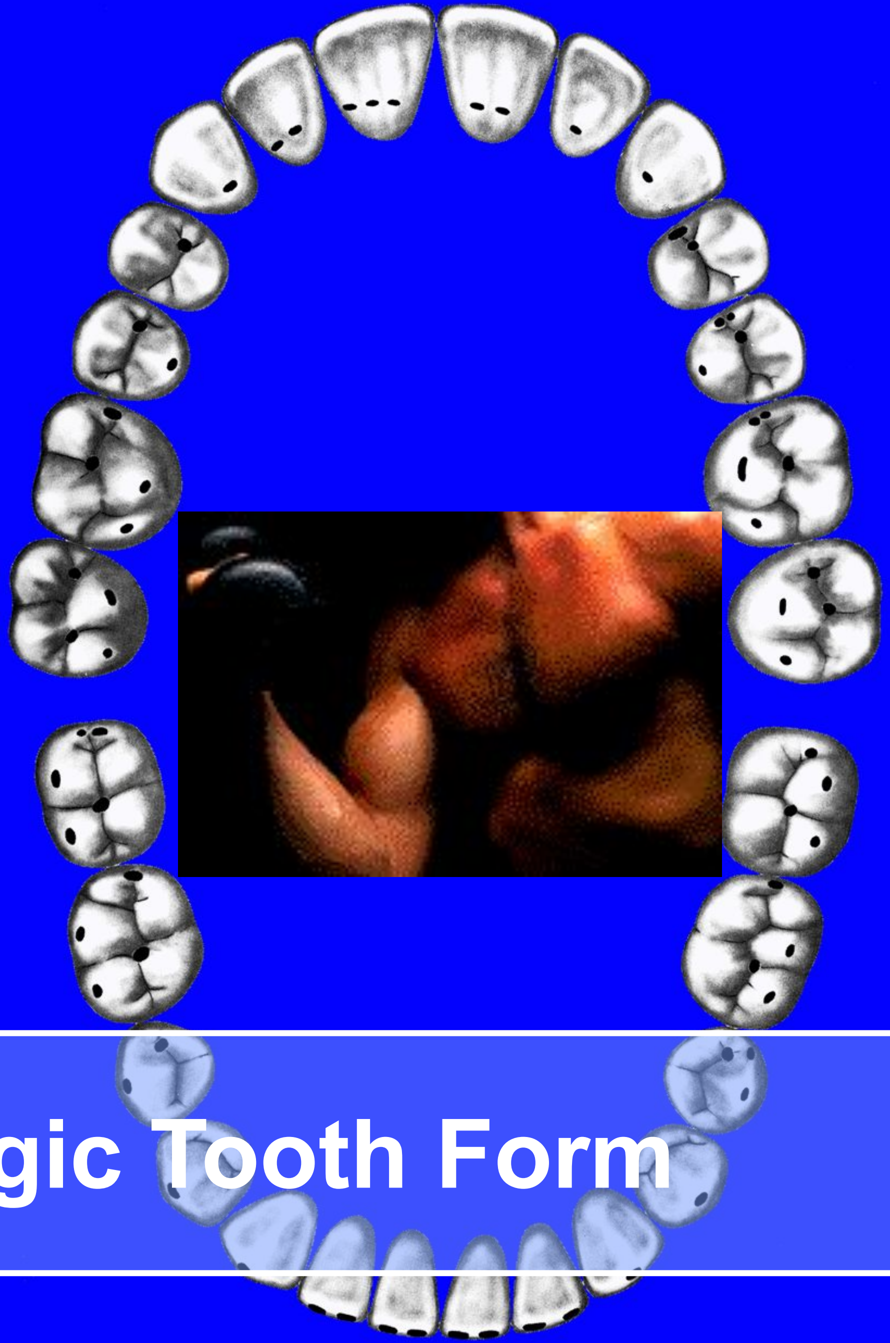


Physiologic Tooth Form

Arch Form

Tooth Alignment

Plane of Occlusion

Proximal Contact Locations

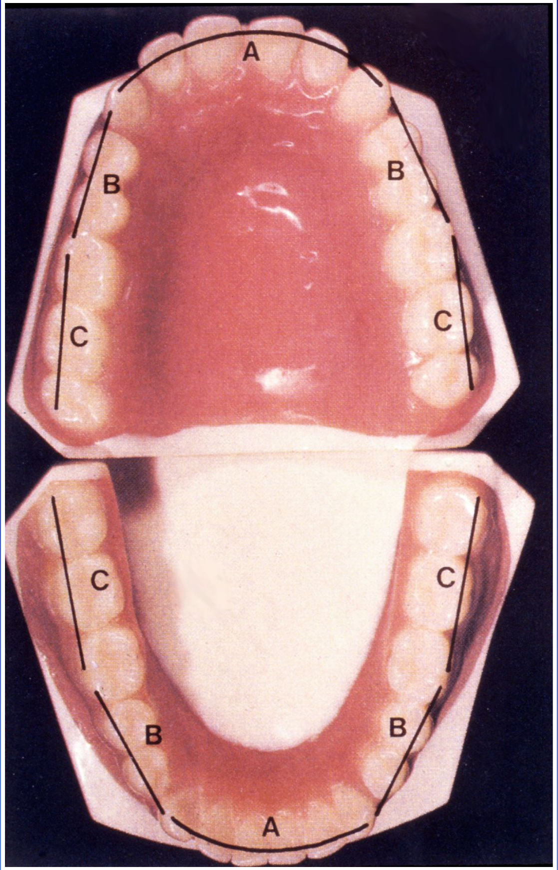
Height of Contour Measurements

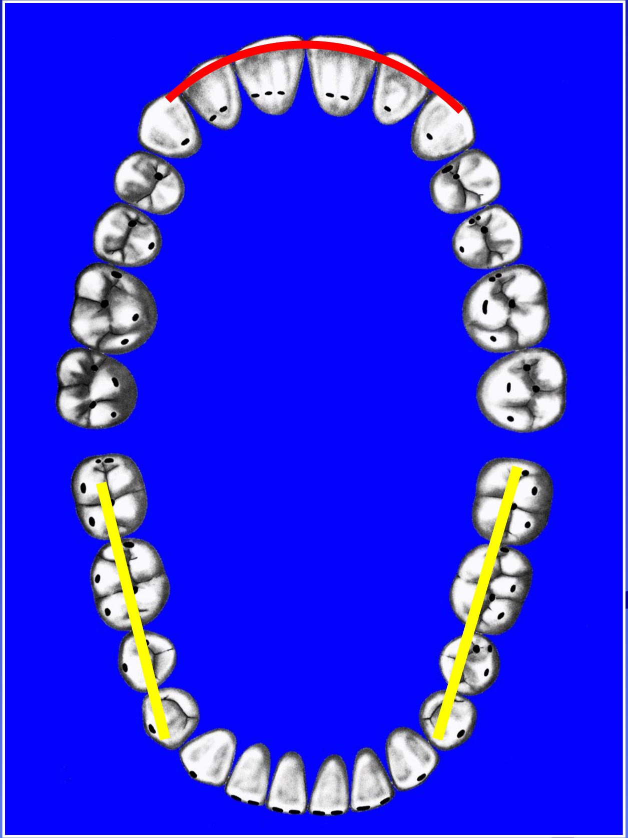
Height of Curvature of the CEJ

Periodontal Tissues & Fibers

rch Form

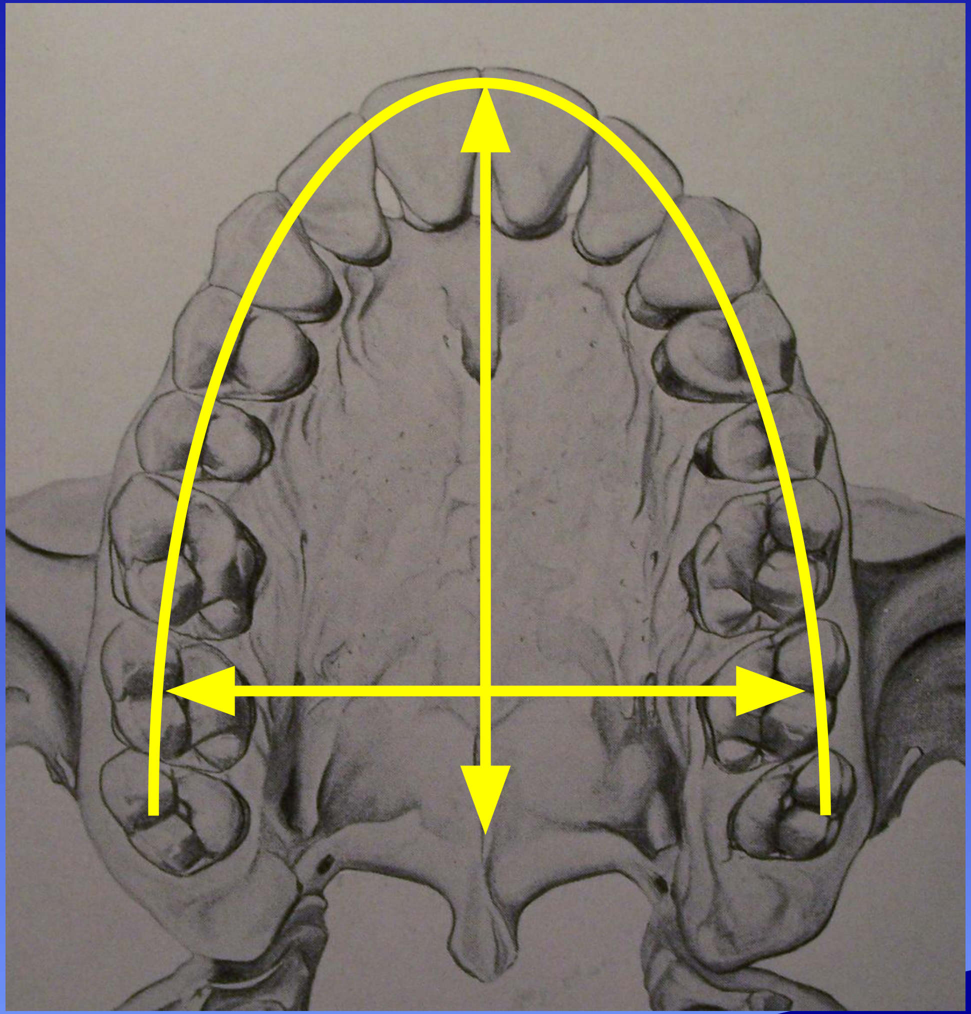
Its Importance for Esthetics *and* Function





Humans have an elliptical-shaped arch with significantly more curvature displayed in the anterior region. The molars have a straighter arch alignment within each quadrant, especially the mandibular molars.

Maxillary Arch Form



Maxillary arch form dominates or influences the mandibular arch form because of the following:

- maxillary arch is **larger** from the distal of one third molar through the middle of each tooth to the other third molar, than in the mandible
- maxillary arch is **wider** (i.e. longer from anterior to posterior) from right to left sides
- esthetics are embedded in the maxillary arch more than the mandible (smile line is evident)

Angulation of Maxillary Teeth

(Sagittal View)

SINGLE-ROOTED TEETH

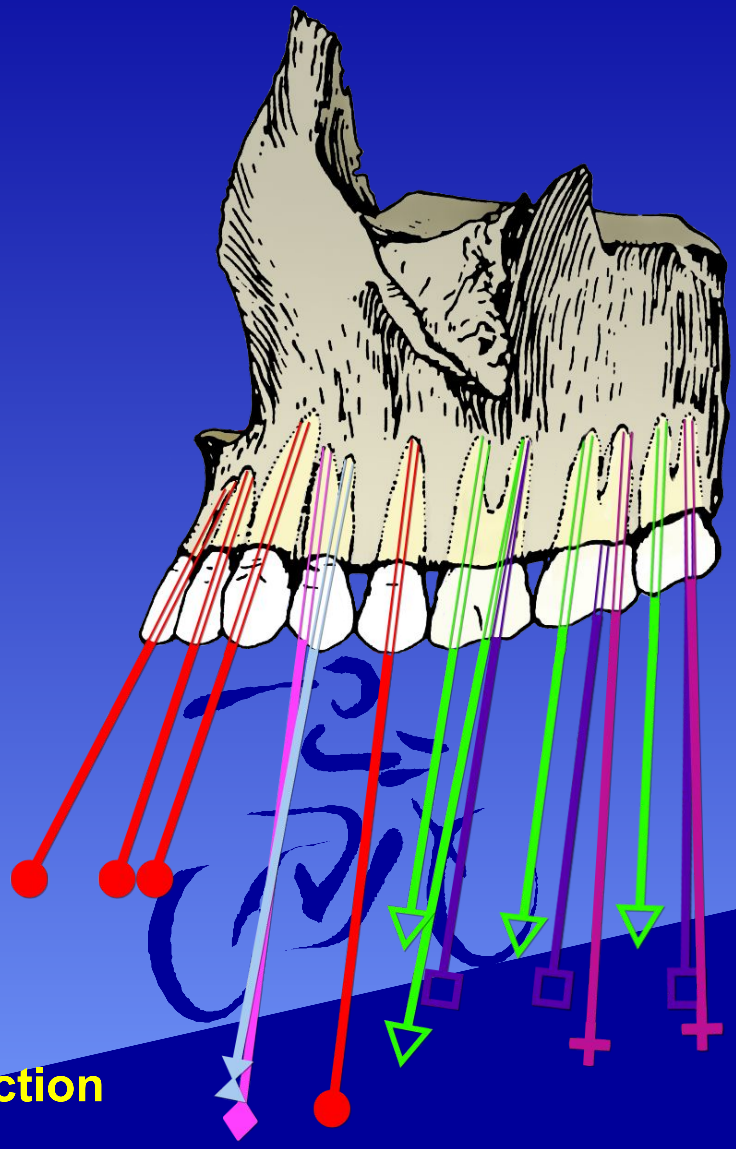
- Incisors
- Canines
- Premolars

FIRST PREMOLAR

- Facial roots
- Palatal roots

MAXILLARY MOLARS

- Mesiofacial roots
- Distofacial roots
- Palatal roots



Notice that the inclination has a predilection toward the anterior region.

Angulation of Maxillary Teeth

(Frontal View)

SINGLE-ROOTED TEETH

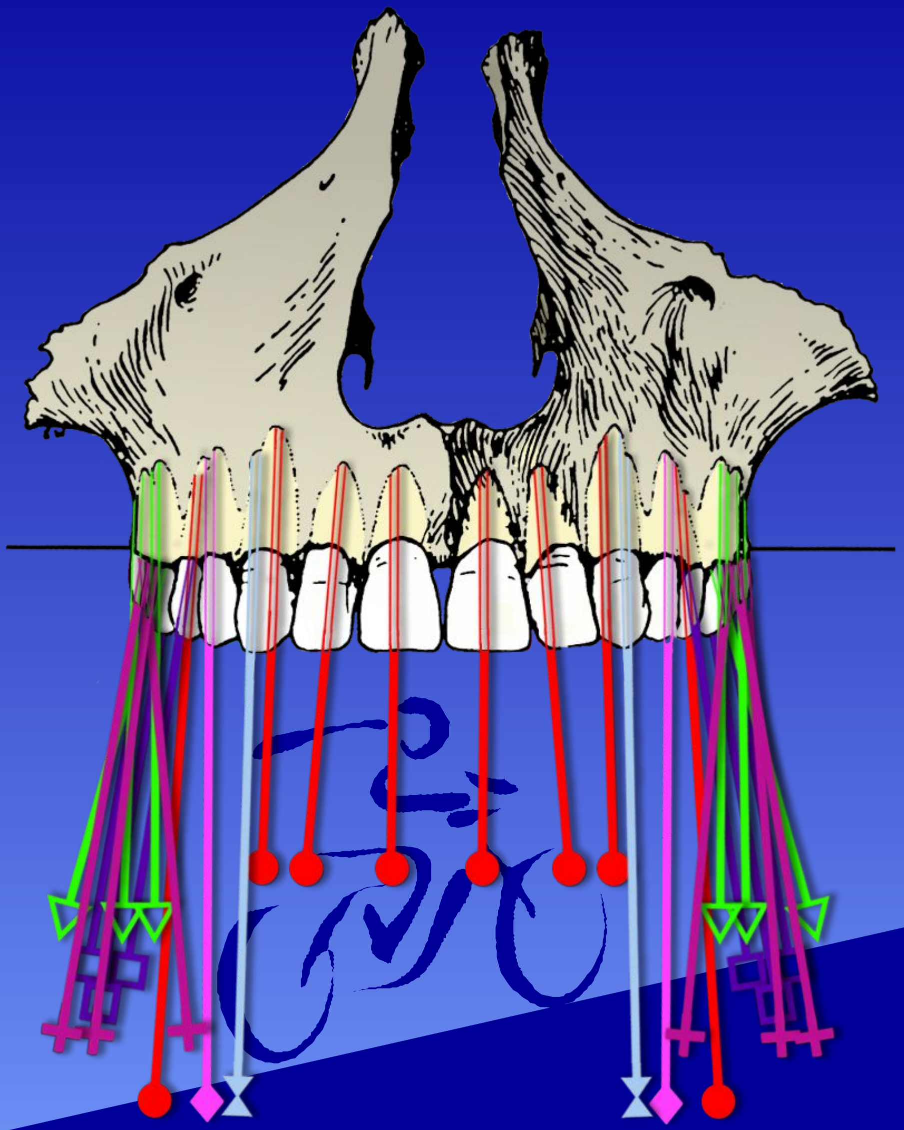
- Incisors
  Canines
  Premolars

FIRST PREMOLAR

- Facial roots
- Palatal roots

MAXILLARY MOLARS

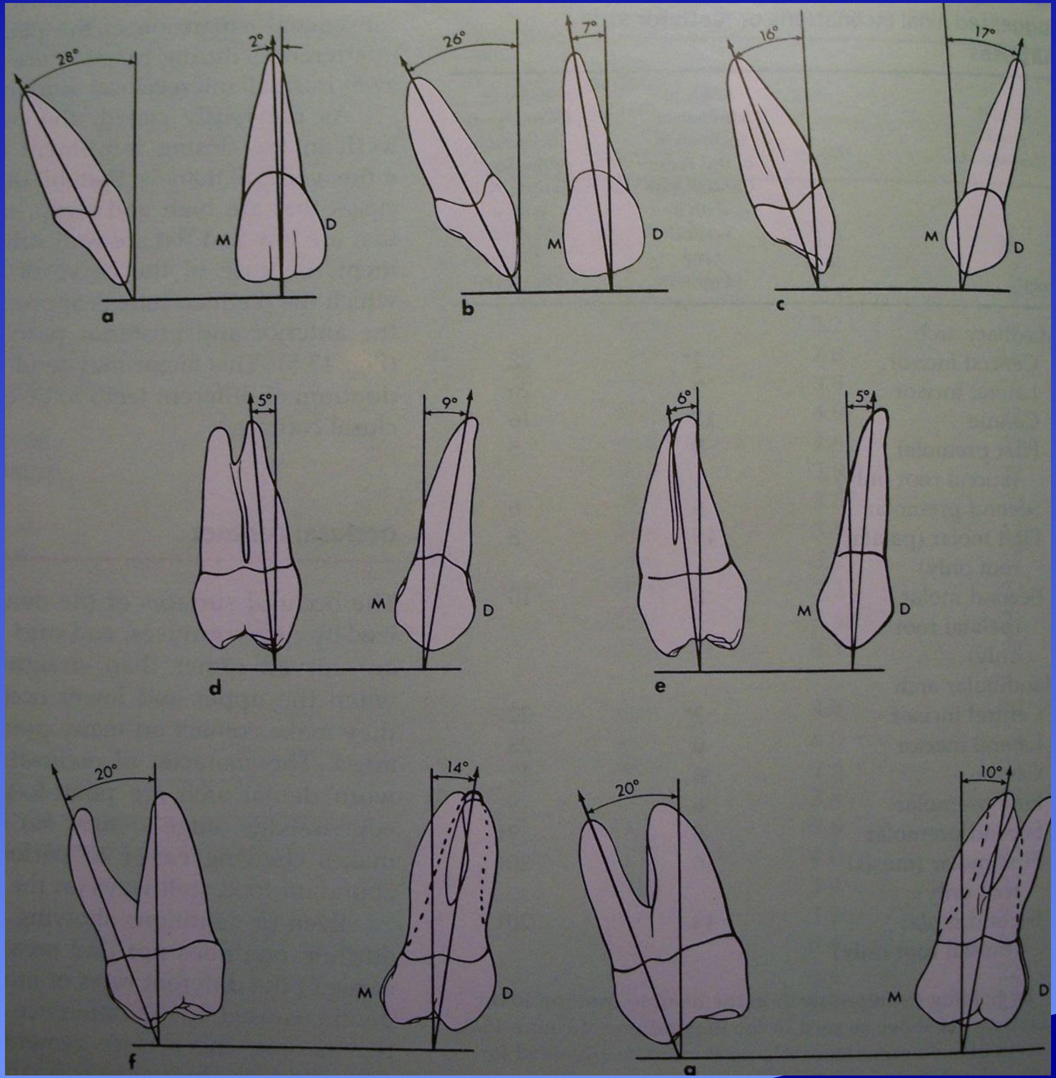
- Mesiofacial roots
- Distofacial roots
- Palatal roots



Anterior teeth and premolars deviate minimally from the vertical axis

Maxillary Teeth Individual Alignment

Measurement of tooth alignment compared to the vertical (longitudinal) axis



Proximal reference angles:

>Central Incisor = 28 degrees

>Lateral Incisor = 26 degrees

>Canine = 16 degrees

>First Premolar = 5 degrees

>Second Premolar = 6 degrees

>First Molar = 20 degrees

>Second Molar = 20 degrees

Facial reference angles:

>Central Incisor = 2 degrees

>Lateral Incisor = 7 degrees

>Canine = 17 degrees

>First Premolar = 9 degrees

>Second Premolar = 5 degrees

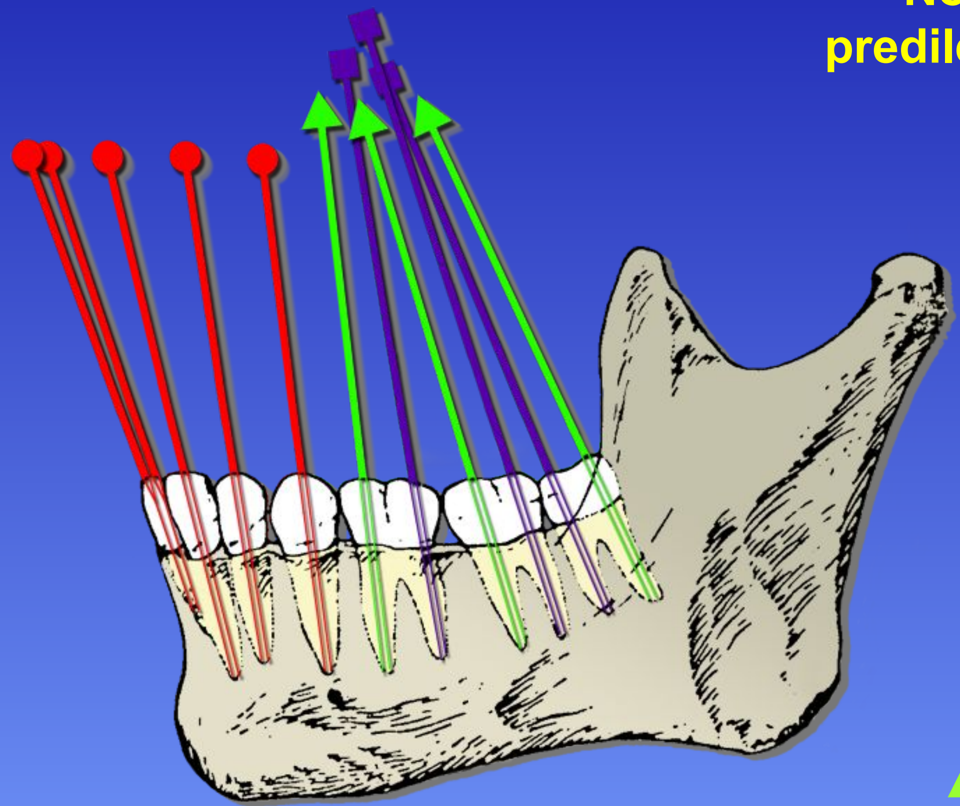
>First Molar = 14 degrees

>Second Molar = 10

Least deviation from vertical = Central Incisor (Facial) & 1st Premolar (Proximal)

Angulation of the Mandibular Teeth

(Sagittal View)



Notice that the inclination has a predilection toward the anterior region

SINGLE-ROOTED TEETH

- Incisors
- Canines
- Premolars

MANDIBULAR MOLARS

- Mesial roots
- Distal roots

Angulation of Mandibular Teeth

Notice that the inclination in the molar region is toward the lingual

(Frontal View)

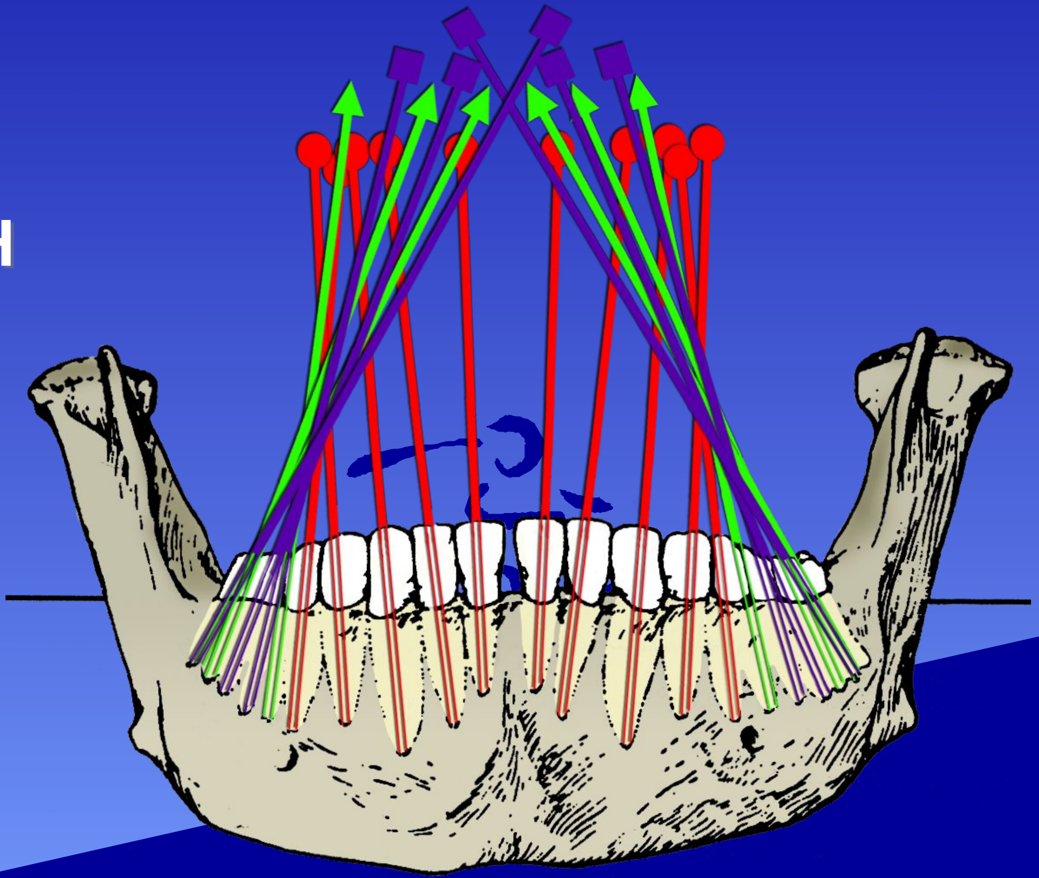
SINGLE-ROOTED TEETH

- **Incisors**
  Canines
  Premolars

MANDIBULAR MOLARS

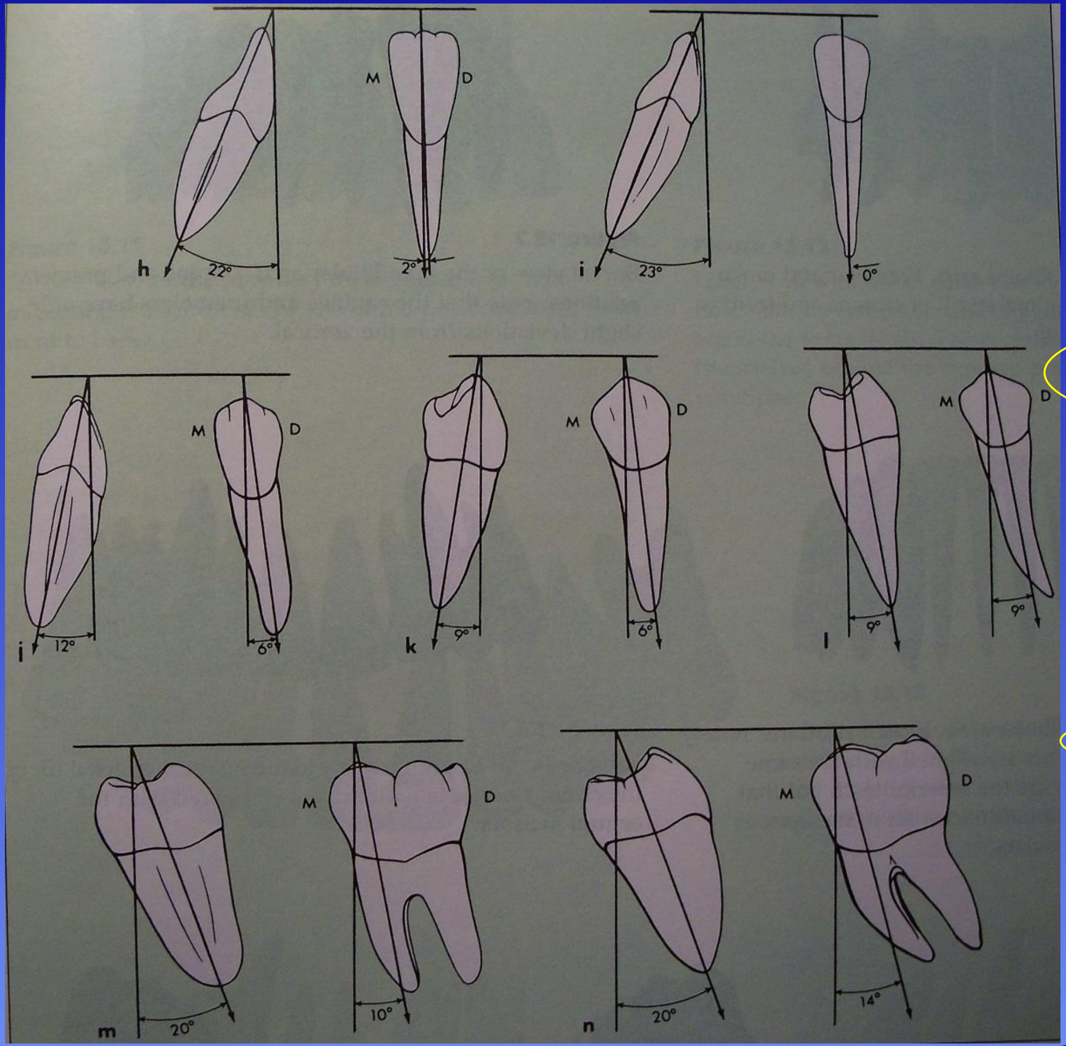
▲ **Mesial roots**

■ **Distal roots**



Mandibular Teeth Individual Alignment

Measurement of tooth alignment compared to the vertical (longitudinal) axis



Proximal reference angles:

>Central Incisor = 22 degrees

>Lateral Incisor = 23 degrees

>Canine = 12 degrees

>First Premolar = 9 degrees

>Second Premolar = 9 degrees

>First Molar = 20 degrees

>Second Molar = 20 degrees

Facial reference angles:

>Central Incisor = 2 degrees

>Lateral Incisor = 0 degrees

>Canine = 6 degrees

>First Premolar = 6 degrees

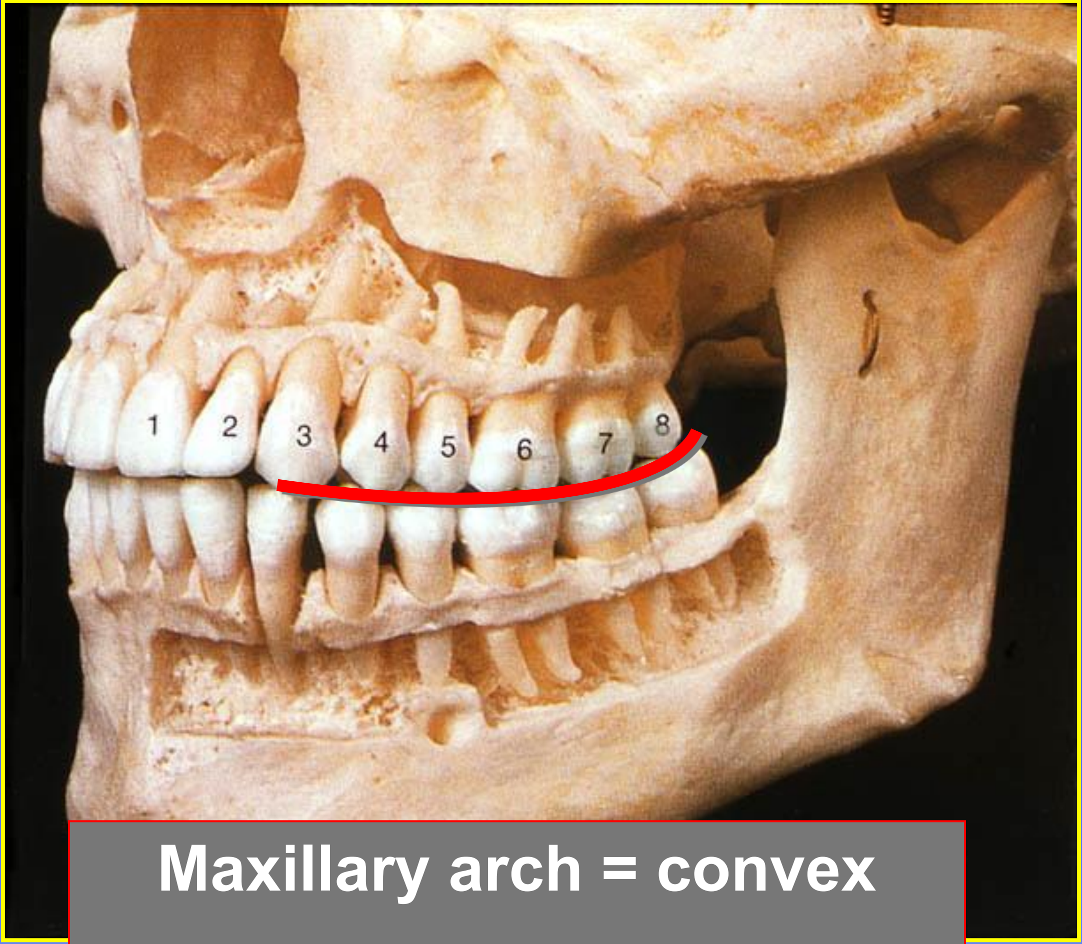
>Second Premolar = 9 degrees

>First Molar = 10 degrees

>Second Molar = 14

Least deviation from vertical = Lateral Incisor (Facial) & Premolars (Proximal)

Curve of Spee



Maxillary arch = convex

Mandibular arch = concave

- "An imaginary line that extends from the cusp tip of the mandibular canine along the facial cusp tips of the posterior teeth resulting in an anteroposterior curve"

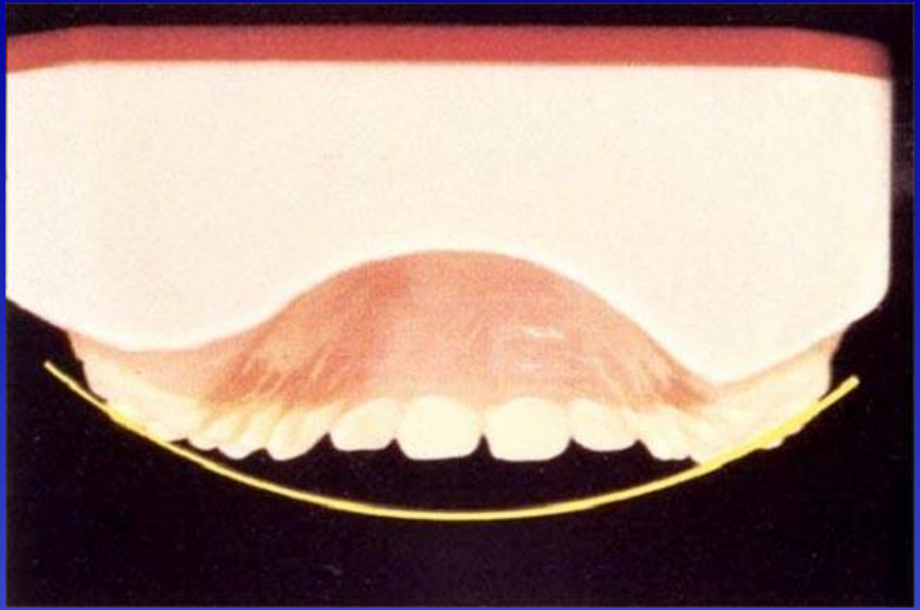
Curve of Wilson

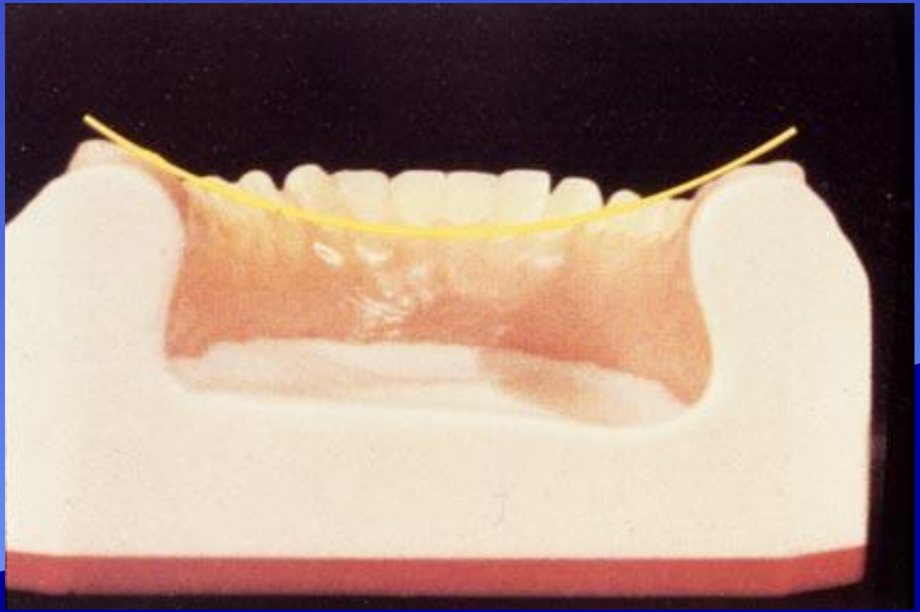
- "From the frontal view, an imaginary curved line connecting the facial and lingual cusps of the posterior teeth"
- Maxillary arch = Convex
- Mandibular arch = Concave

Curve of Wilson

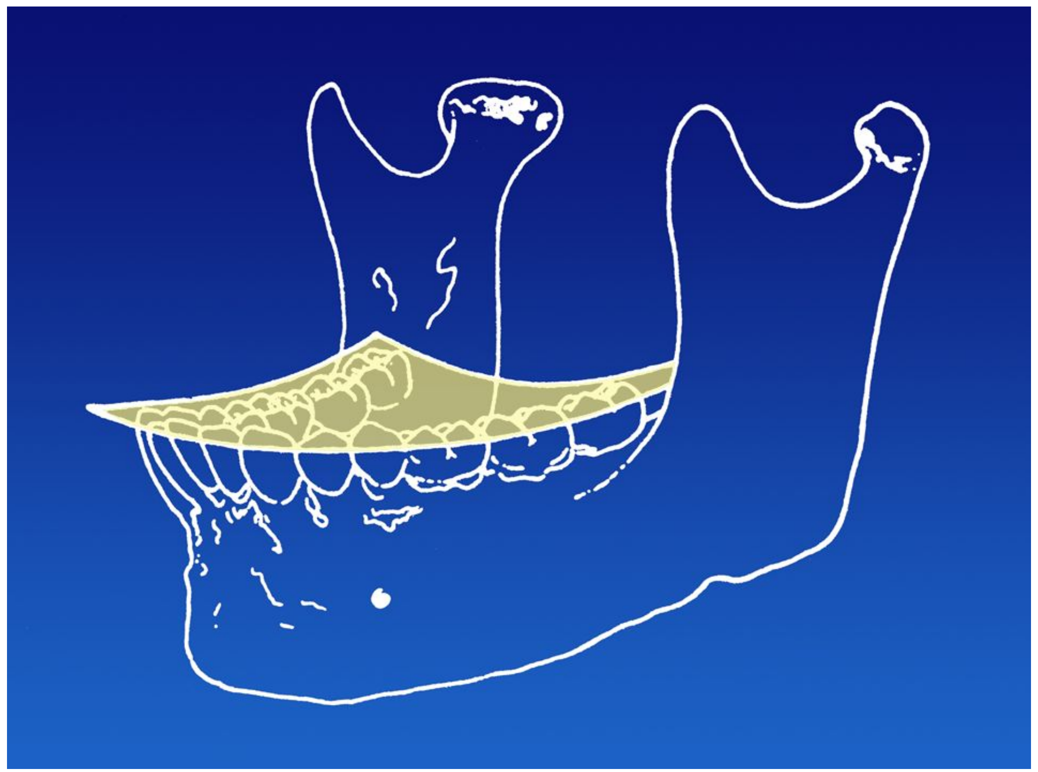
- Convex form of the maxilla

- Concave form of the mandible





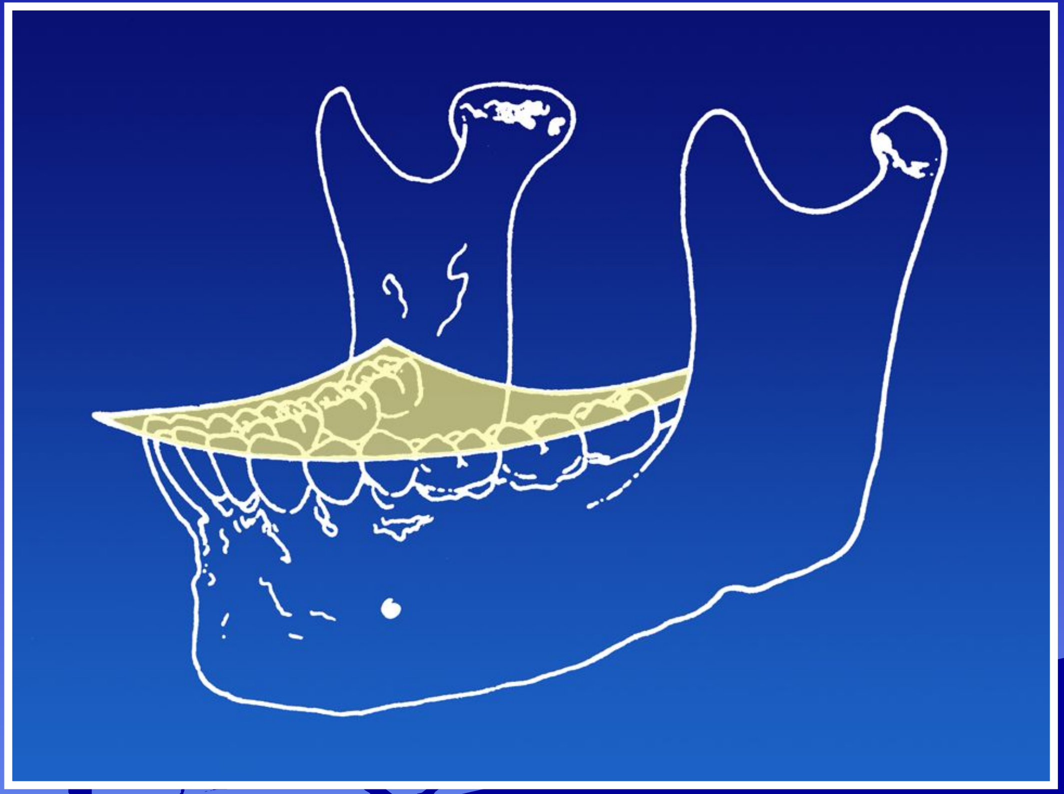
Plane of Occlusion *aka* "Plane of Occlusal Curvature"



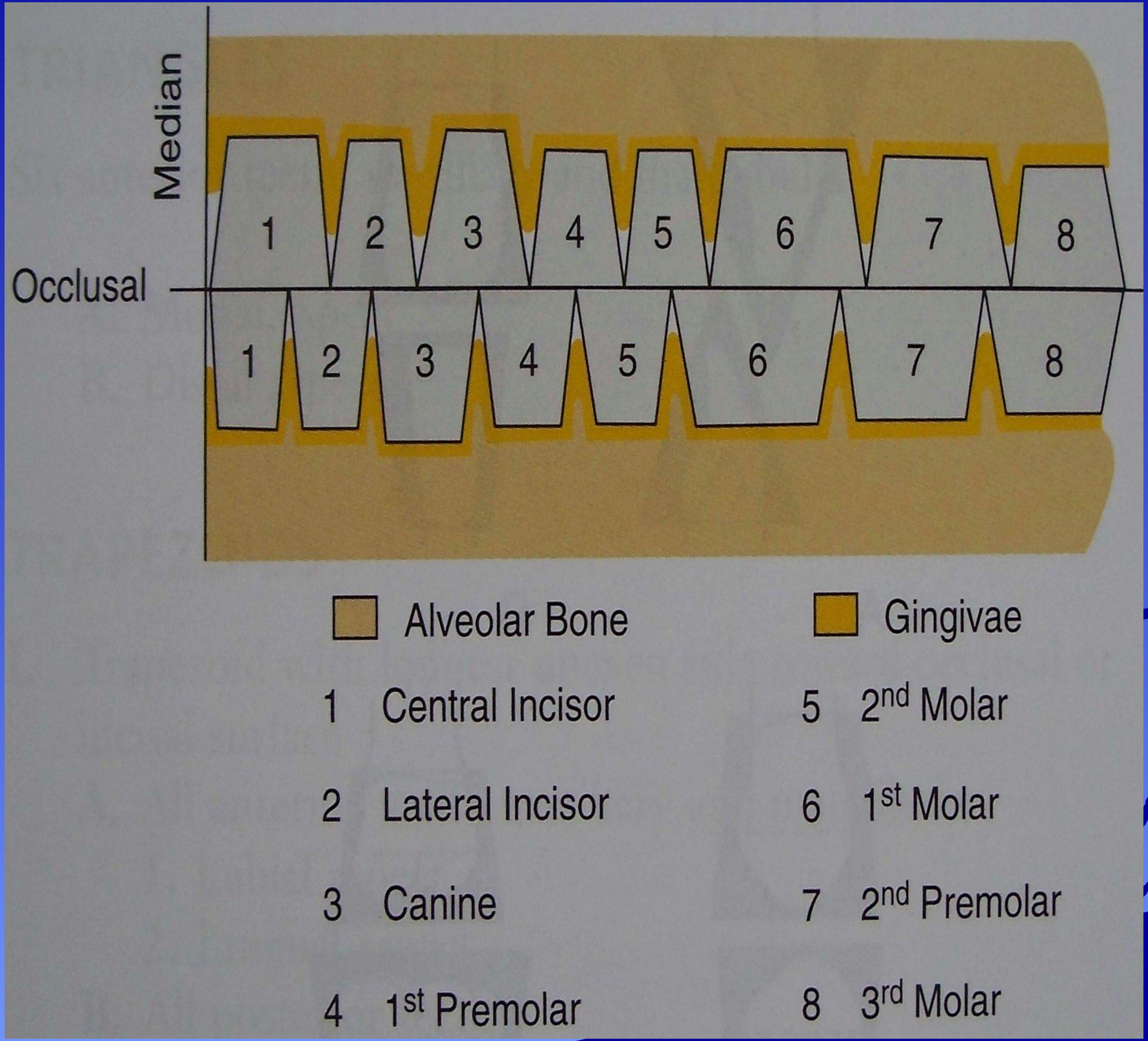
Combination of the Curves of Wilson & Spee

Plane of Occlusion

- The plane is not flat
- A flat plane will not permit a simultaneous functional contact in more than one area of the dental arch
- A curved plane permits maximum use of tooth contacts during function
- Teeth are positioned in the arches at varying degrees of inclination (see individual angulations of teeth as they deviate from the vertical)



Occlusal Antagonists in the Opposing Arch



Items to note:

>teeth typically contact their **namesakes** in the opposing arch

>the maxillary third molars and mandibular central incisors have only **ONE** antagonist in the opposing arch

Proximal Contact Areas

- Proximal contact areas
  - Allow for the maintenance of proper arch form
  - They prevent the accumulation of food debris and impaction between teeth
  - Occlusal view of proximal contacts display convex contacts with adjacent teeth

Proximal Contact Areas

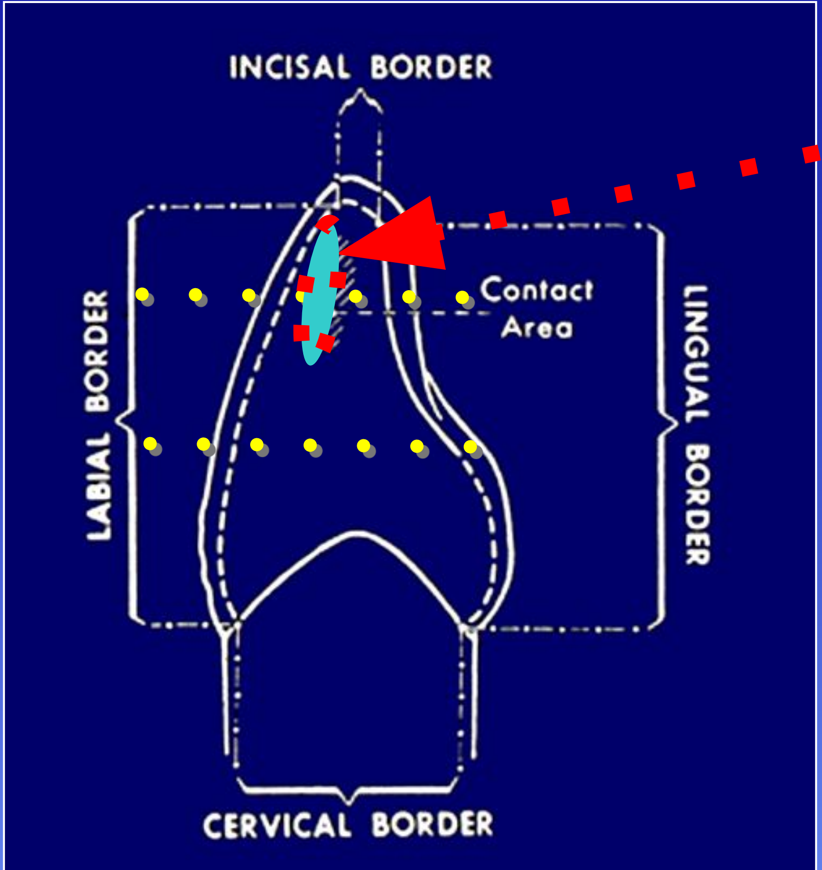


FIG. 6. DISTAL SURFACE

Anterior Teeth: The shape of the contact is described as an incisogingival ovoid

Allows for the construct of the incisal & gingival embrasures

Proximal Contact Areas

What does a properly formed proximal contact look like?

What size should it have? Should it be light, heavy or does it not matter?

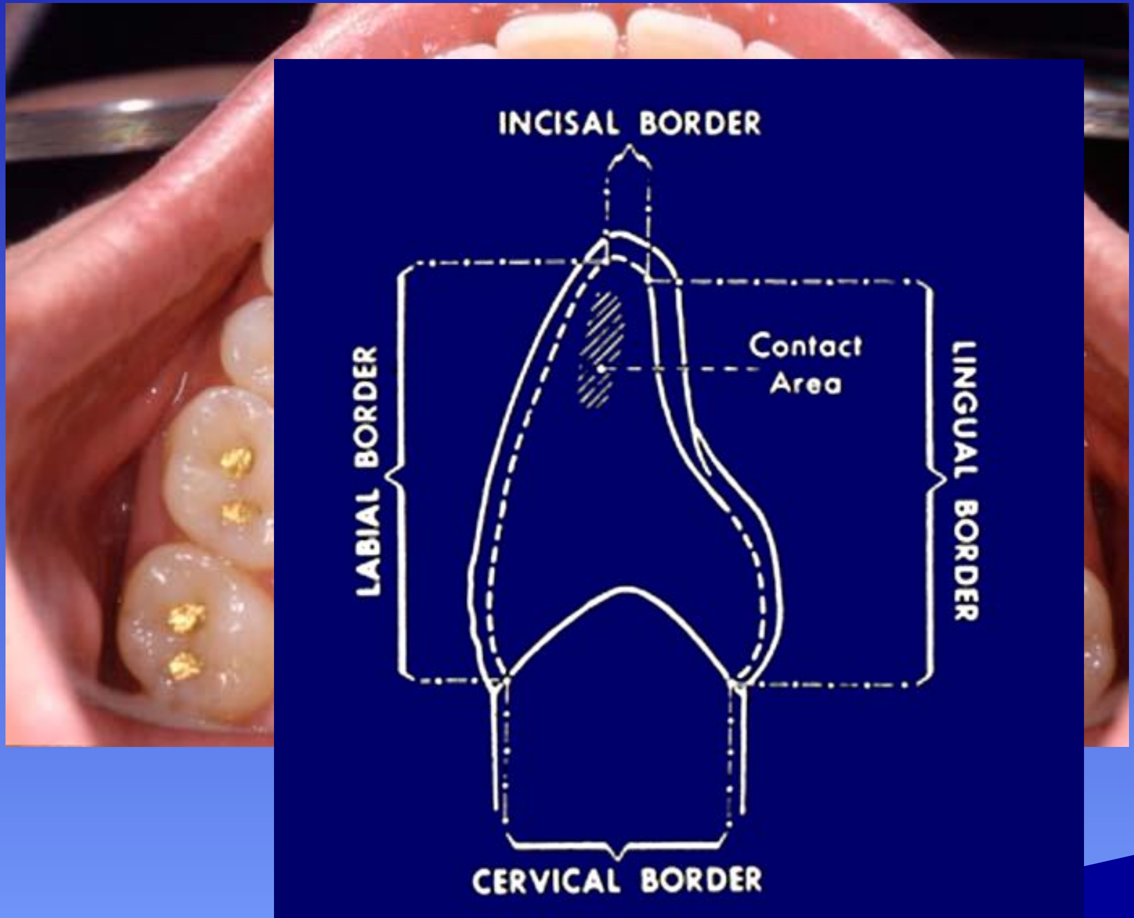
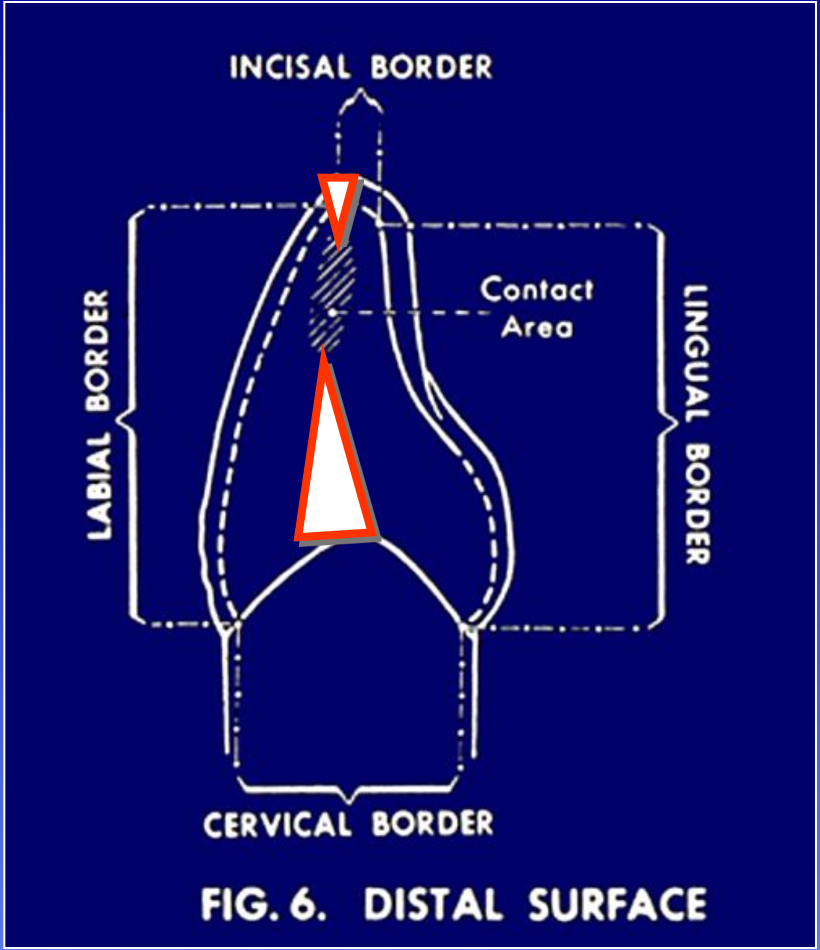


FIG. 6. DISTAL SURFACE

Proximal Contact Areas



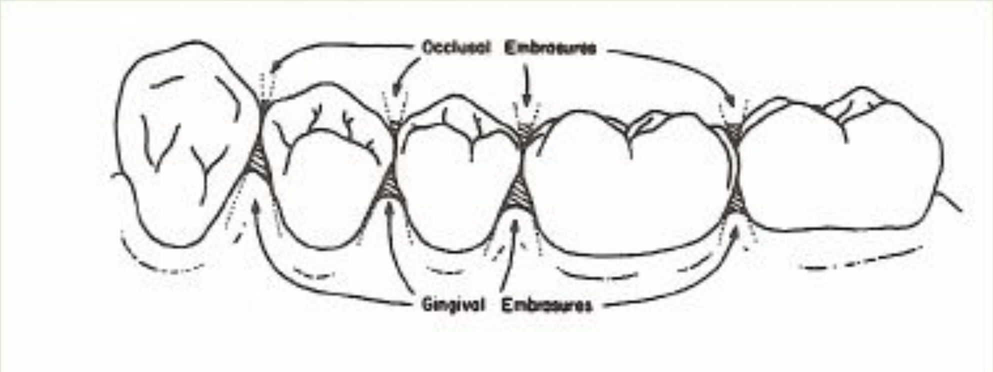
Anterior Teeth: character of the contact is ovoid, from an incisogingival perspective

Allows for incisal embrasure & gingival embrasure

Note that the gingival embrasure is larger in surface area than the incisal embrasure. Why is this so?

TERMINOLOGY REVIEW - EMBRASURES

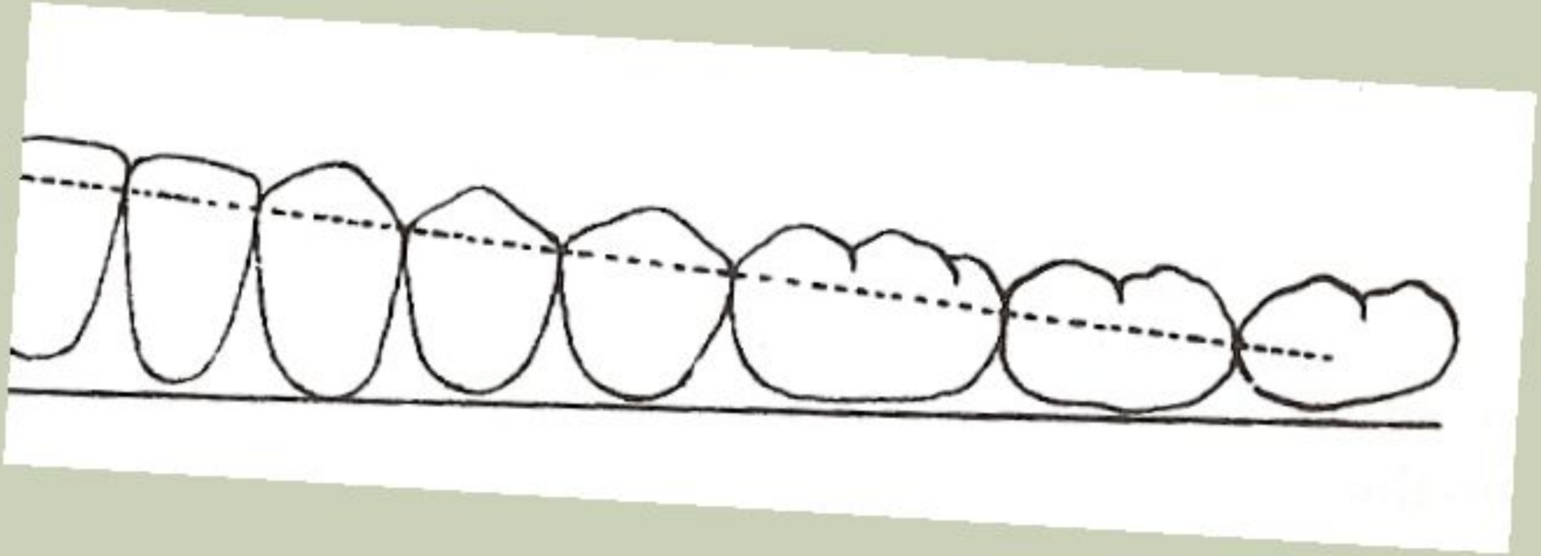
- The open space (negative space) between the proximal surfaces of two adjacent teeth in the same arch.



- **There are four types:** Buccal, Lingual, Occlusal and Gingival

TERMINOLOGY REVIEW – CONTACT AREA

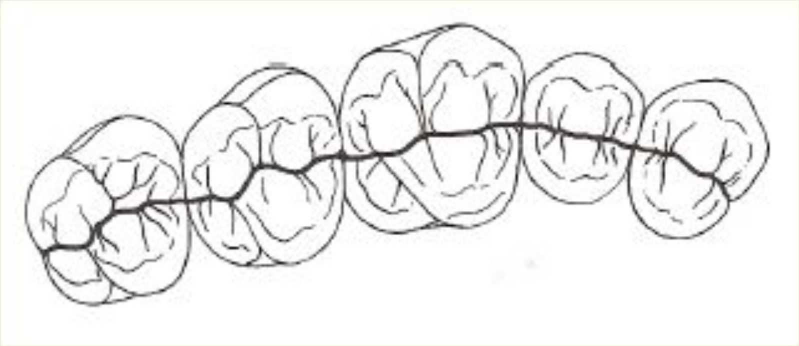
- The area on a proximal surface of the crown of a tooth that contacts the adjacent tooth.



- The contacts appear ***lower*** on each tooth as you move posteriorly.

Marginal Ridges – Height

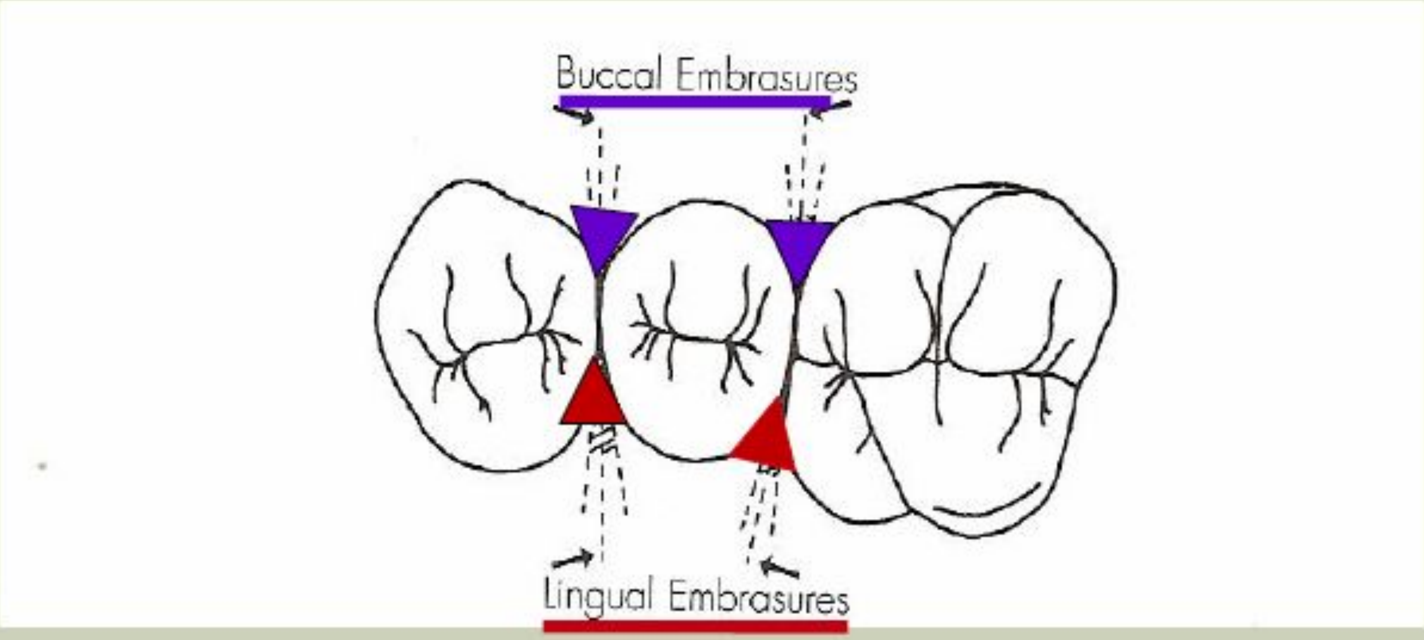
The ***height*** of the marginal ridges of ***adjacent teeth*** should be at the ***same level*** (unless the teeth are malpositioned)



Note how the central grooves are CONTINUOUS

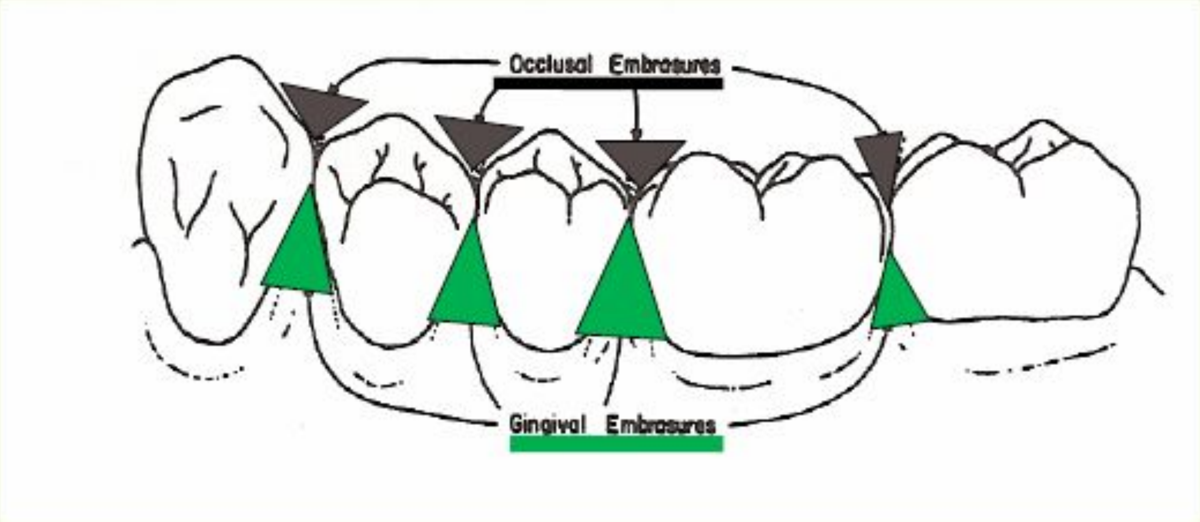
TERMINOLOGY REVIEW – INTERPROXIMAL EMBRASURES

- Viewed from the ***occlusal*** aspect.
- There are **buccal** and **lingual** embrasure spaces.

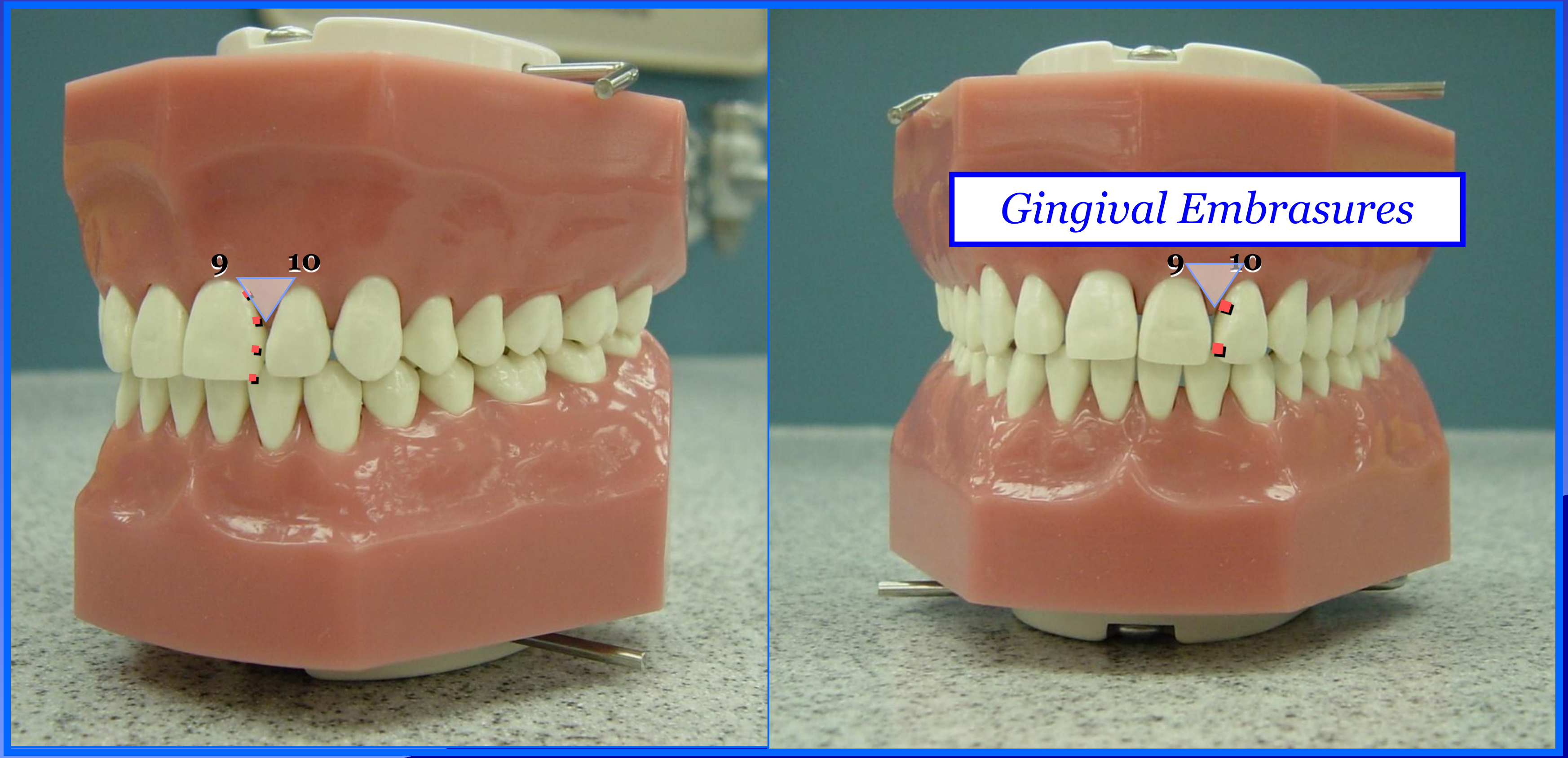


TERMINOLOGY REVIEW - OCCLUSAL & GINGIVAL EMBRASURES

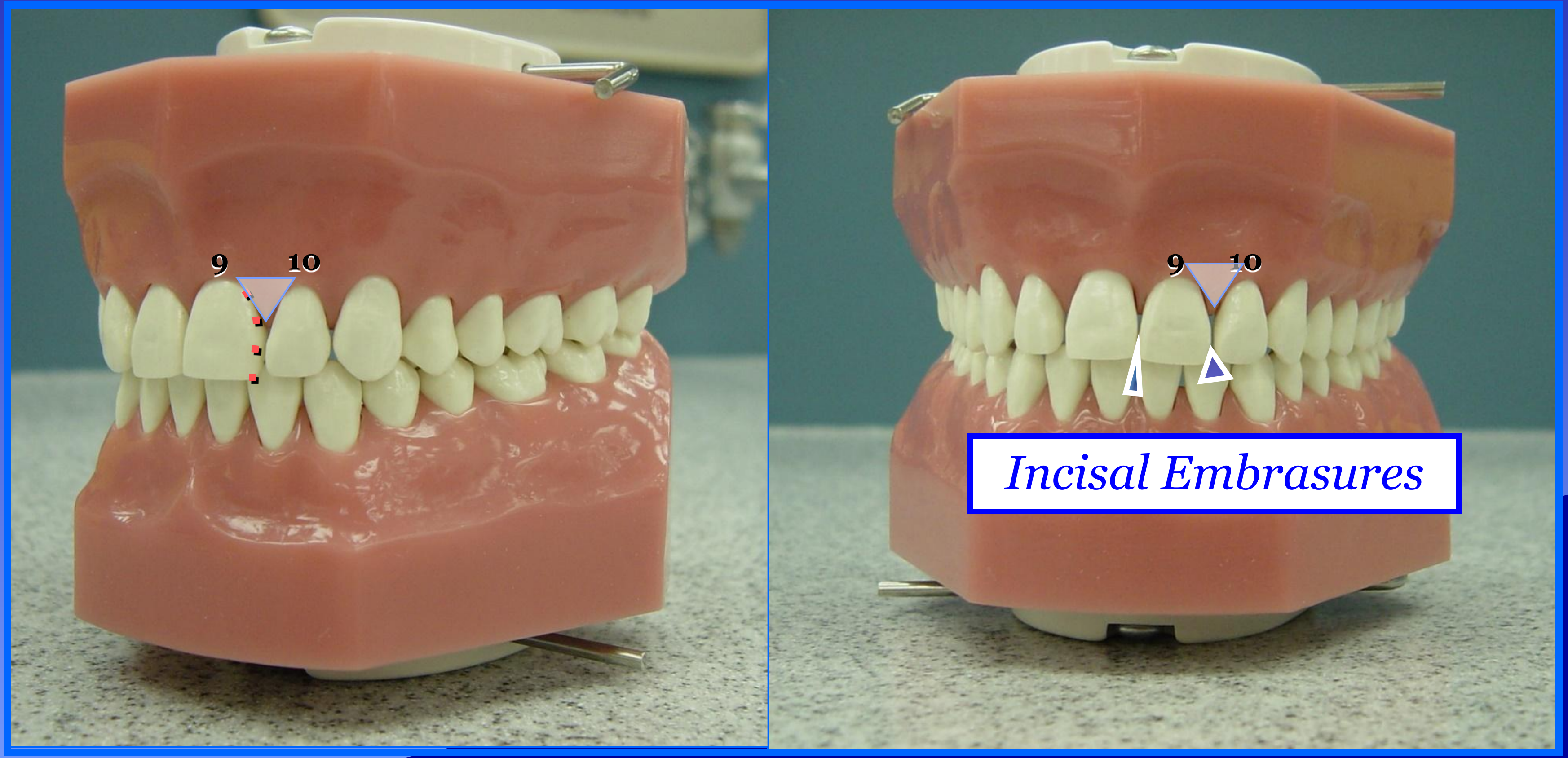
- The space between the teeth as viewed from the direct ***Buccal*** ***AND*** ***Lingual*** aspects.



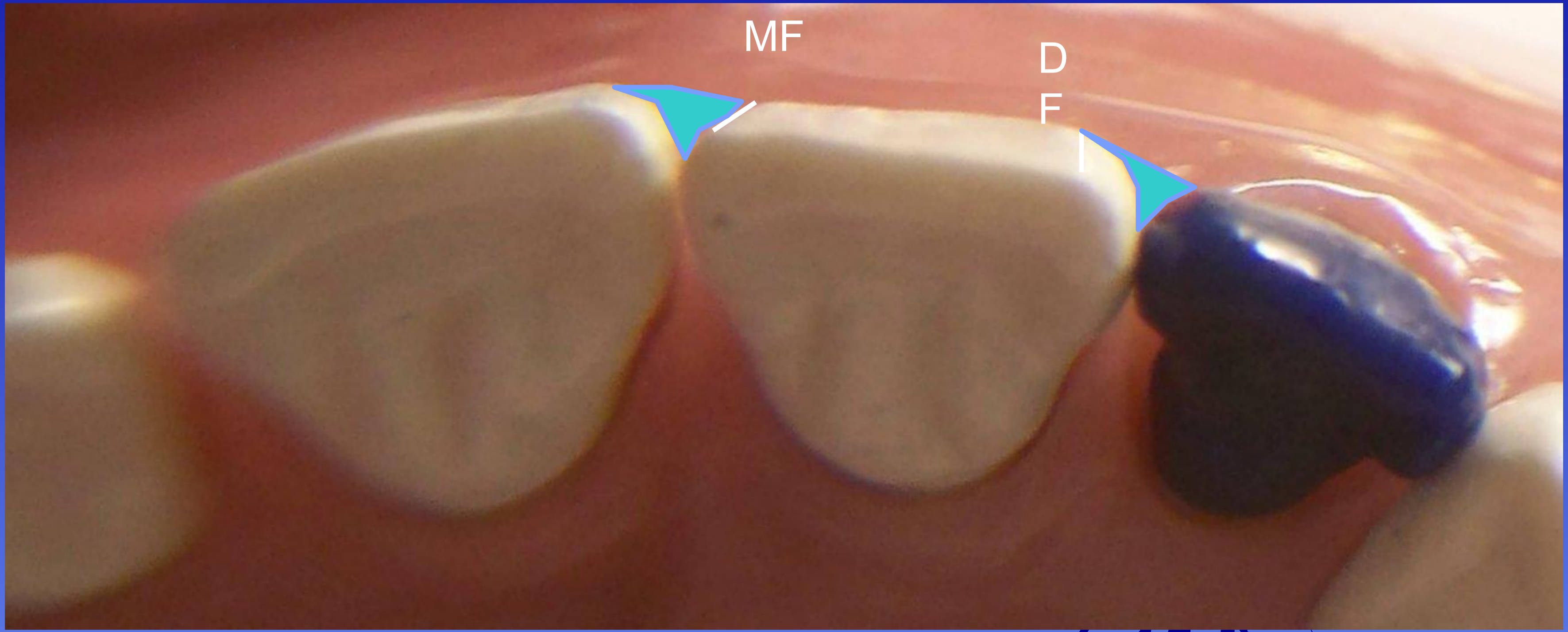
GINGIVAL Embrasures



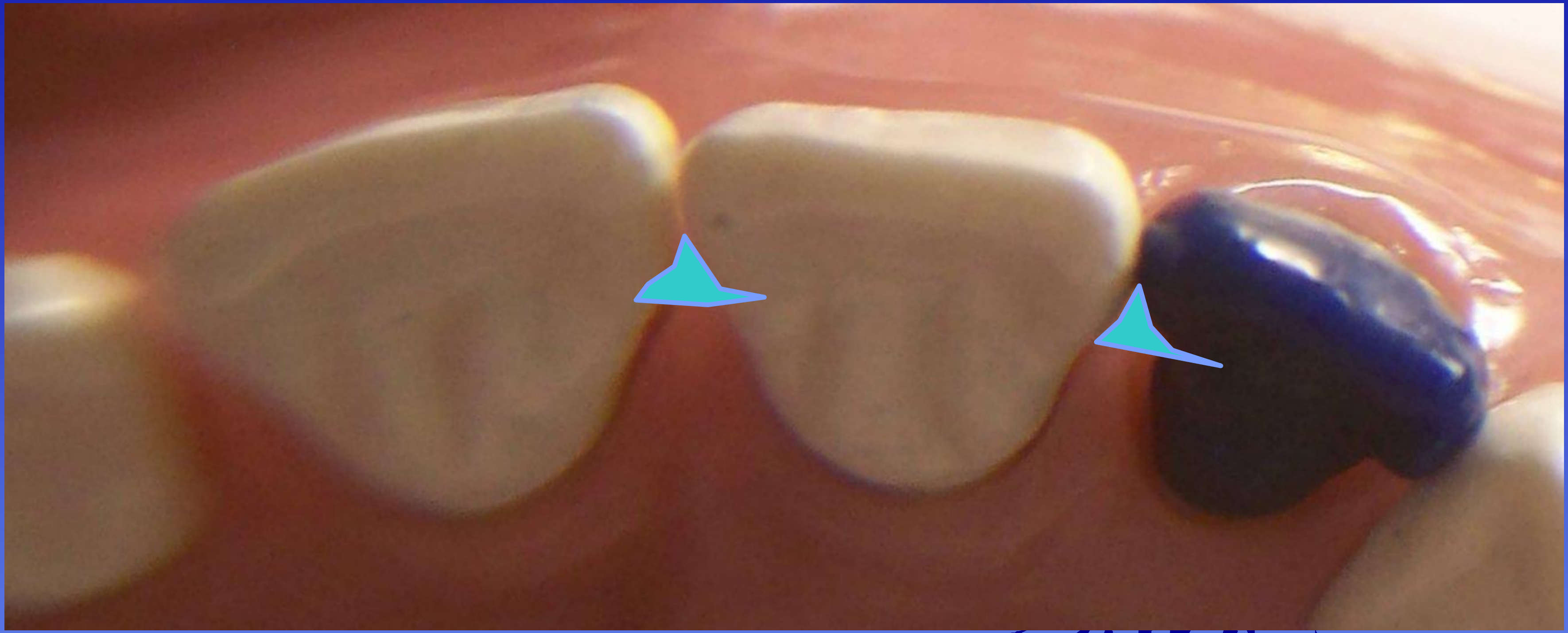
INCISAL EMBRASURES



Incisal View – Facial Embrasures



Incisal View – lingual Embrasures



Proximal Contact Areas

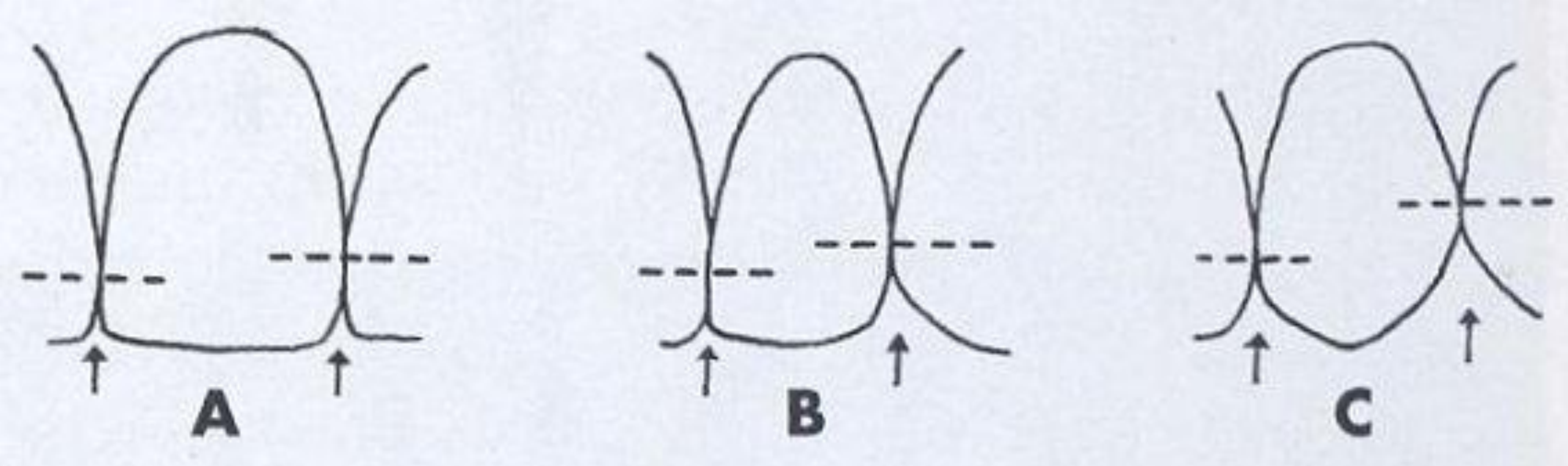
- Proximal contact areas
  - Provides for the maintenance of proper arch form
  - Prevents the accumulation of food debris and impaction
  - Occlusal view of proximal contacts of anterior teeth: they are located in the middle 1/3 (from a faciolingual perspective)
  - On posterior teeth, the contacts are located slightly facial of center (faciolingually)

Open Contact Area

Posterior Teeth: Contacts are located slightly facial of center, although there are a few exceptions

Proximal Contact Areas

Facial Perspective of Maxillary Anterior Teeth

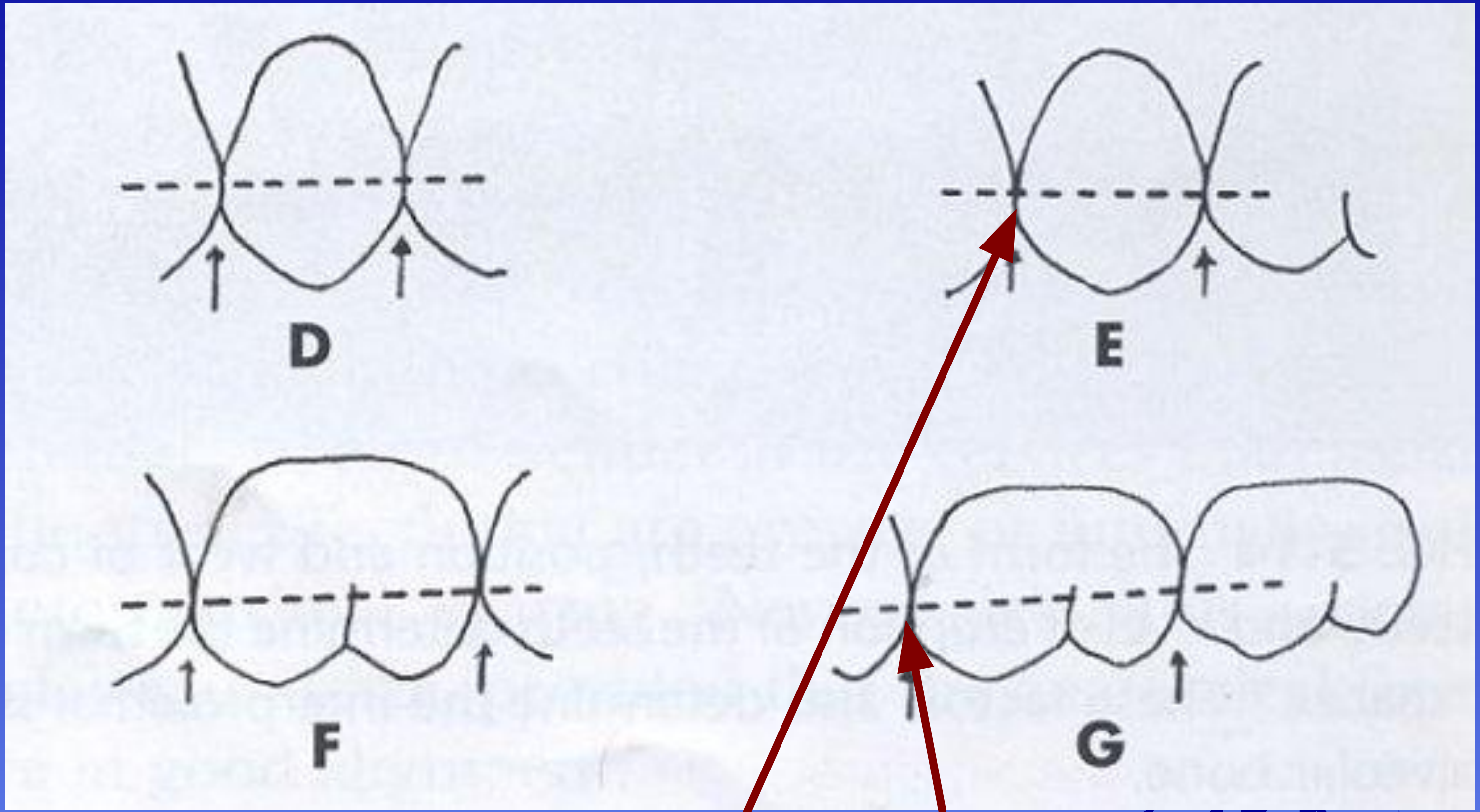


Location of Proximal Contacts:

- Midline: Incisal third
- Distal of Central: Junction of incisal and middle thirds
- Mesial of Lateral: Junction of incisal and middle thirds
- Distal of Lateral: Middle third
- Mesial of Canine: Junction of incisal and middle thirds
- Distal of Canine: Middle third

Proximal Contact Areas

Facial Perspective of Maxillary Posterior Teeth

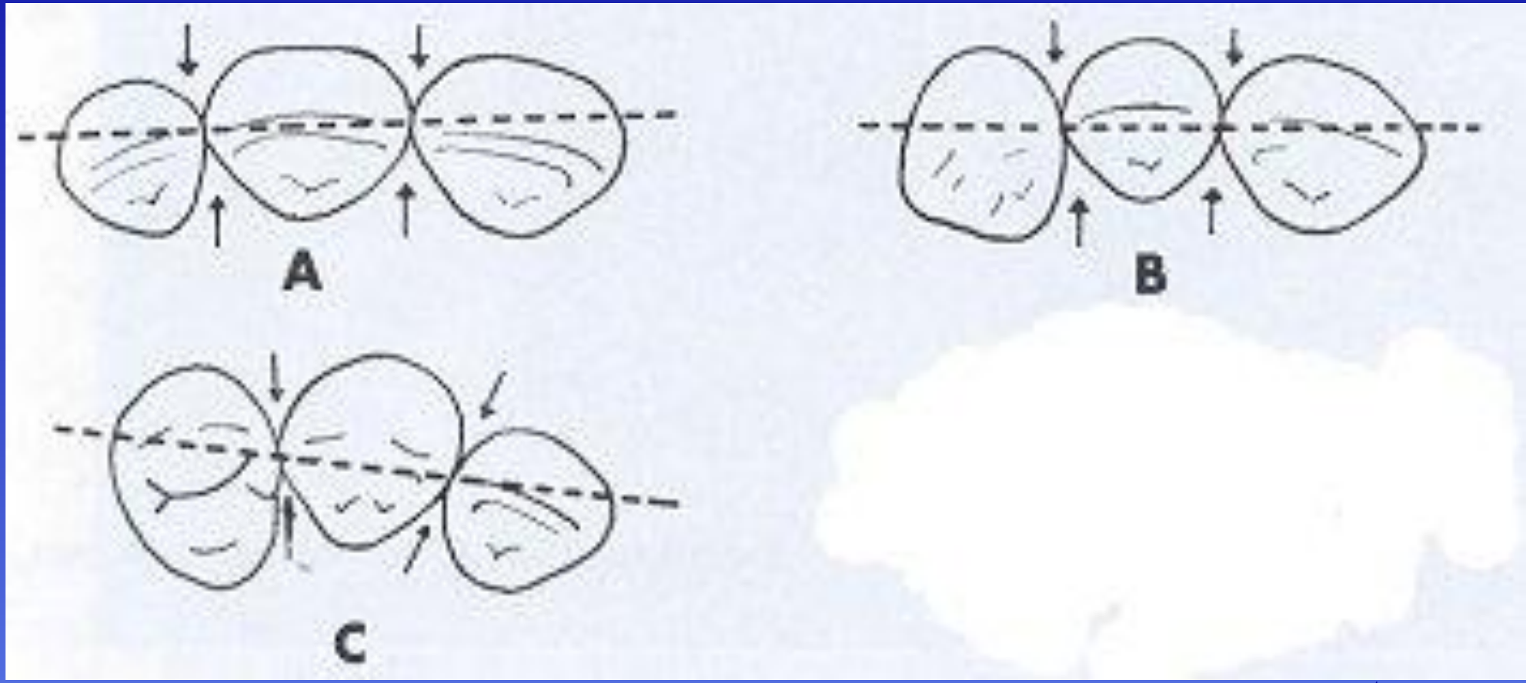


Location of Proximal Contacts:

- All Posterior Teeth: Contacts are located within the middle third
- Longer crowns on the premolars vs. the molars displays a molar proximal contact that is still within the middle third, but at a more occlusal level than what is evident on the premolars

Proximal Contact Areas

Incisal Perspective of Maxillary Anterior Teeth

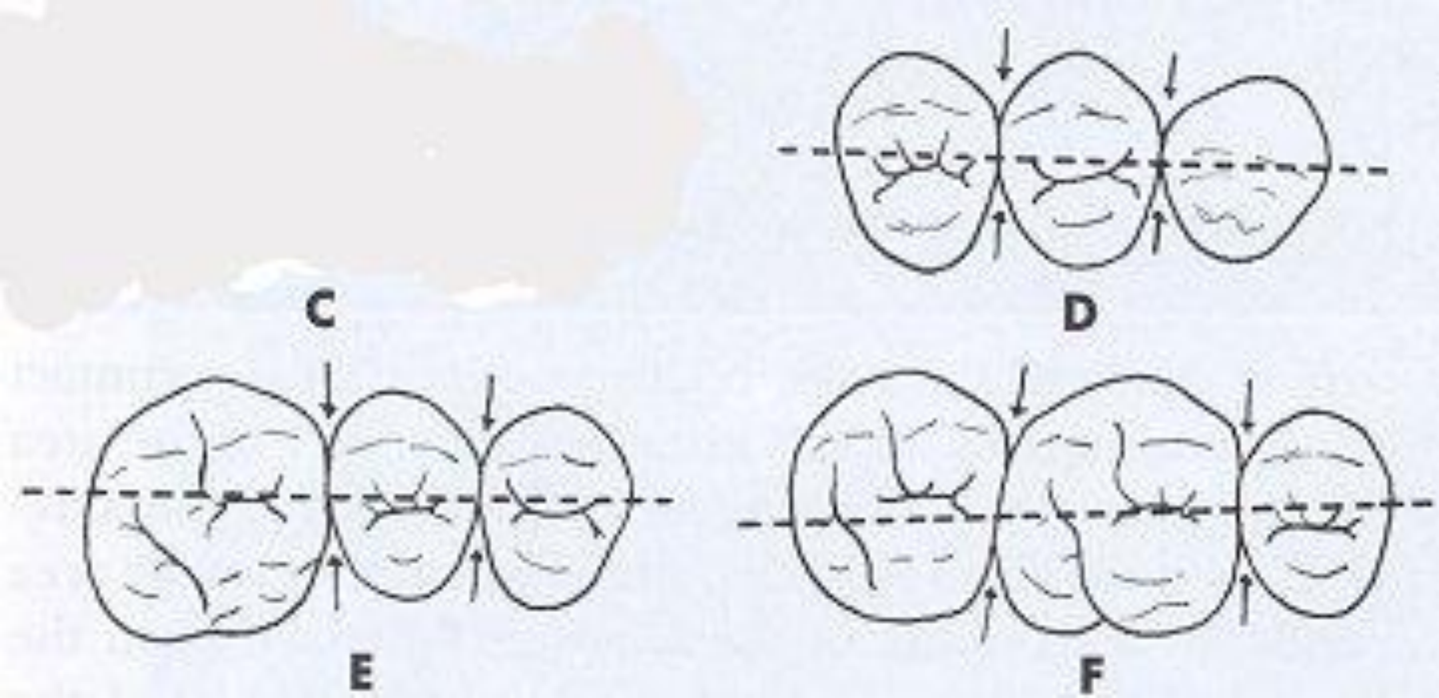


Location of Proximal Contacts:

- All Maxillary Anterior Teeth: Contacts are centered faciolingually, or considered equidistant from the facial and lingual profiles

Proximal Contact Areas

Occlusal Perspective of Maxillary Posterior Teeth



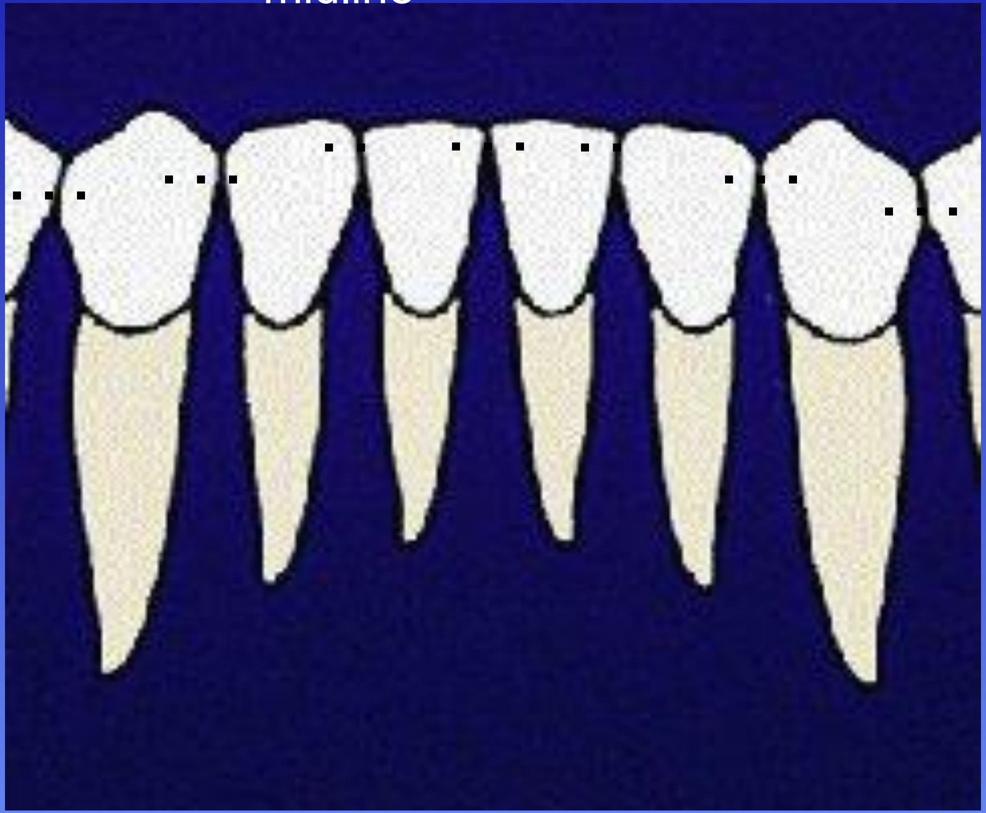
Location of Proximal Contacts:

- All Maxillary Posterior Teeth: Contacts are slightly facial to the line that divides the tooth in half, from a faciolingual perspective. This is more evident on the premolars than the molars. The contact between the first molar and second molar is almost centered faciolingually.

Proximal Contact Areas

Facial Perspective of Mandibular Anterior Teeth

The anterior proximal contact is most incisal at the midline

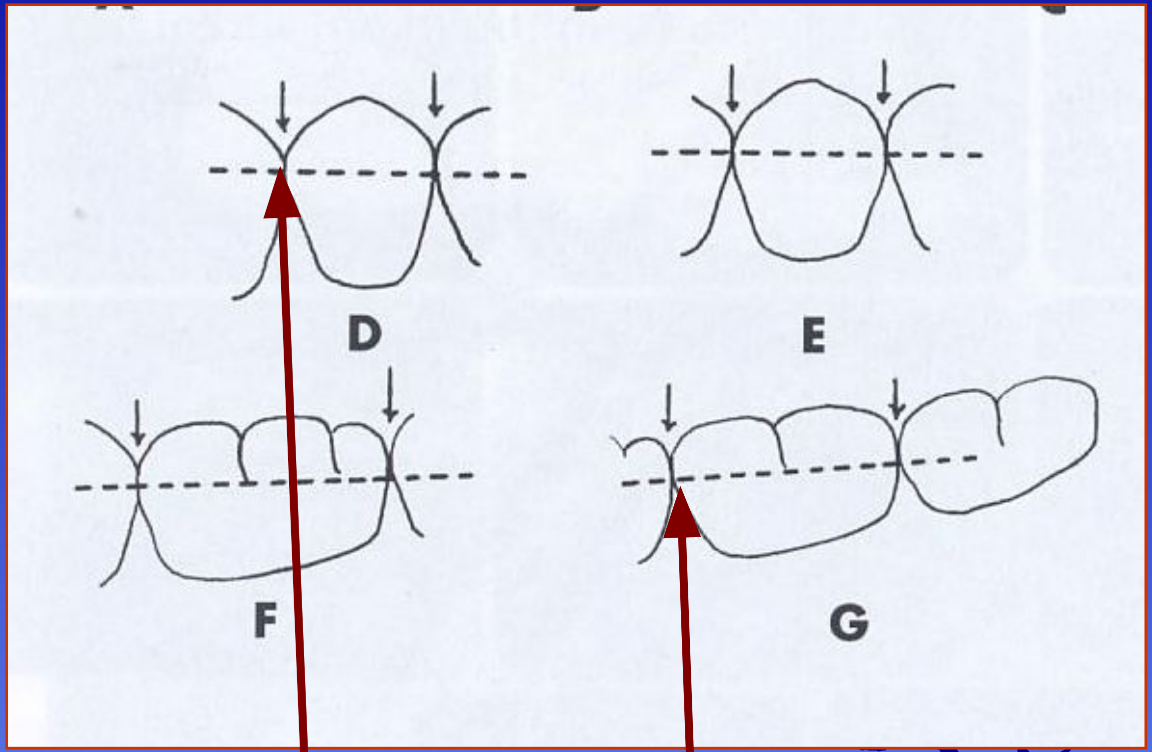


Location of Proximal Contacts:

>Midline: Incisal third
>Distal of Central: Incisal third
>Mesial of Lateral: Incisal third
>Distal of Lateral: Incisal third
>Mesial of Canine: Incisal third
>Distal of Canine: Middle third

Proximal Contact Areas

Facial Perspective of Mandibular Posterior Teeth



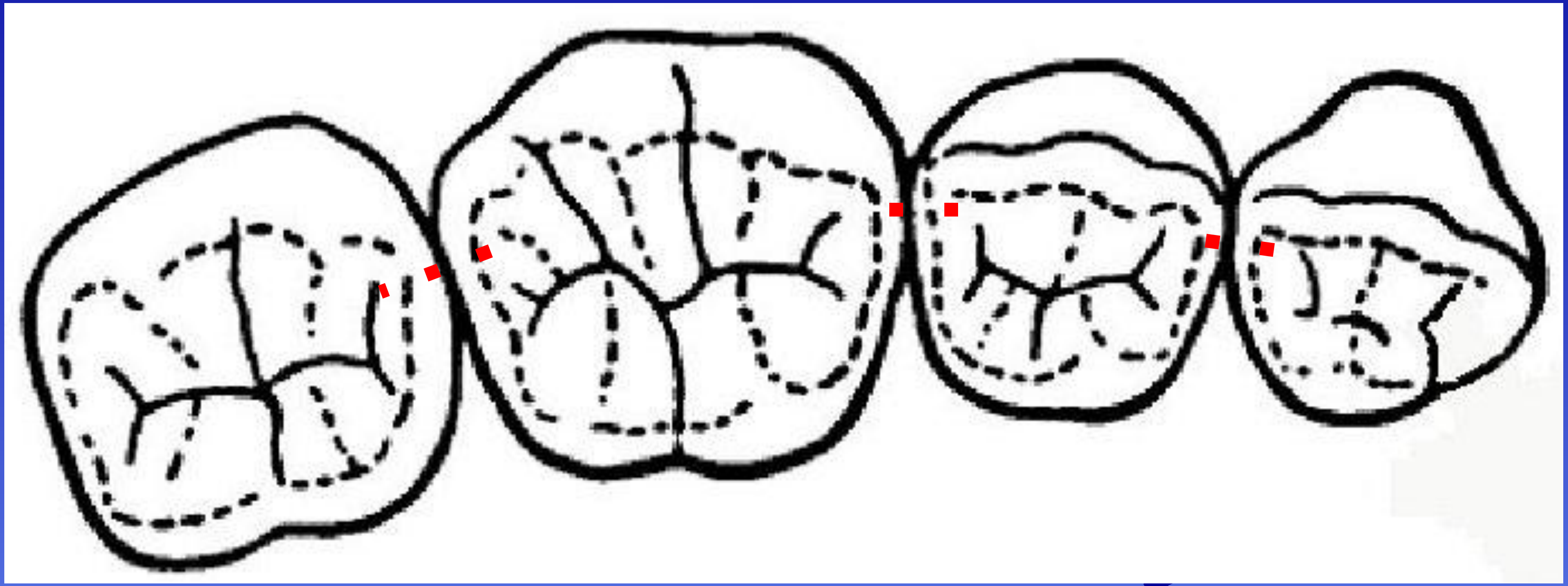
Location of Proximal Contacts:

>All Posterior Teeth: Contacts are located within the middle third

>Longer crowns on the premolars vs. the molars displays a molar proximal contact that is still within the middle third, but at a more occlusal level than what is evident on the premolars

Proximal Contact Areas

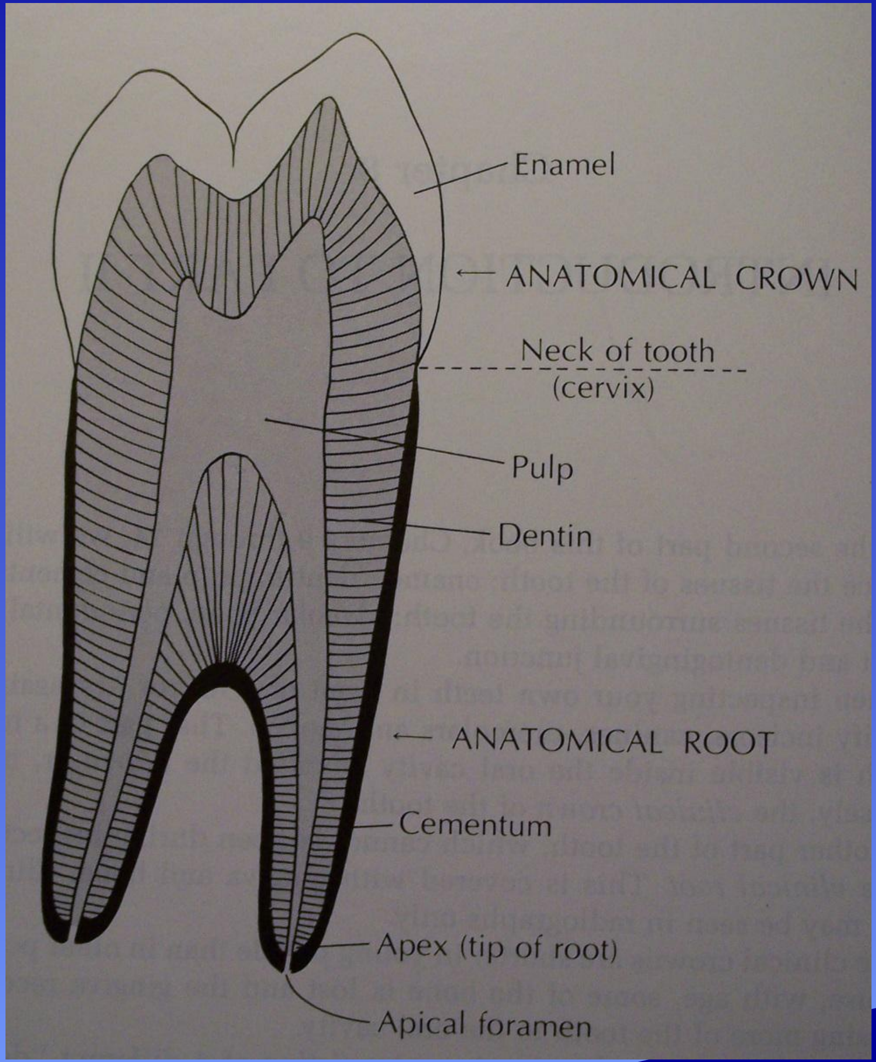
Occlusal Perspective of Mandibular Posterior Teeth



Location of Proximal Contacts:

>All Mandibular Posterior Teeth: Contacts are slightly facial to the line that divides the tooth in half, from a faciolingual perspective. The distal contact on the mandibular first molar is comprised completely of the distal cusp on the first molar. Mandibular molar contact area locations are slightly more facial than their maxillary counterparts.

Anatomical Structures of Teeth

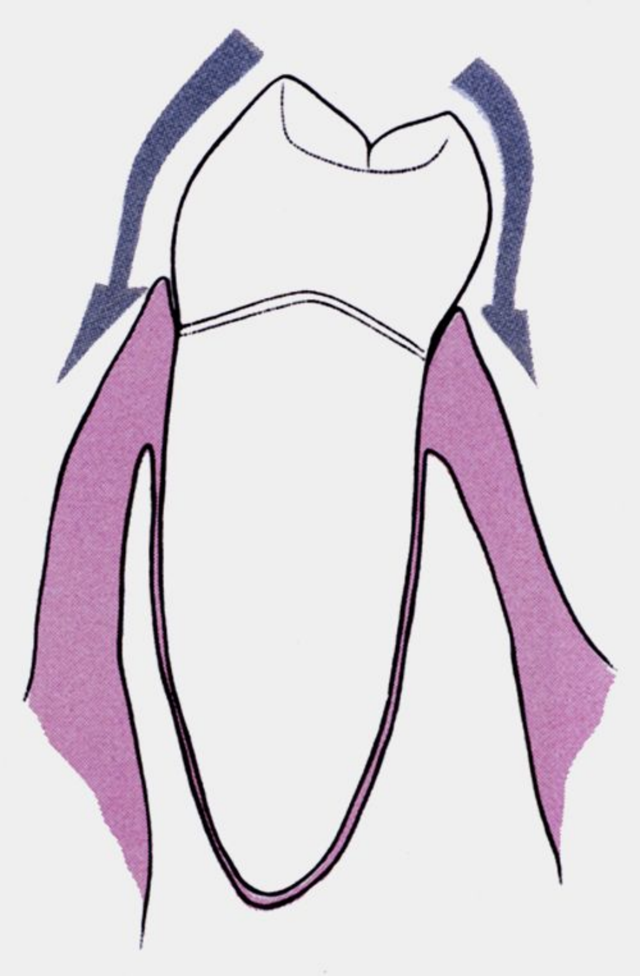


Anatomical entities to consider:

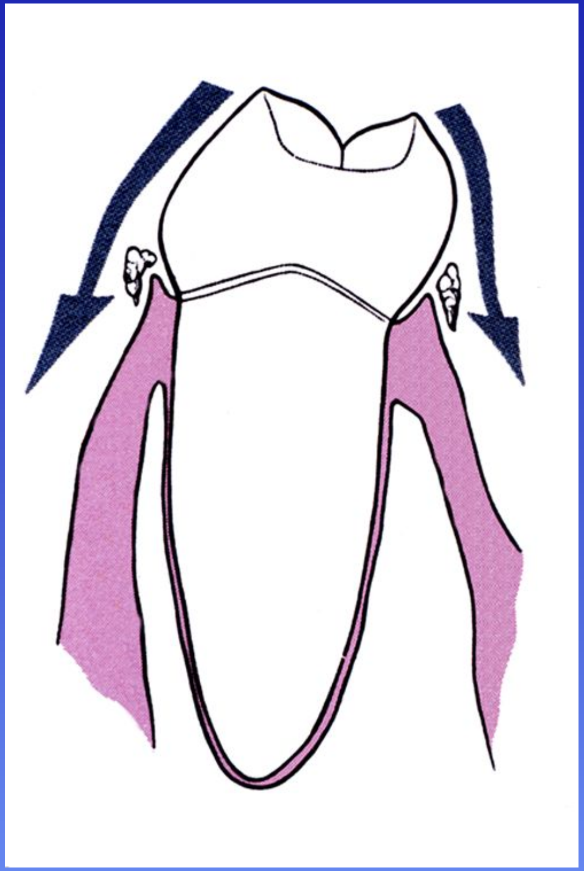
Enamel
Dentin
Pulp
Cementum
Apical Foramen
Cementoenamel Junction (CEJ)
Pulp Horn
Pulp Chamber
Pulp Canal

Height of Contour

- This is seen from the proximal view
- It is designated as the most protruded or bulkiest location on the facial and lingual surfaces
- It additionally serves a protective function for the gingival tissues and emergence profile feature for the tooth



Effects of Improper Height of Contour

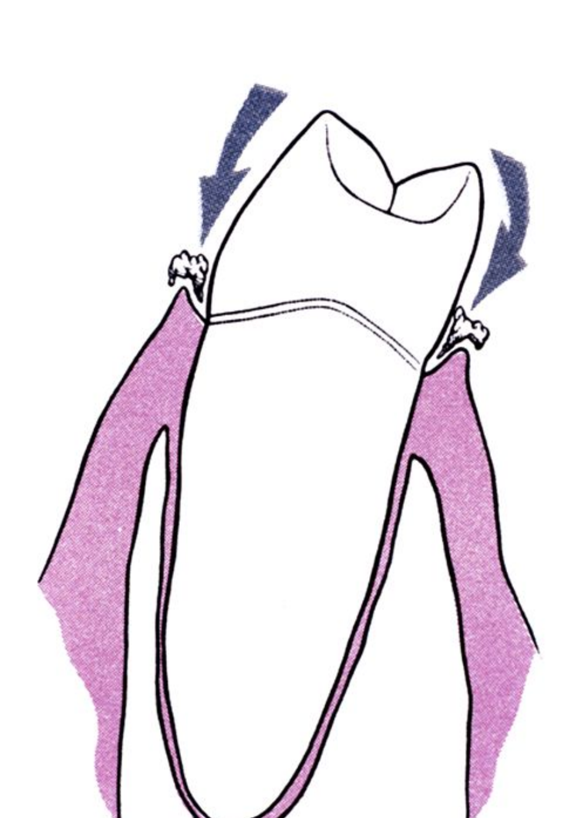


Characteristics of Healthy and Unhealthy Gingival Tissue

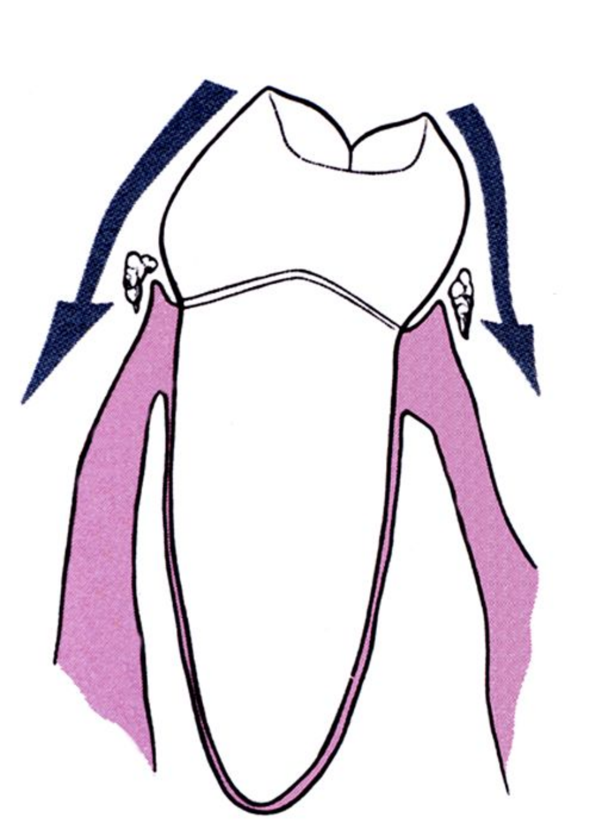
Healthy Gingival Tissues

Unhealthy Gingival Tissues

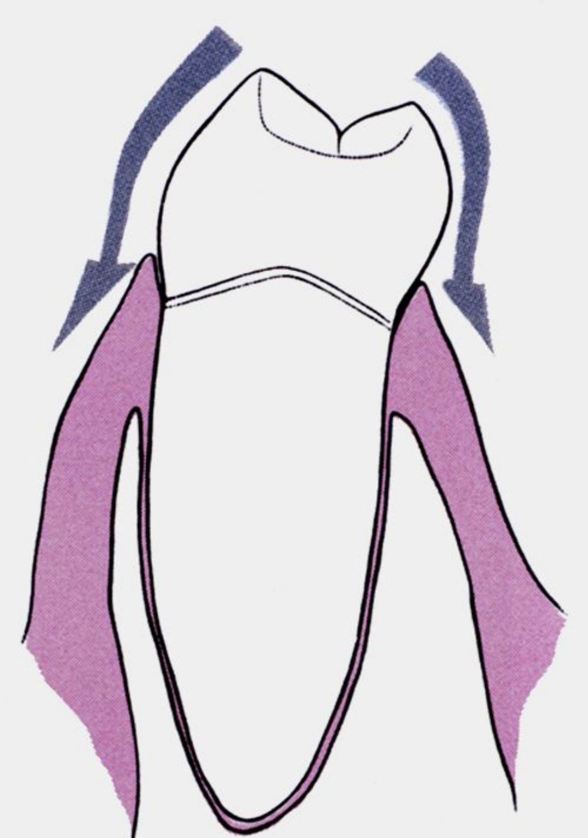
Proper Height of Contour



Insufficient Contour or Undercontoured

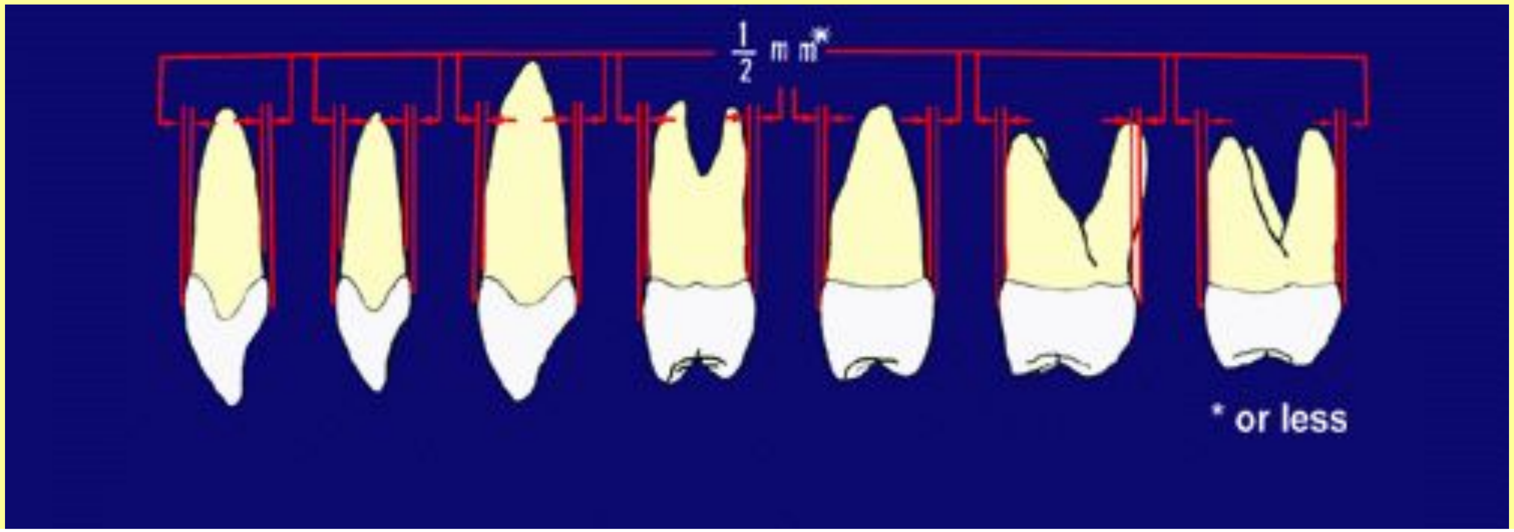


Over-contoured or bulky



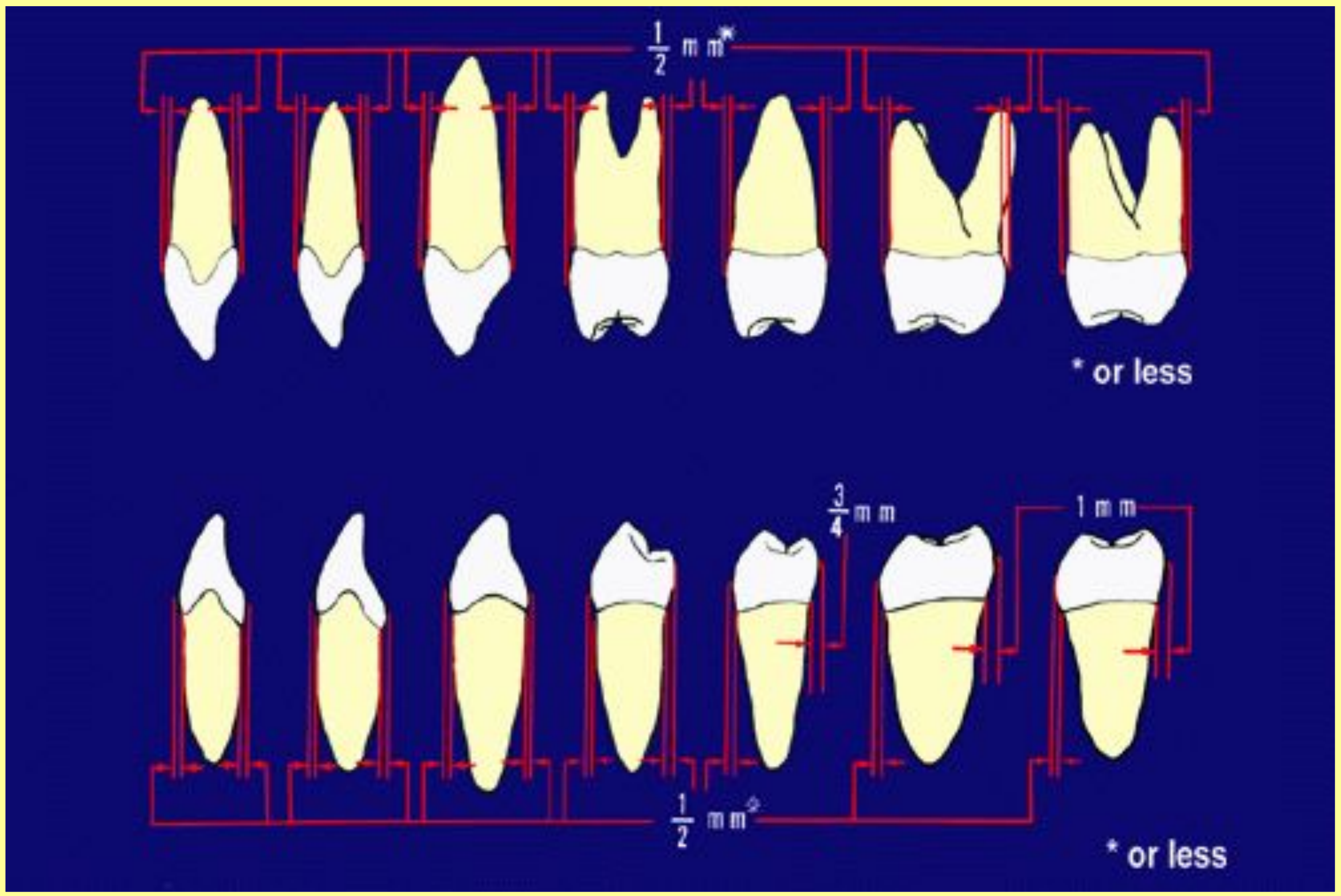
Just right!

Height of Contour- *Maxillary*



Maxillary Anterior Teeth	Amount of Contour
Facial surface	½ mm located in the cervical 1/3
Lingual surface	½ mm located in the cervical 1/3
Maxillary Posterior Teeth	**Amount of Contour**
Facial Surface	½ mm located in the cervical 1/3
Lingual Surface	½ mm located in the middle 1/3

Measurements of the Heights of Contour

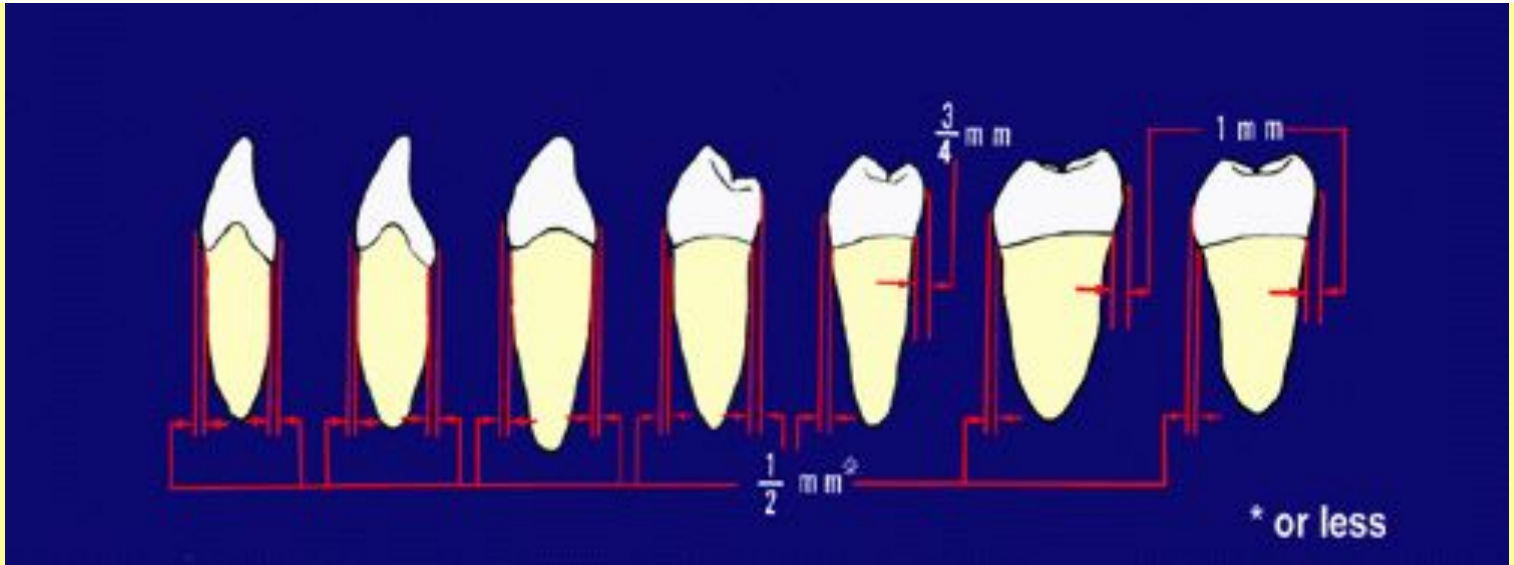


The most developed point on any crown surface

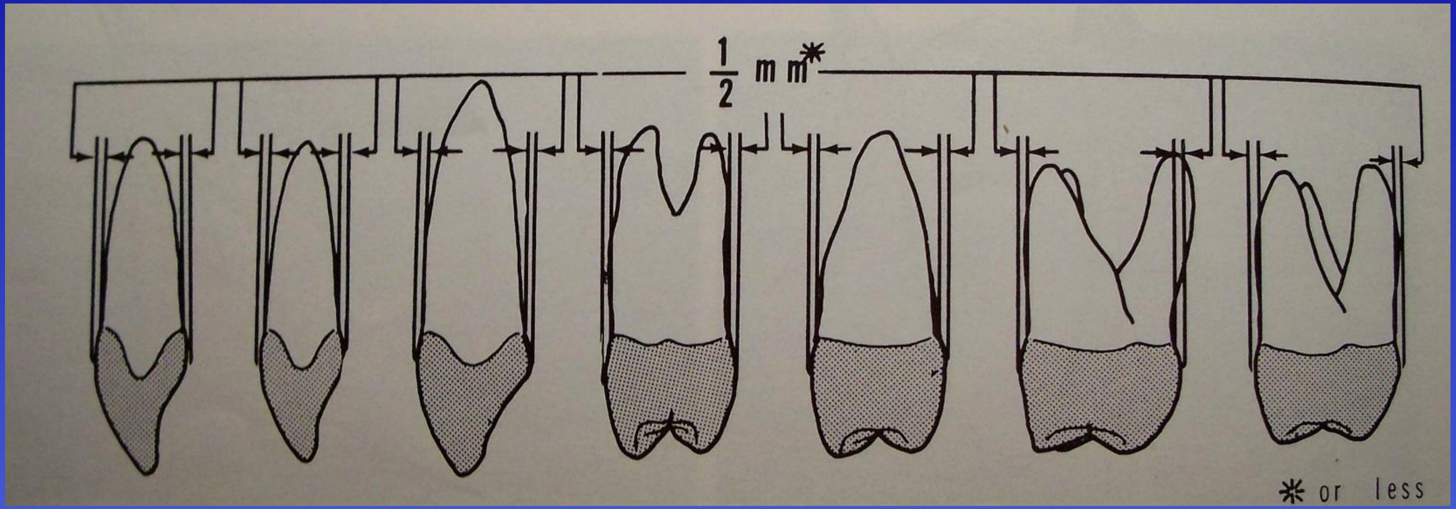
Height of Contour- Mandibular

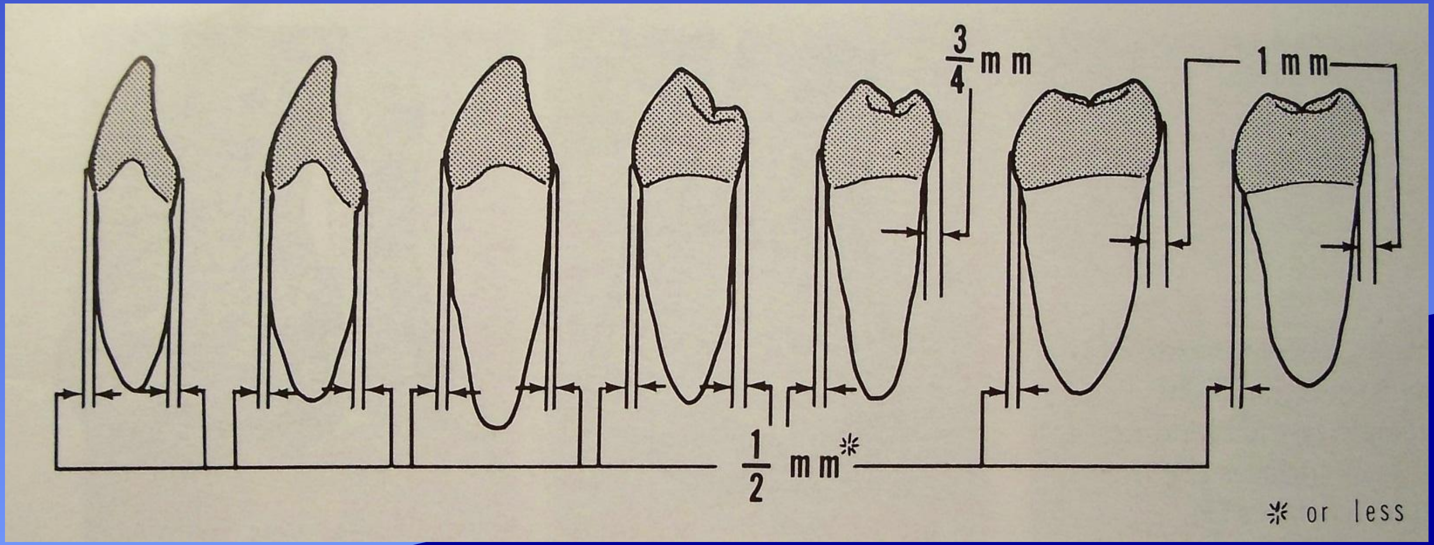
All facial surfaces ½ mm located in the cervical 1/3

Mandibular Anterior Teeth	Amount of Contour
Lingual surface	½ mm located in the cervical 1/3
Mandibular Premolars	Amount of Contour
Lingual surface	½ mm located in the middle 1/3
Mandibular Molars	Amount of Contour
Lingual surface	1 mm located in the middle 1/3



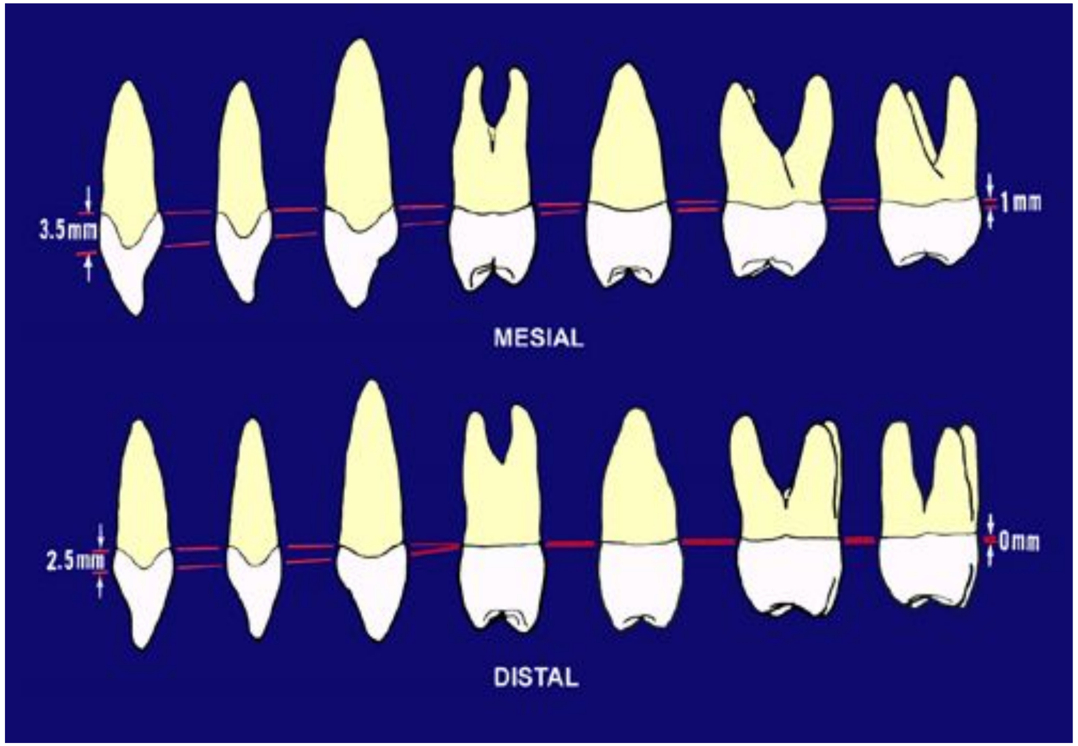
Comprehensive Measurements of the Crests or Heights of Contour



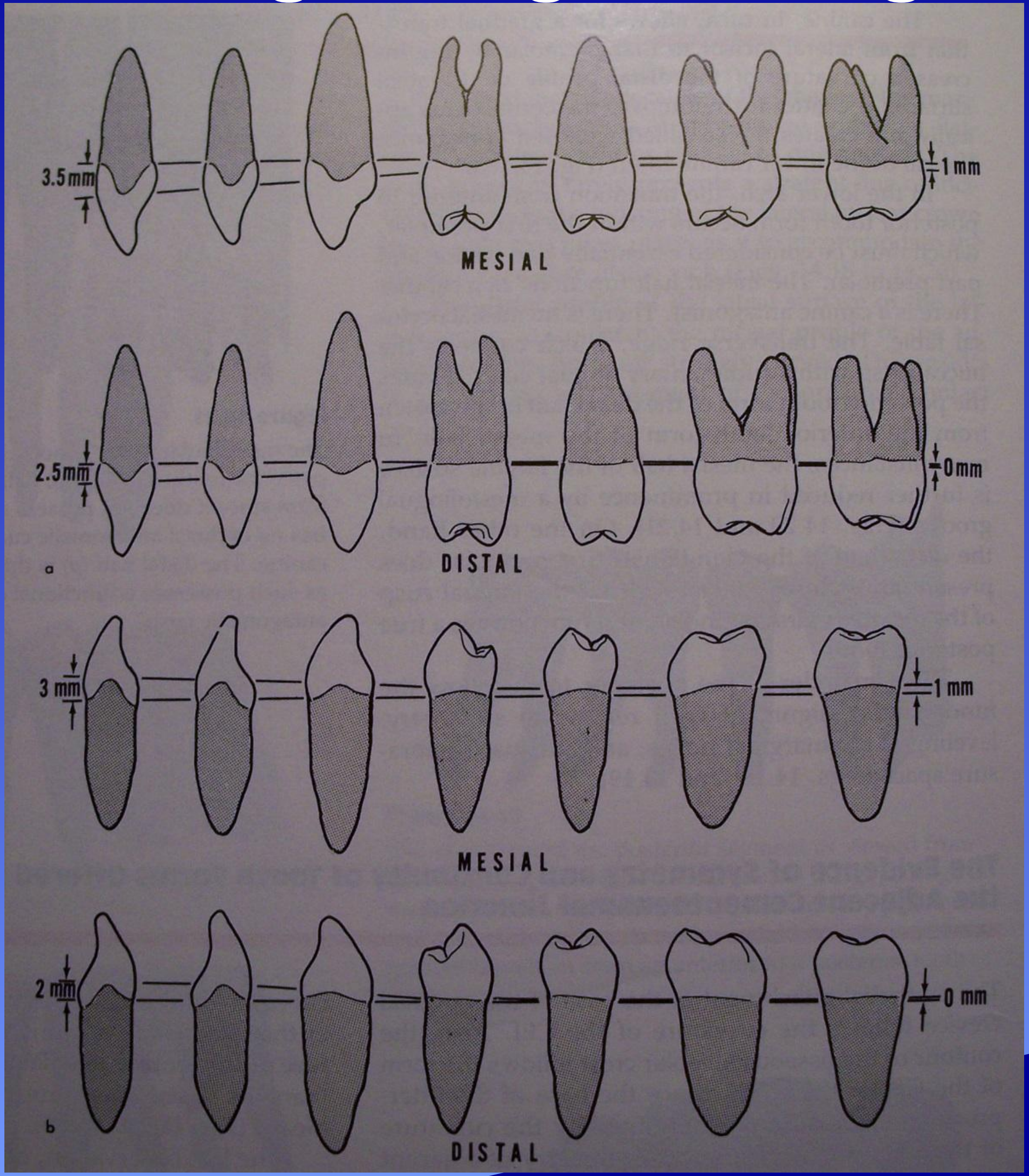


Measurements of the Height of Curvature of the Cervical Line

- The CEJ curves in a coronal direction on the proximal surface
- Anterior teeth have the most measurable curvature
- Molars have little or no curvature
- On an individual tooth, the curvature is always greatest on the mesial
- Mesial of the Maxillary Central is 3.5 mm (largest)
- Distal of Maxillary Central is 2.5 mm



Correlations Regarding the Height of Curvature of CEJ

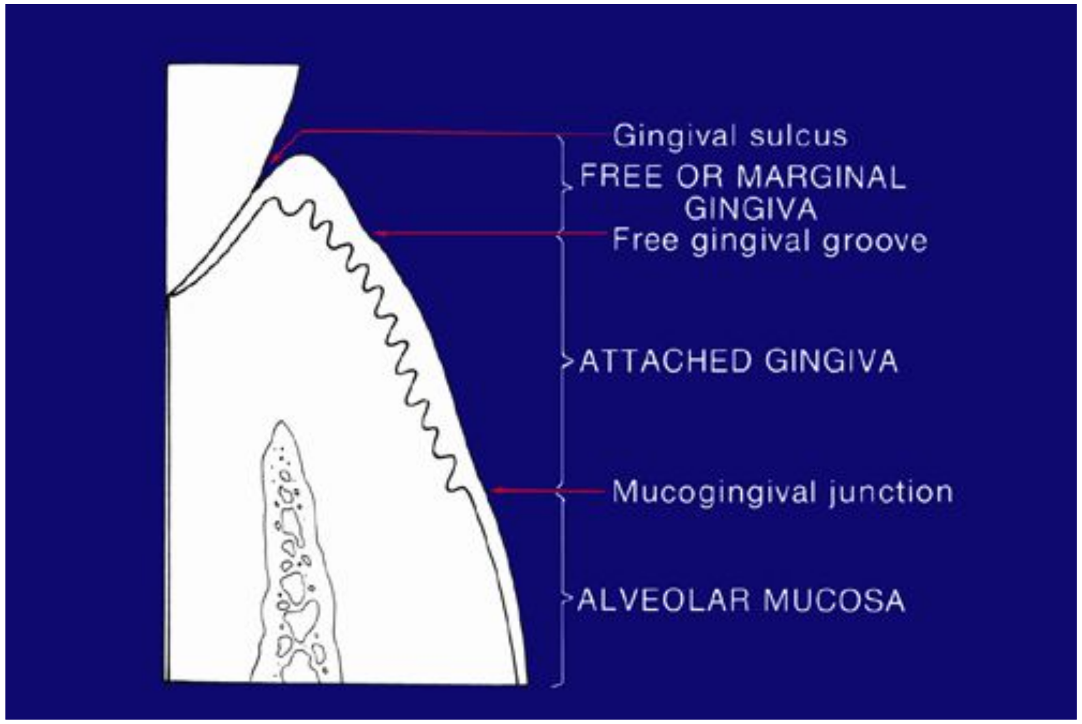


>Maxillary teeth have more height of curvature of the CEJ than mandibular teeth

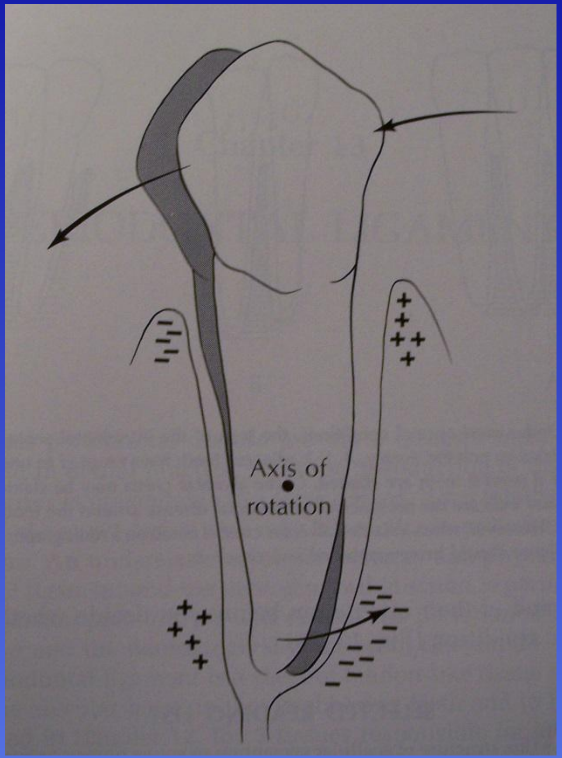
>There is a correlation between the height of curvature of the cervical line, the size of the cervical and occlusal/incisal embrasures, and the cervico-incisal or cervico-occlusal height of the contact area

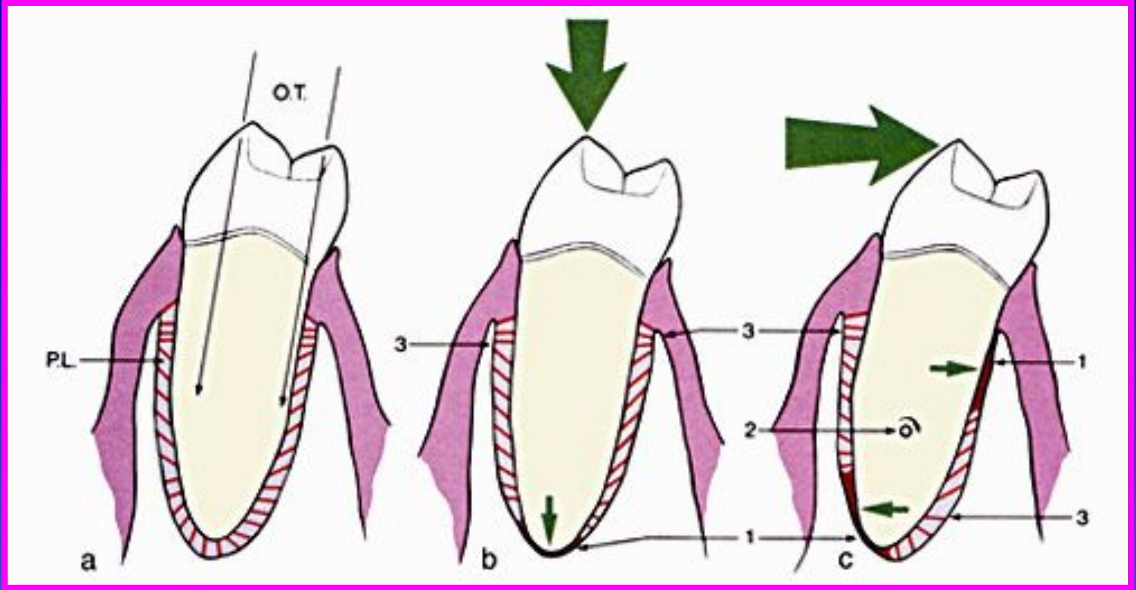
Gingival Tissues

- Gingiva
  - Forms a seal around the entire circumference of each tooth
  - Attached gingiva and free gingiva produce the seal
  - 3 mm pocket depth is ideal
  - Optimum Oral hygiene is dependent upon the integrity of these tissues, long-term



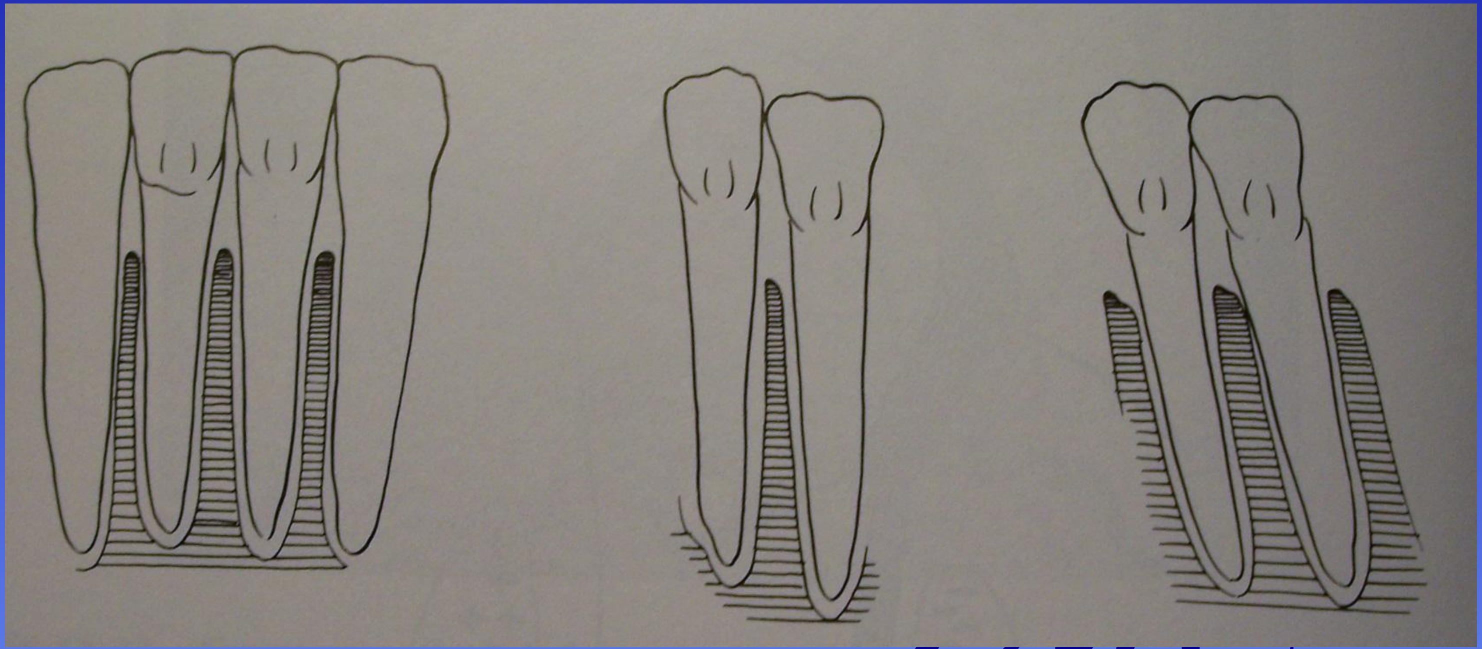
Supporting Structures of the Dentition





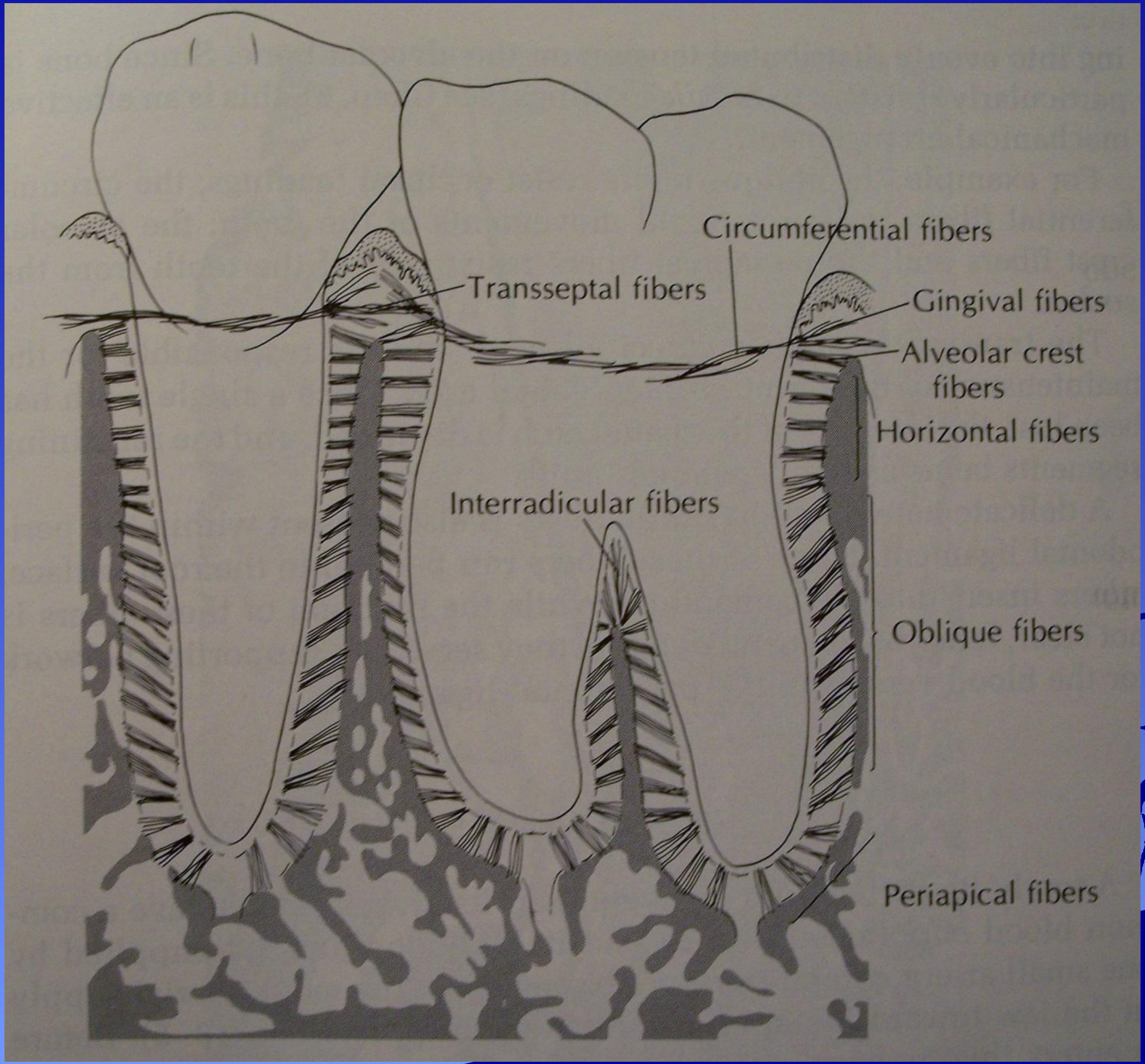
- **Periodontal Ligament**
  - Attaches tooth to alveolar bone
  - Acts as a shock absorber
  - Innervated
  - Longitudinal vs. lateral forces
  - Orthodontic forces applied to crown and the effect on periodontal fiber and osseous tissue in the alveolus

Changes in Interseptal Bone Integrity because of Arch Alignment



As the arch alignment is altered (either Orthodontically or because of occlusal imbalances) the teeth will shift positions and the integrity of the height of the alveolar bone may be affected

Location of Periodontal Fibers

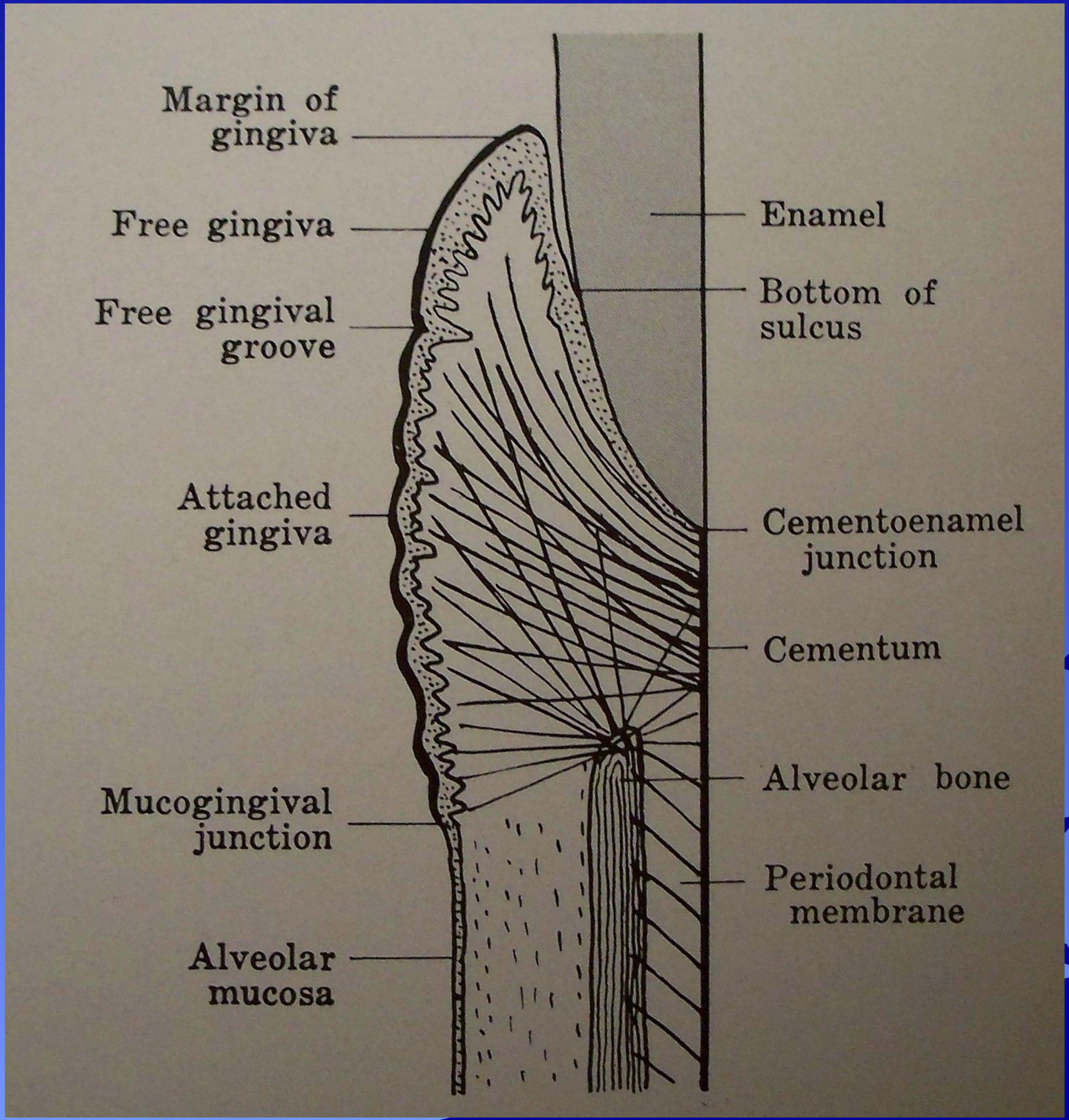


Names of Fibers:

>Gingival
>Alveolar crest
>Horizontal
>Oblique
>Periapical
>Interradicular
>Transseptal
>Circumferential

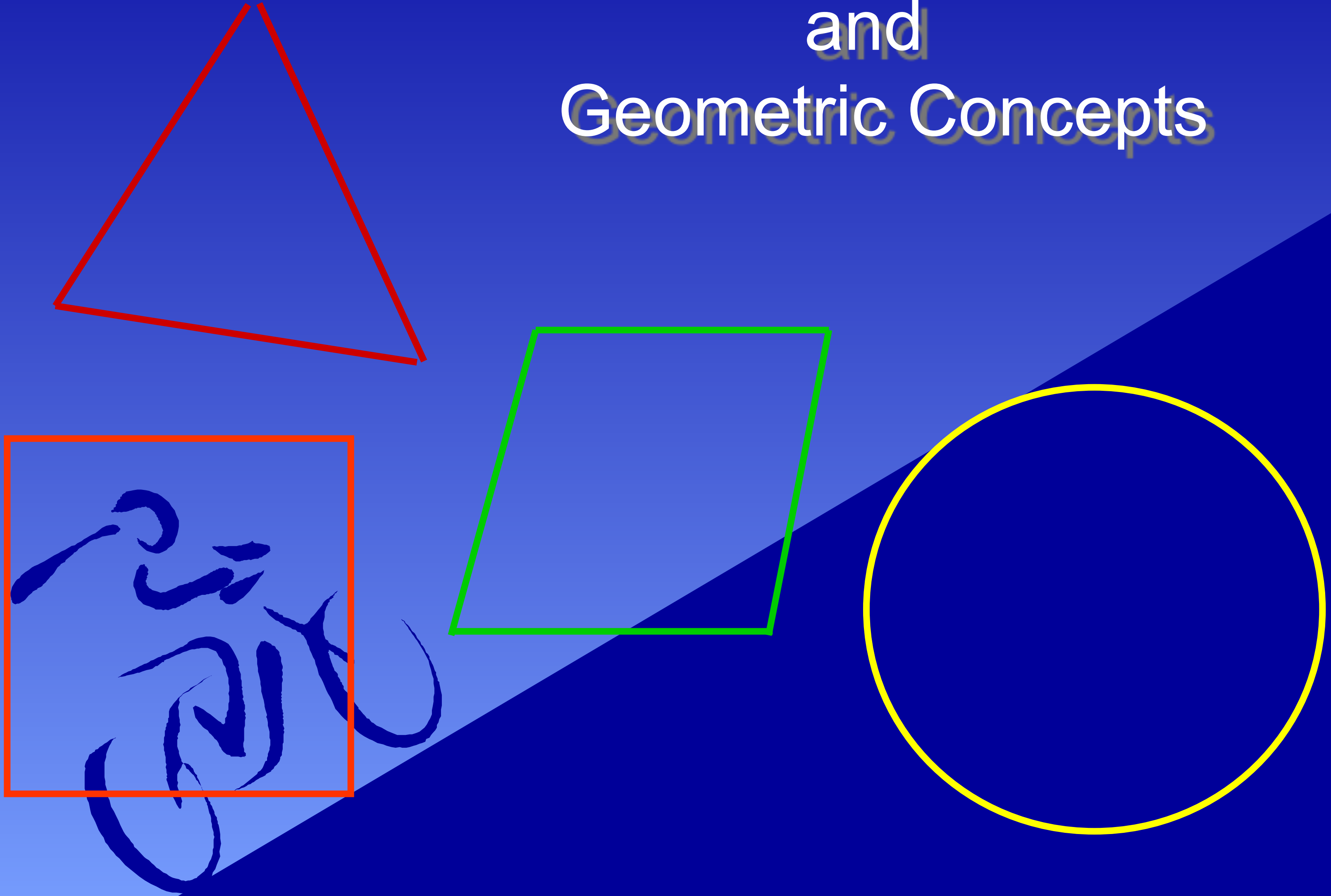
Oblique fibers are known to resist forceful impaction of a tooth into the alveolus, consistent with many types of trauma.

Periodontal Fibers at the CEJ



Periodontal disease will affect the integrity of the periodontal fibers at level of the CEJ (circumferential, alveolar crest, transseptal and horizontal) and lead to a degradation of the crest of the alveolar bone

Dento-osseous structures
Lobe Forms
and
Geometric Concepts



Learning objectives

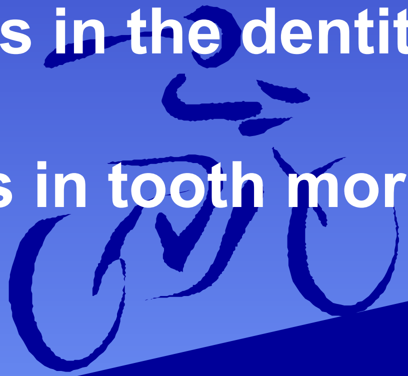
By the end of this lecture, students will be able to apply the concept of Lobes In tooth morphology

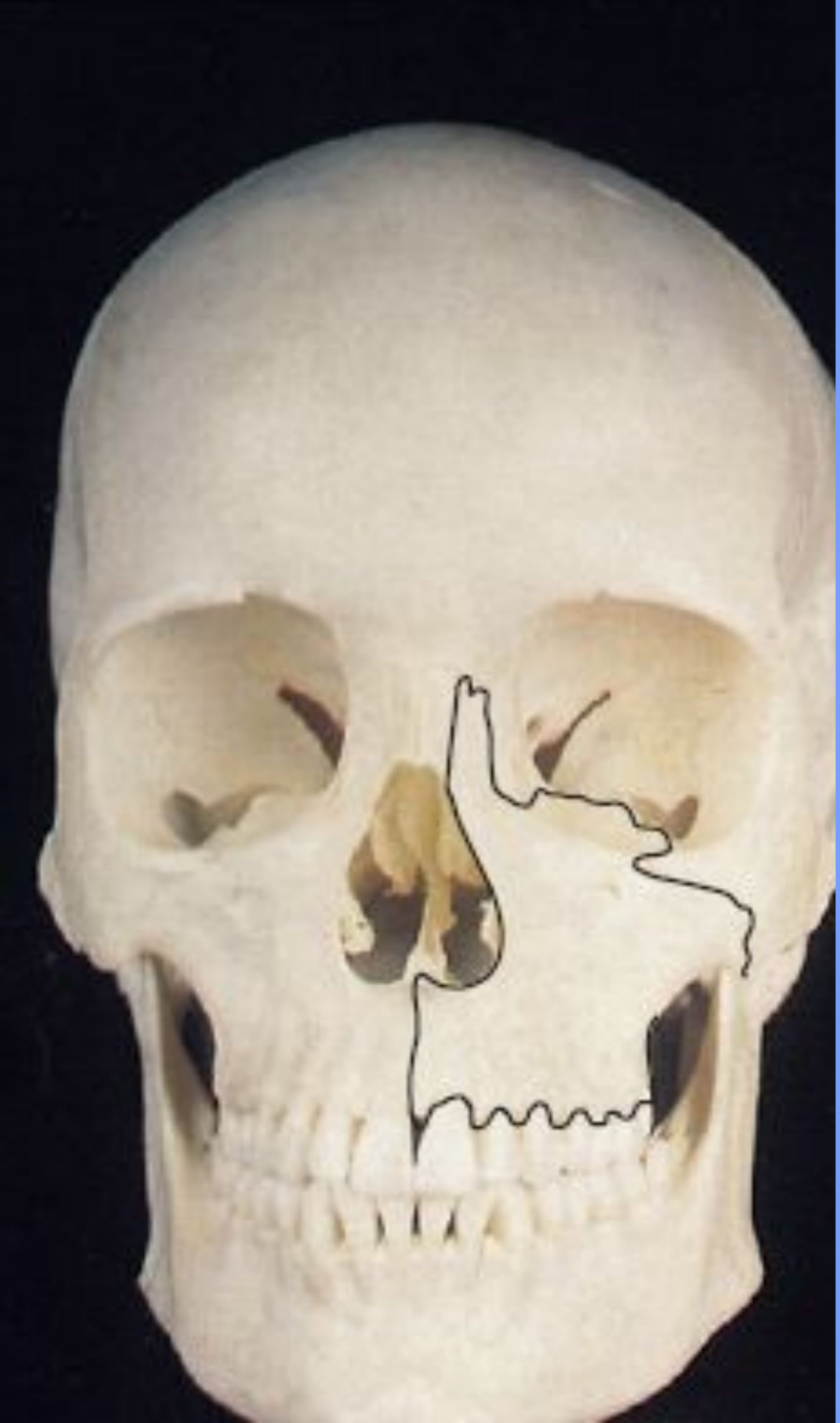
Relate osteology of maxillae & mandible to

Understand and relate Tooth Geometry to the morphology of teeth

Relate and locate Embrasures in the dentition

Apply Tooth Form Generalizations in tooth morphology





B

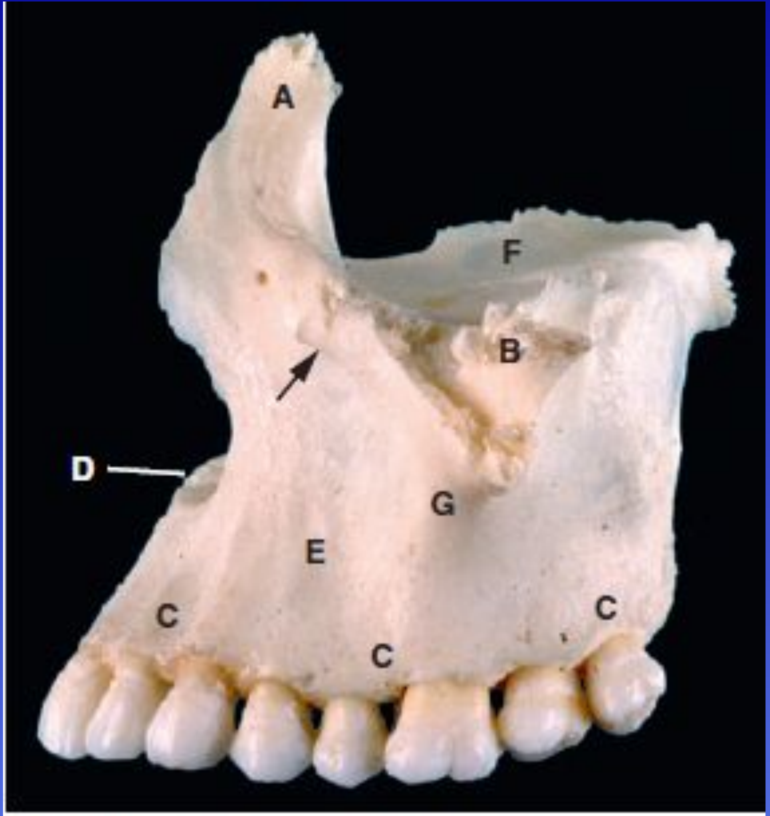


Fig. 2.2 Lateral aspect of the maxilla. A = frontal process; B = zygomatic process; C = alveolar process; D = site of anterior nasal spine; E = canine fossa; F = orbital plate; G = jugal crest. The infraorbital foramen is arrowed.

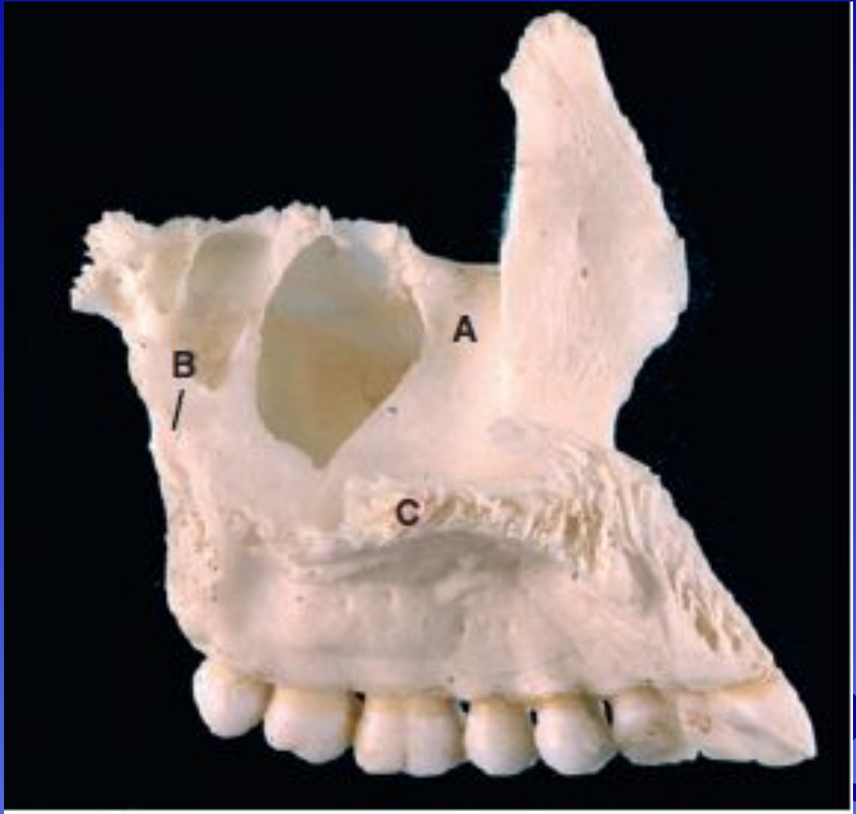


Fig. 2.3 Medial aspect of the maxilla. A = lacrimal groove; B = palatine groove; C = palatine process of maxilla. Note the large opening into the maxillary sinus.

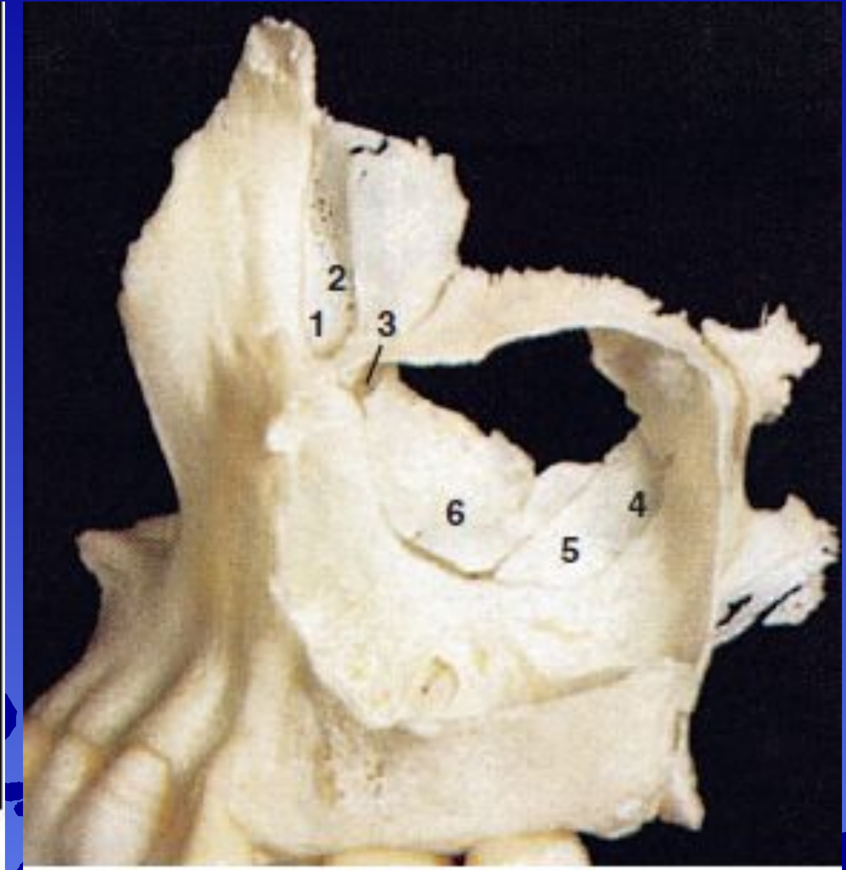
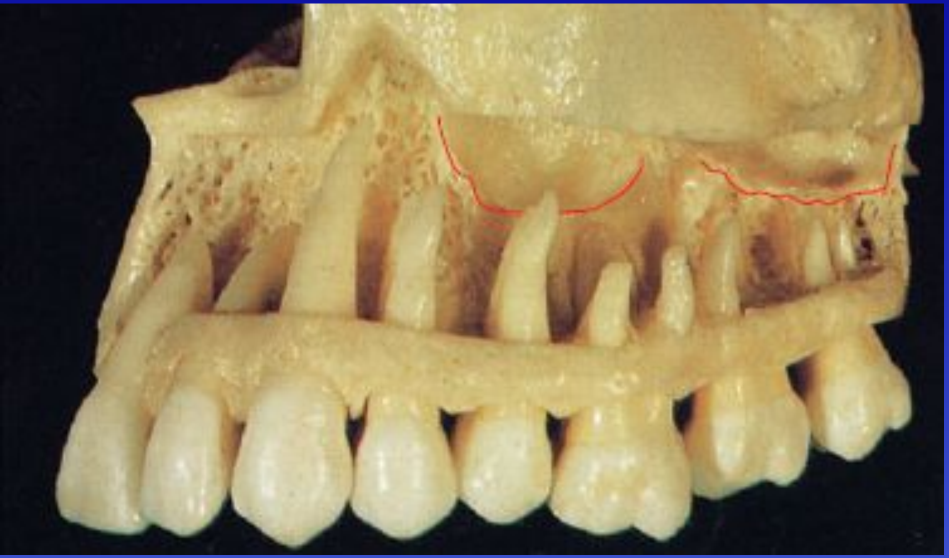
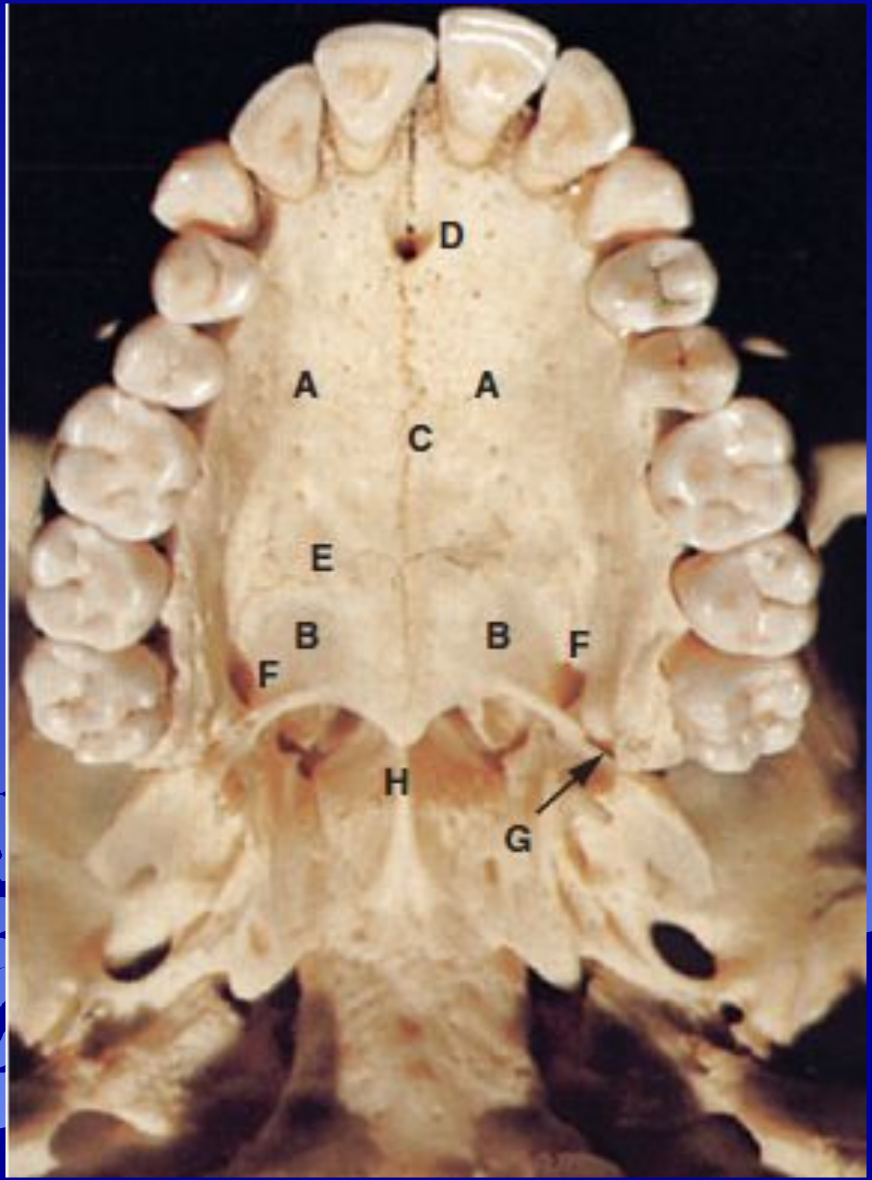


Fig. 2.4 Osteology of the maxillary air sinus showing adjacent bones reducing the size of the ostium. 1 = lacrimal groove of maxilla; 2 = lacrimal groove; 3 = lacrimal bone; 4 = ethmoid bone; 5 = palatine bone; 6 = inferior nasal concha. Courtesy Prof. R M H McMinn.




A
B
C
D


D
A
A
C
E
B
B
F
F
H
G

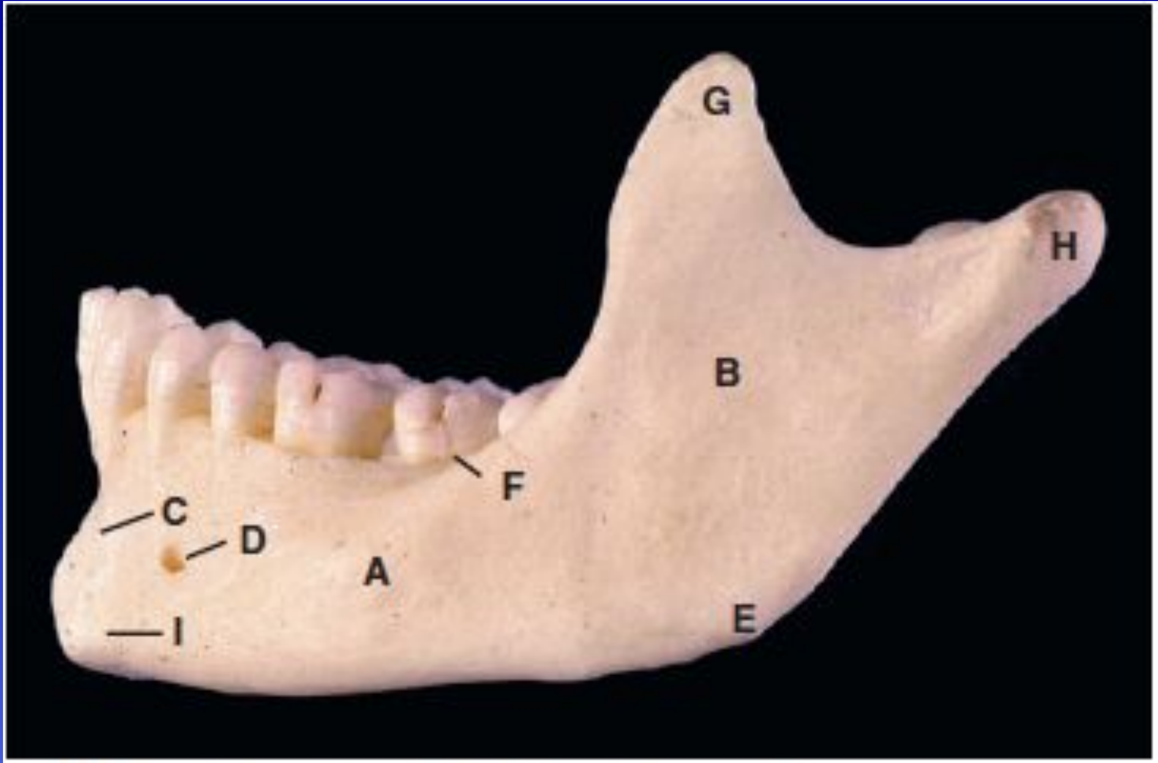


Fig. 2.8 Lateral aspect of the mandible. A = body; B = ramus; C = incisive fossa; D = mental foramen; E = angle; F = external oblique line; G = coronoid process; H = condyle; I = mental protuberance.

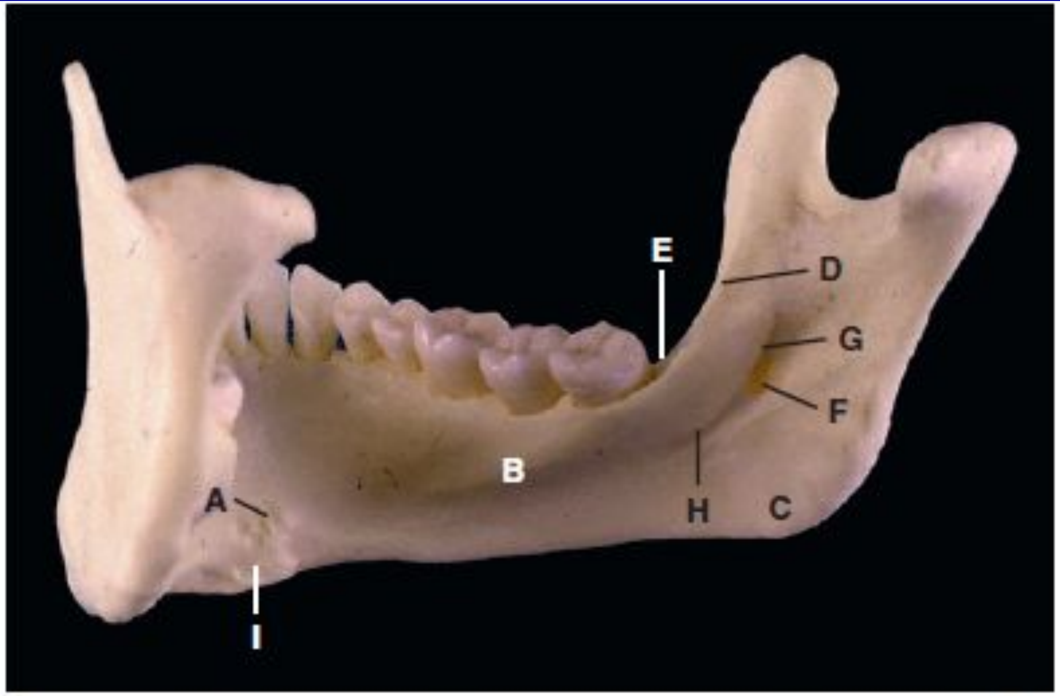


Fig. 2.9 Inner (medial) surface of the mandible. A = genial spines (tubercles); B = internal oblique ridge (mylohyoid ridge); C = attachment area for medial pterygoid muscle; D = temporal crest; E = retromolar triangle; F = mandibular foramen; G = lingula; H = mylohyoid groove; I = digastric fossa.

Fig. 2.11 The mandibular alveolus and the arrangement of the tooth sockets. Note that the left secon
mandibular molar has previously been extracted and the socket has healed.

Fig. 2.10 Lateral view of the mandible, showing the roots of the teeth and the relationship to the m
canal (A). B = mandibular foramen.

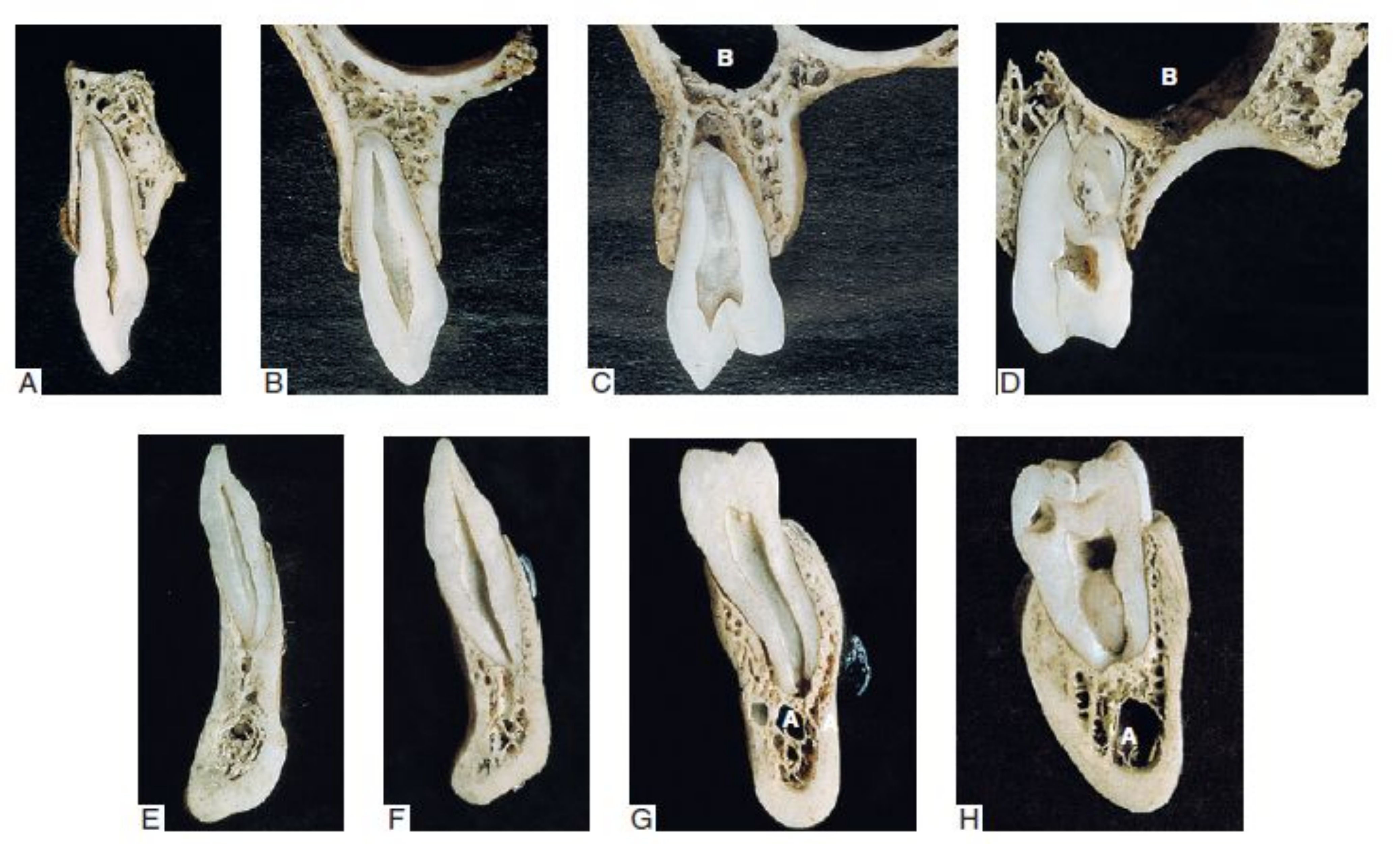
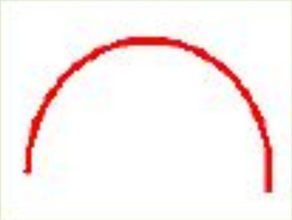


Fig. 2.12 Buccolingual sections through the maxilla and mandible demonstrating the distribution of alveolar bone in relation to the roots of the teeth. A = maxillary incisor region; B = maxillary canine region; C = maxillary premolar region; D = maxillary molar region; E = mandibular incisor region; F = mandibular canine region; G = mandibular premolar region; H = mandibular molar region. Note the relationship of the mandibular cheek teeth to the mandibular canal (A) and of the maxillary cheek teeth to the maxillary sinus (B). Courtesy the Royal College of Surgeons of England.

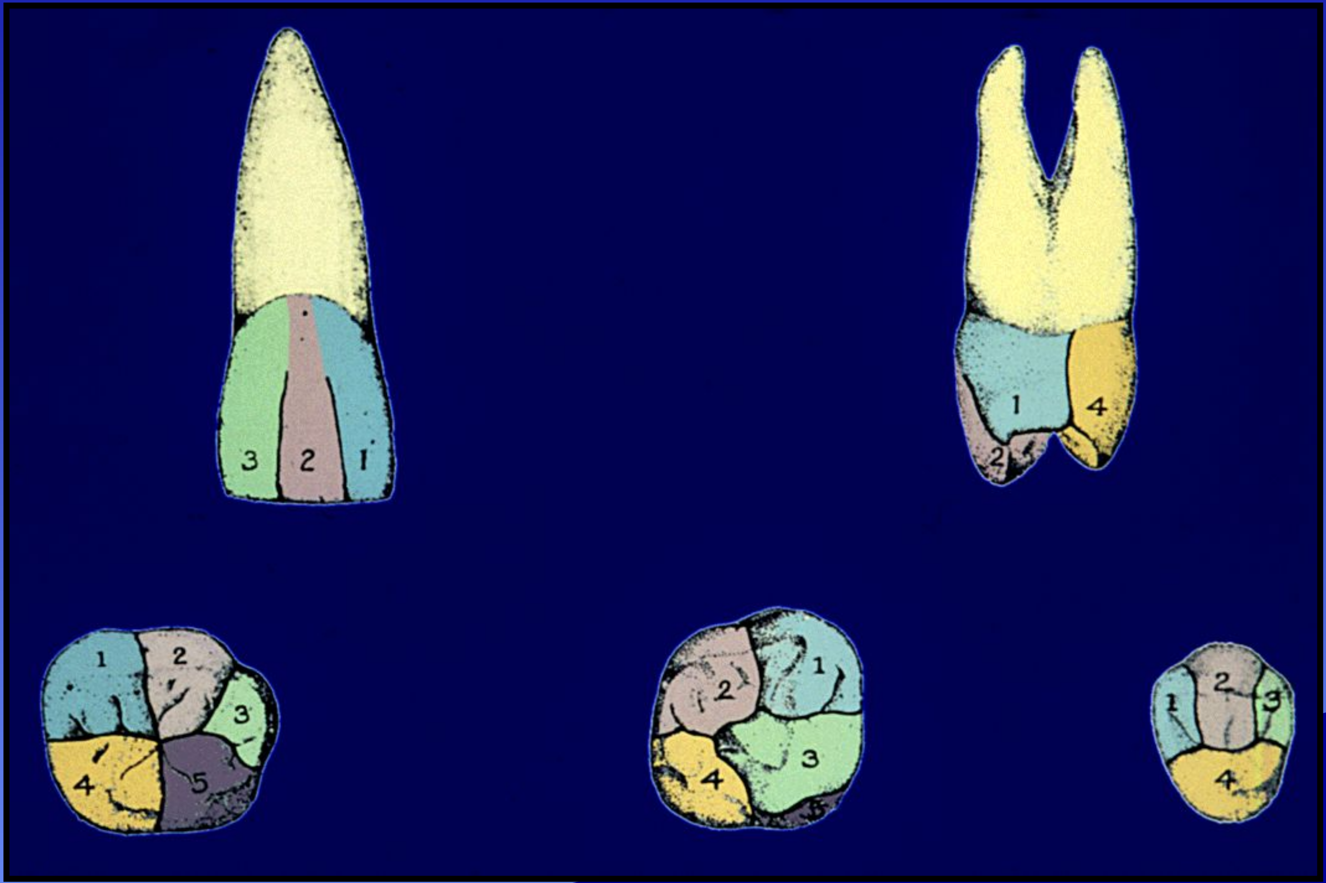
- **Convex:**
  - curving or bulging outward
  - rounded outward



- **Concave:**
  - curving inward like a cave or the inside of a spoon.
  - hollowed out

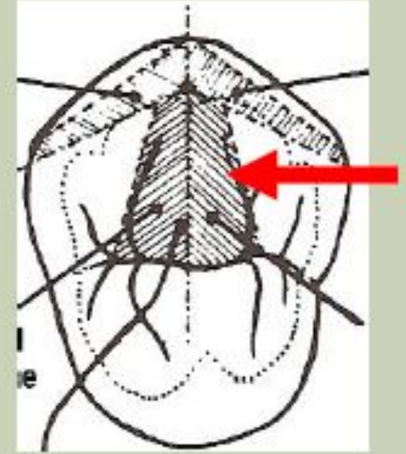
Developmental Depression

A defined, recessed, or concave area on the surface of a tooth found between two primary parts of a crown or root



Triangular Ridges

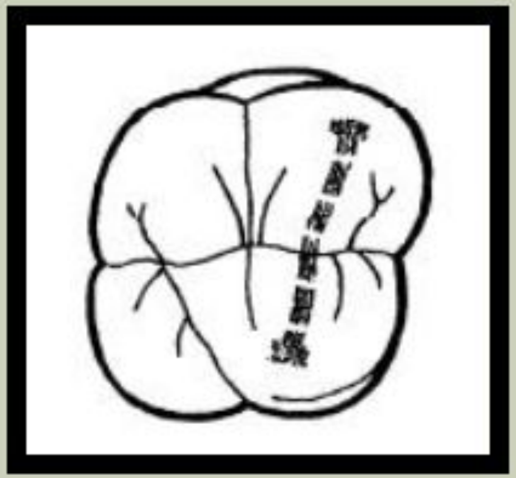
Convex Linear ridges which descend ***from*** the **CUSP TIPS** of posterior teeth ***toward*** the **central groove** of the occlusal surface.

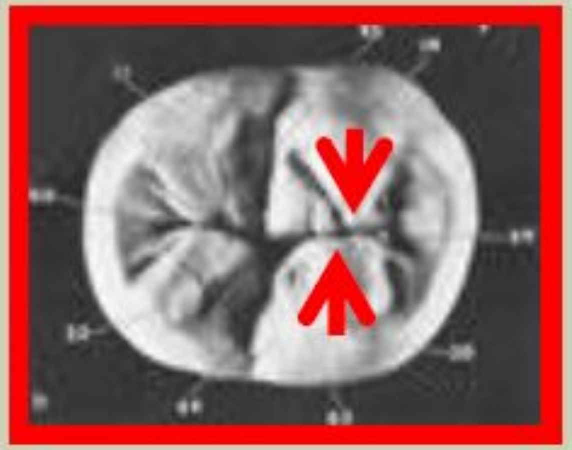


In cross section they are more or less triangular in shape.

TRANSVERSE RIDGES

A ***transverse ridge*** is a ***union*** of
2 triangular ridges that ***face each other; and are parallel with the marginal ridges*** on the occlusal table of a posterior tooth.

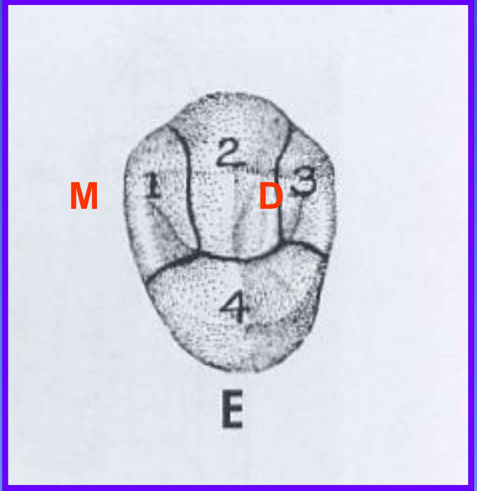


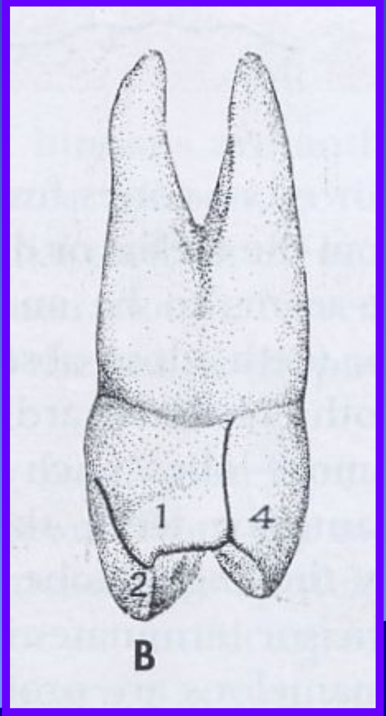


Permanent Teeth

- **Develop from 4 or more centers of calcification ("lobes")**

- **Note the developmental grooves or depressions marking the division of the lobes as they coalesce**

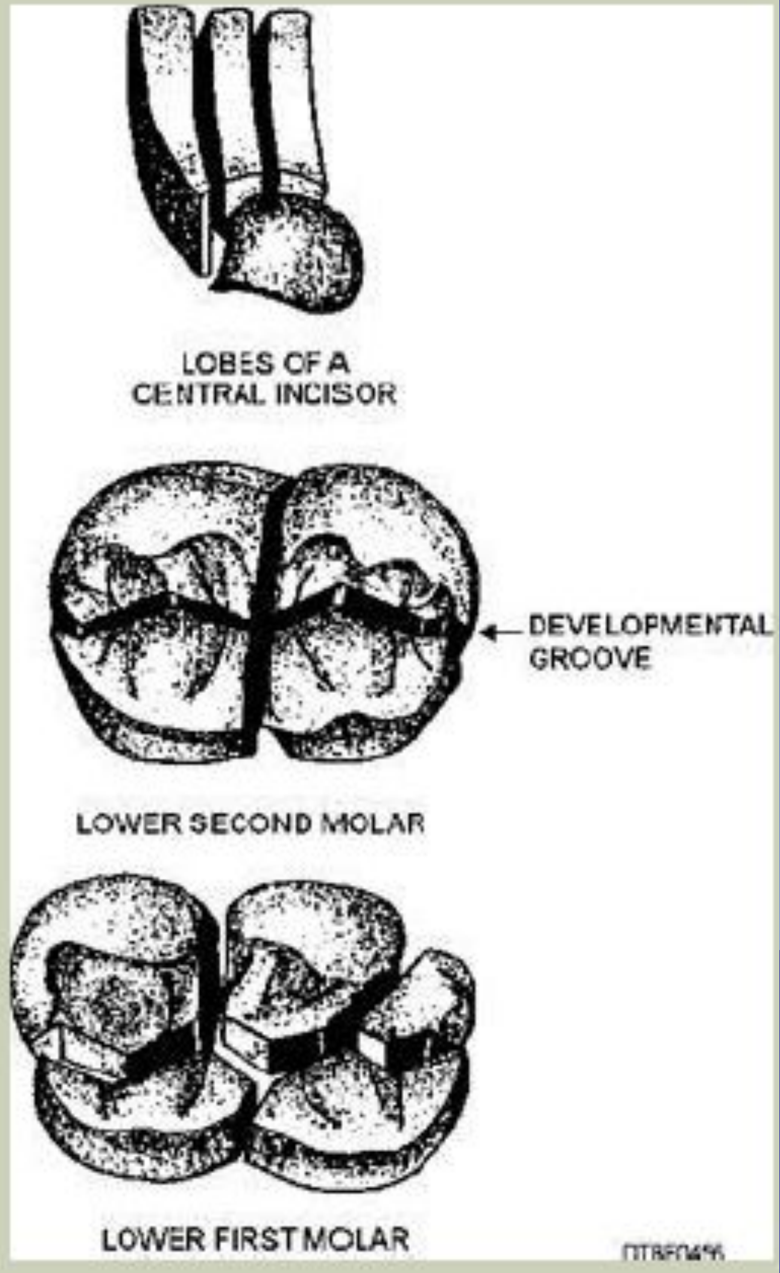




Names of lobes on Anteriors & Premolars:

1 = mesial or mesiofacial
2 = middle
3 = distal or distofacial
4 = lingual

- **Lobe** — one of the **primary divisions** of a crown; all teeth **develop** from four or five lobes.
- Lobes are usually separated by readily identifiable **developmental grooves**



Permanent Teeth

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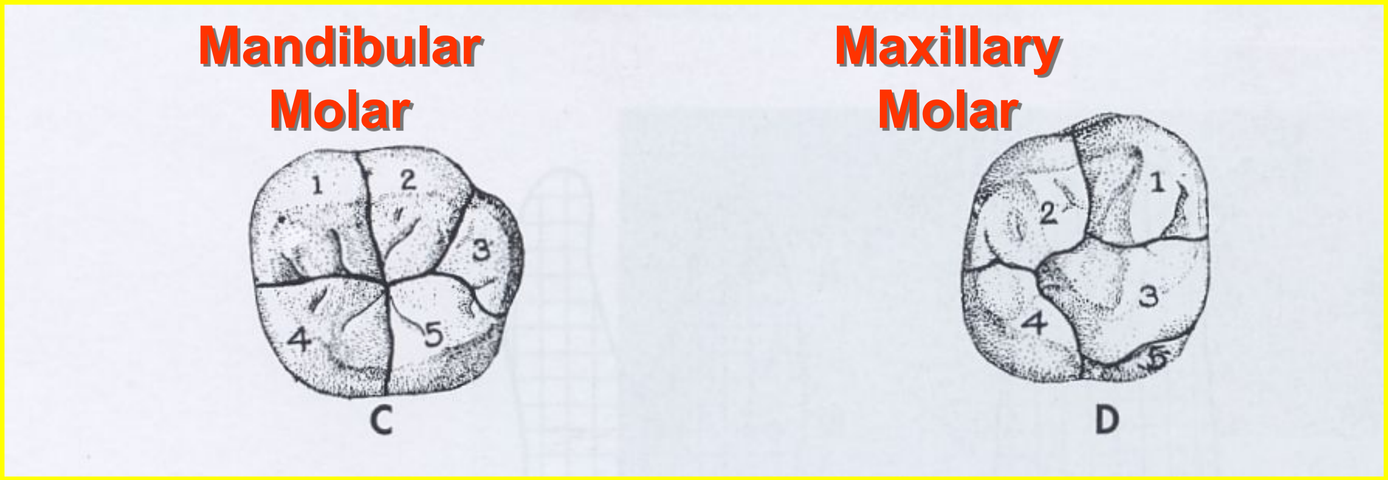
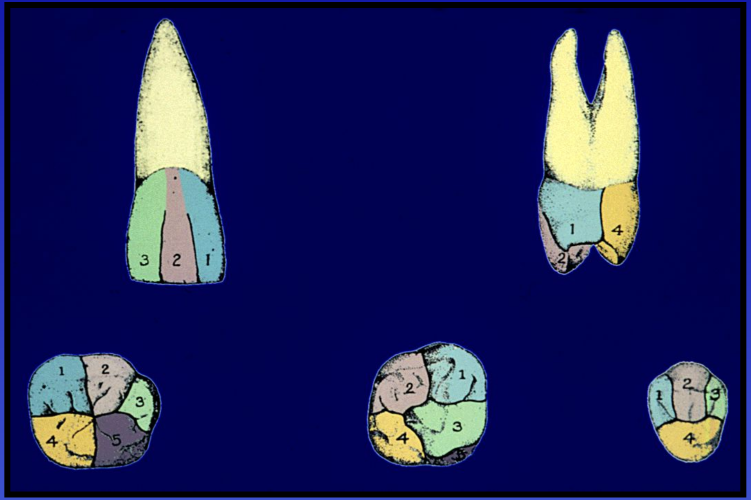


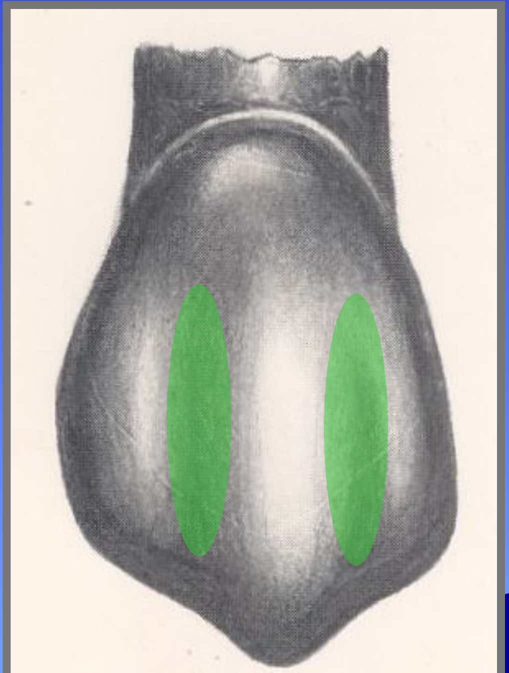
Names of lobes on Mandibular Molars:
1 = mesiofacial
2 = distofacial
3 = distal
4 = mesiolingual

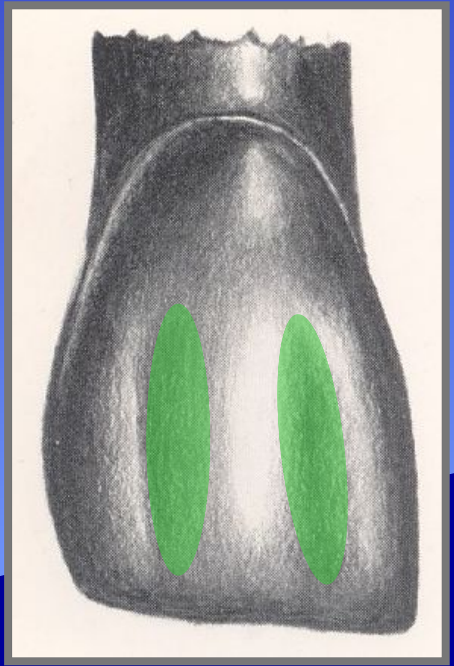
Names of lobes on Maxillary Molars:
1 = mesiofacial
2 = distofacial
3 = mesiolingual
4 = distolingual

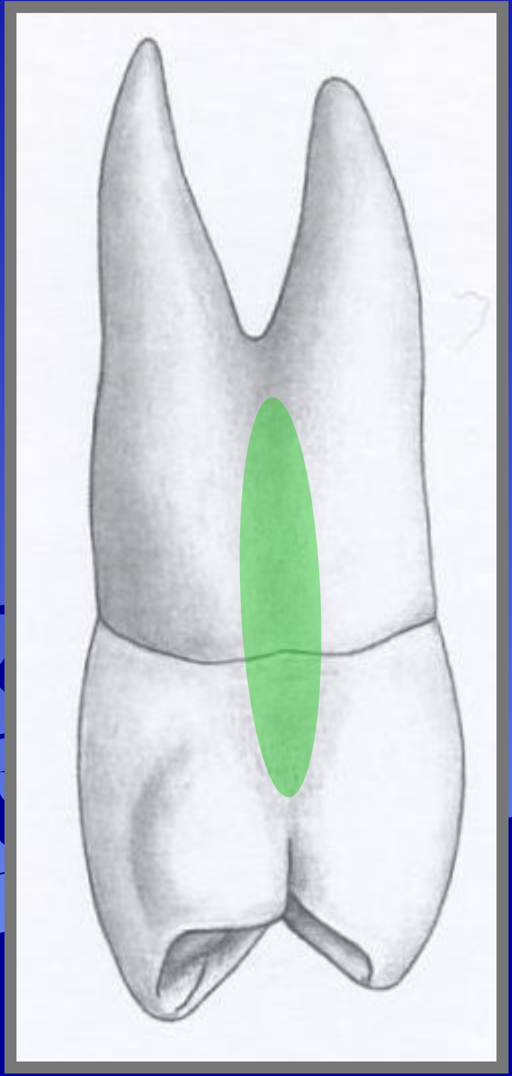
Lobe Divisions

Developmental depressions are typically exhibited as lobe divisions

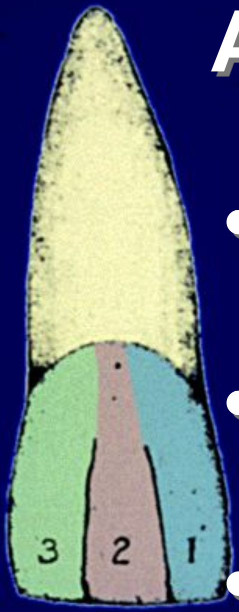






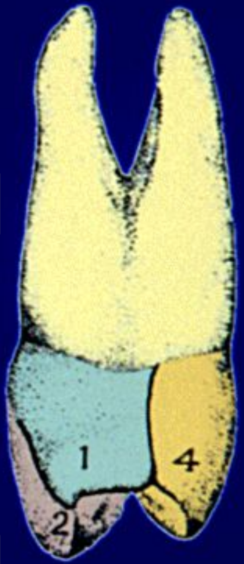


The Negative Anatomy of Teeth

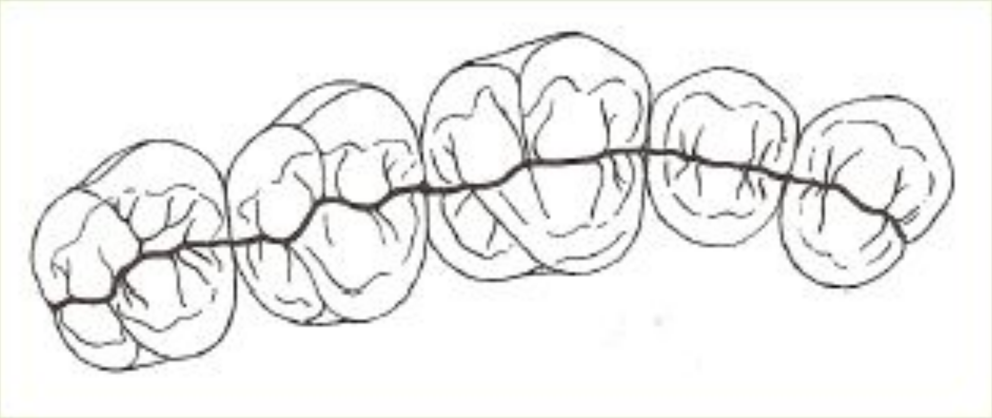


All grooves are either:

- Developmental Grooves
- Supplemental Grooves
- Developmental Depressions

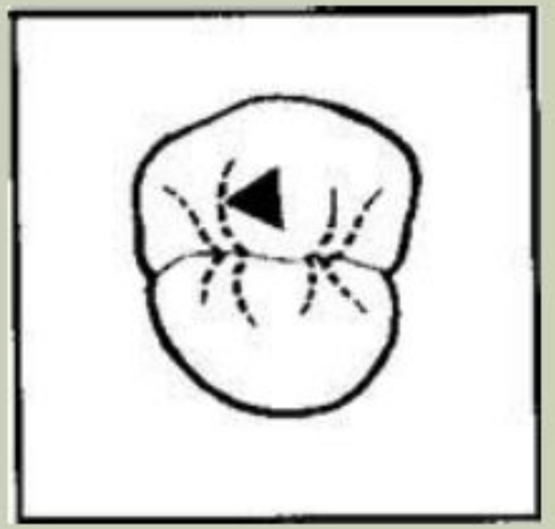


- The central developmental groove of posterior teeth aligns into a continuous valley dividing the teeth approximately in half.



TERMINOLOGY REVIEW - SUPPLEMENTAL GROOVE

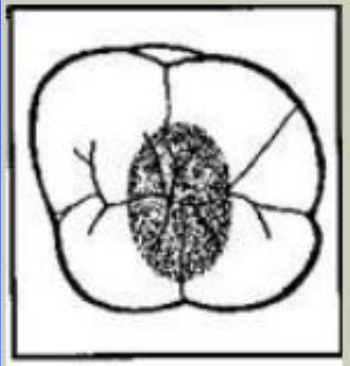
- An auxiliary groove which branches from a primary developmental groove.
  - Its location is **not related** to the junction of primary tooth parts, and is normally not as deep as a developmental groove.



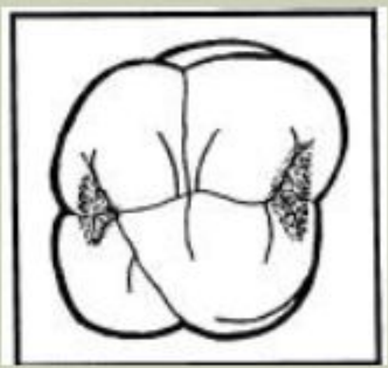
TERMINOLOGY REVIEW – FOSSA/FOSSAE

- An irregular, depression or concavity, on the occlusal table of a tooth.

  - **Premolars**: normally have **two**: mesial & distal
  - **Molars**: have **three**: mesial, distal, & central



central



distal / mesial

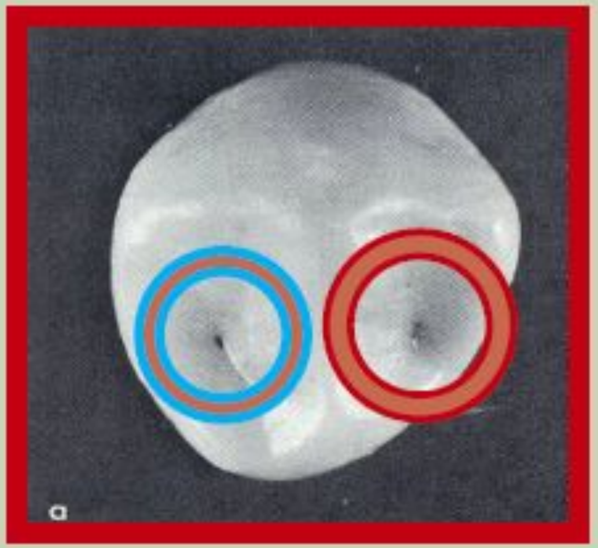
distal/mesial

distal/central/mesial

TERMINOLOGY REVIEW – PIT

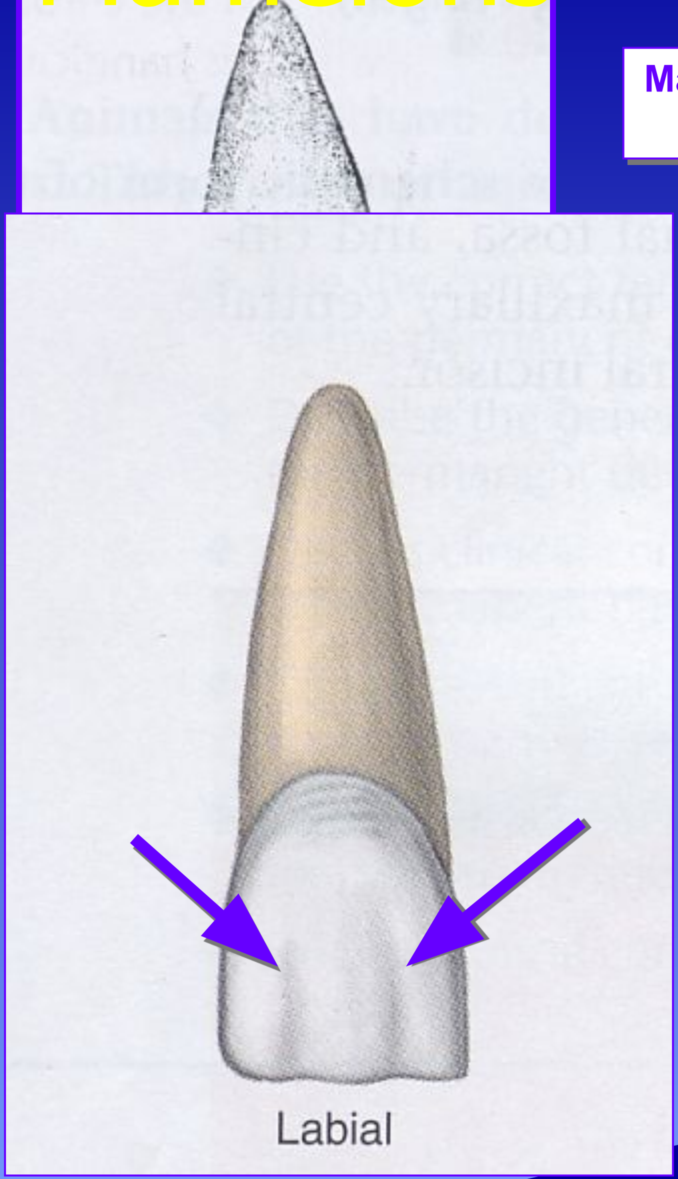
- A pit is the deepest portion of a fossa.
- A small depressed area where developmental grooves/lobes often join or end.

Distal pit

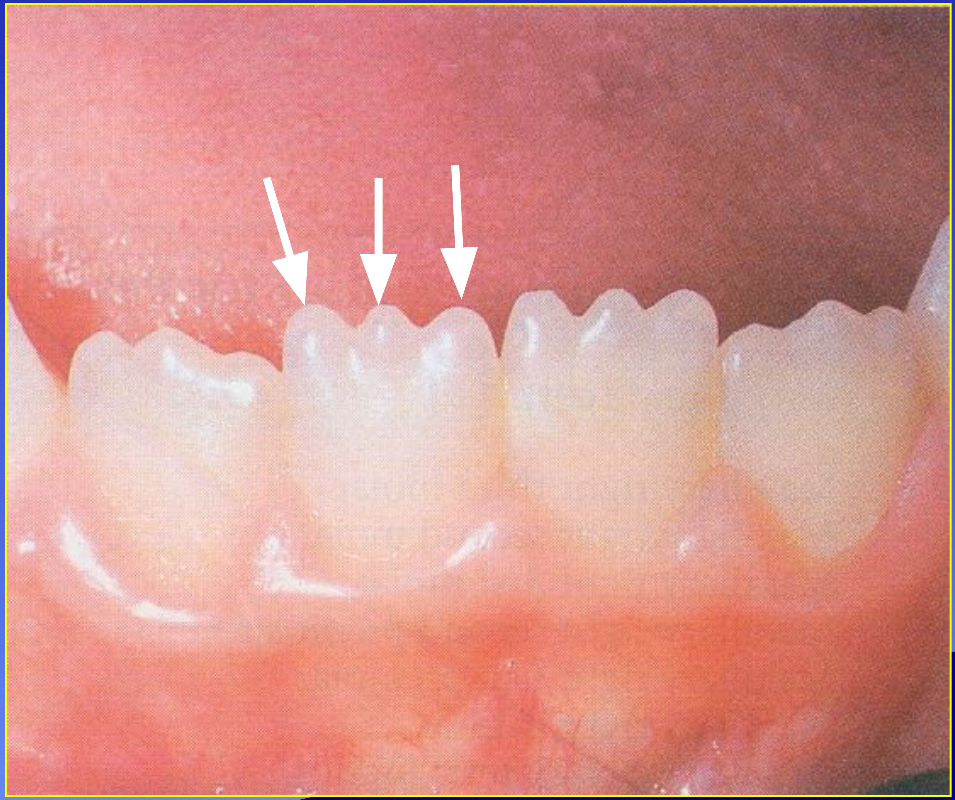


Mesial pit

Mamelons



Manifestations of developmental depressions are exhibited as Mamelons in the incisal third of permanent incisor teeth



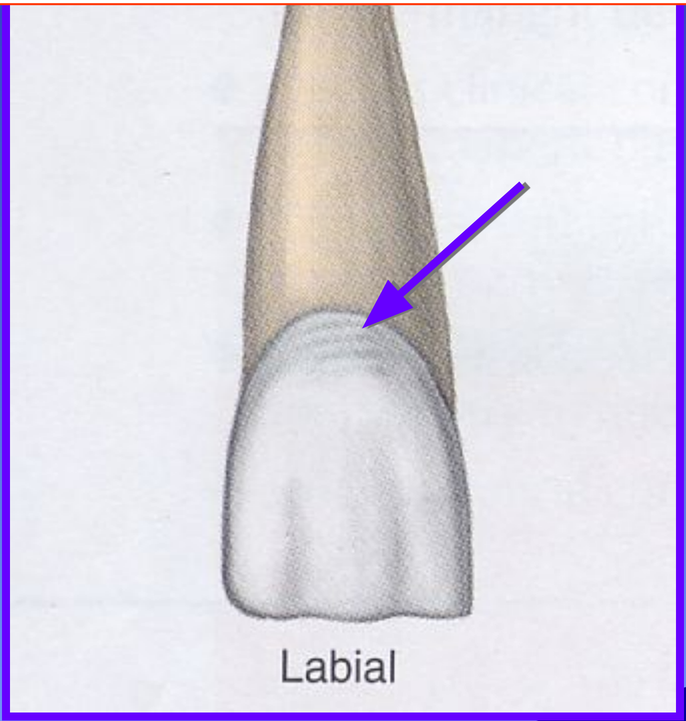
Imbrication Lines: (negative anatomy)

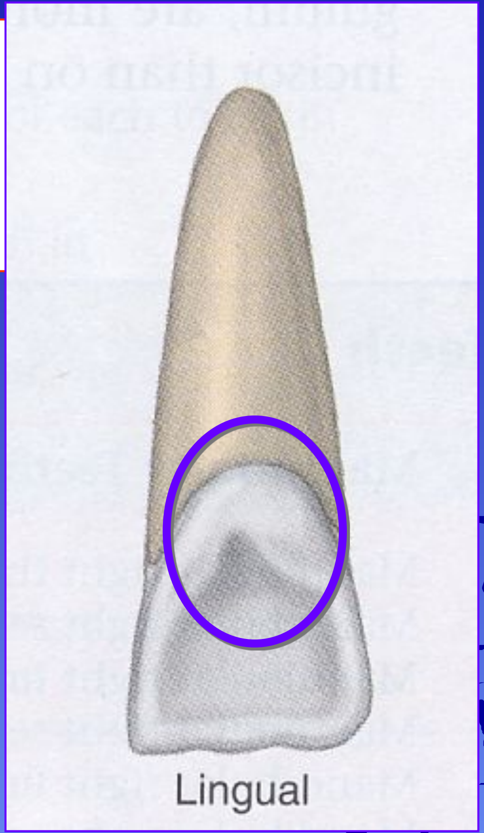
Result from an overlapping of enamel at the cervical 1/3. Lines run parallel to the CEJ. Mostly noted on Maxillary Central Incisors

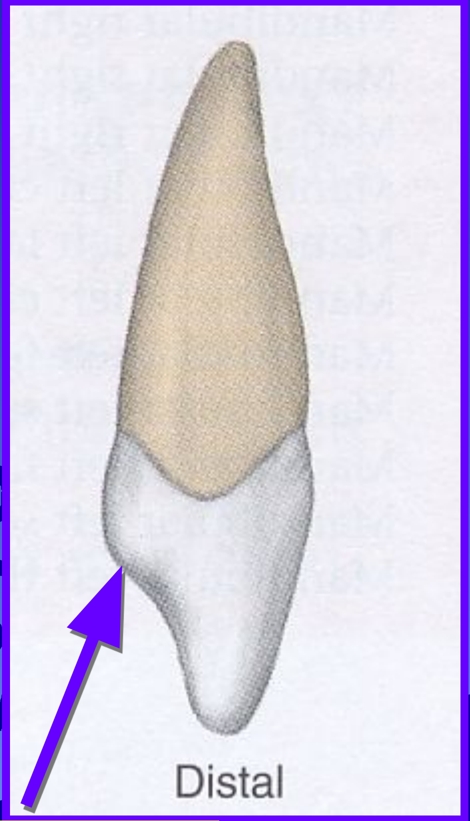
Imbrication Lines & Perikymata

Positive Anatomy

Perikymata







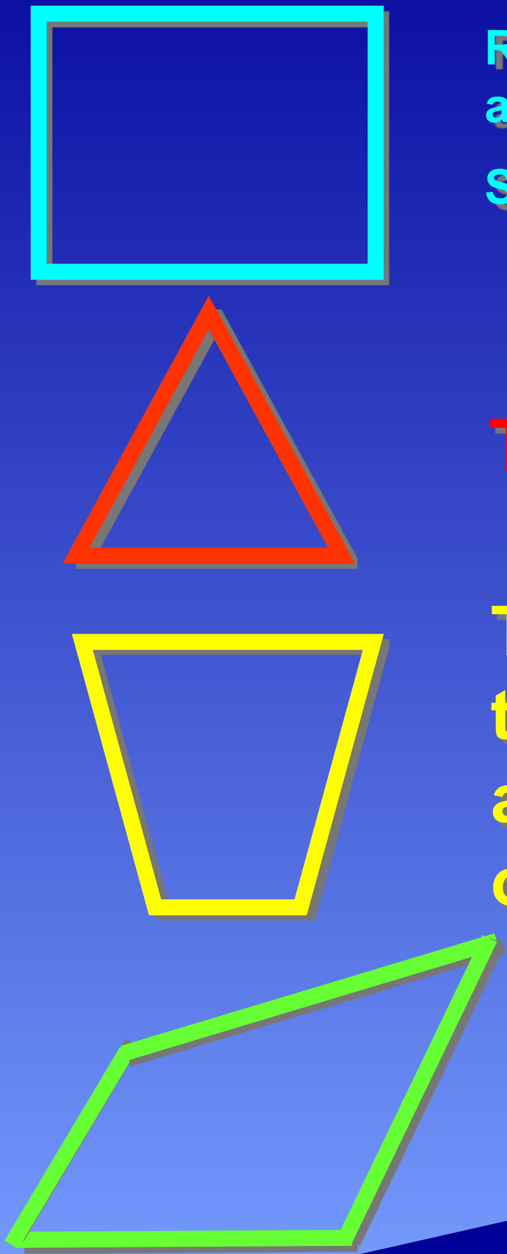
Cingulum

4th Lobe of Anterior Teeth?

Formation of the Primary Dentition

- **Primary anterior teeth form from a <u>SINGLE CENTER</u> of calcification.**
  - **no mamelons**
  - **no developmental depressions**

Geometry of the Teeth



Rectangle: 4 sides, opposing sides = length, 4 right angles

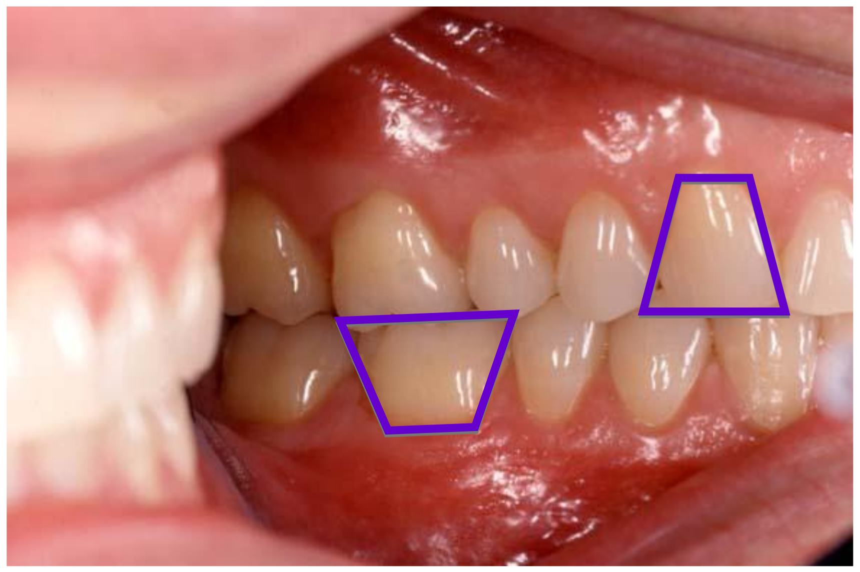
Square: 4 sides of equal length & 4 right angles

Triangle: 3 sides & 3 angles

Trapezoid: 4 sides, two are parallel to one another and two are not, 4 angles, two acute opposite two obtuse

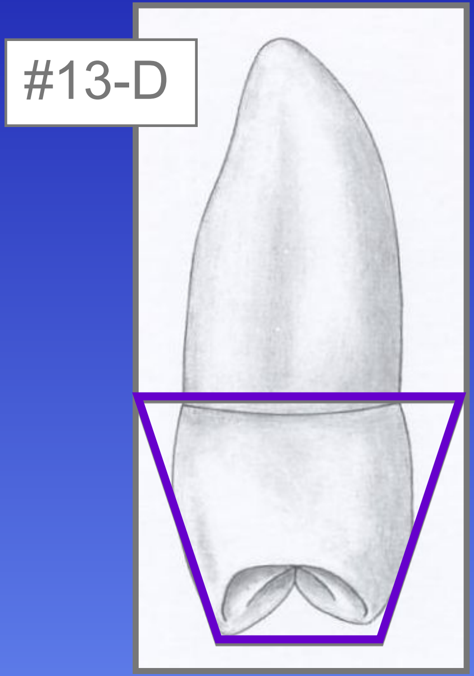
Rhombus (rhomboid): 4 sides, two are parallel to one another, 4 angles, two are acute & two are obtuse

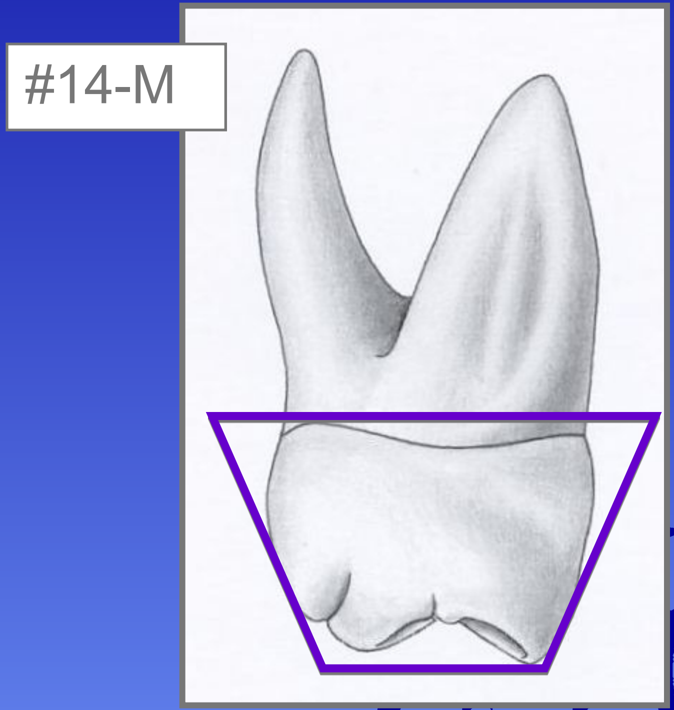
All crowns from the Facial or Lingual aspect are Trapezoidal



The shorter of the parallel sides will be gingival, the longer will be incisal/occlusal

Maxillary posterior teeth from the proximal aspect are Trapezoidal

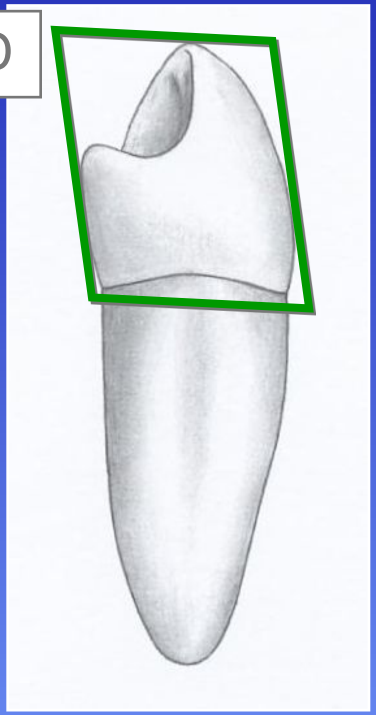


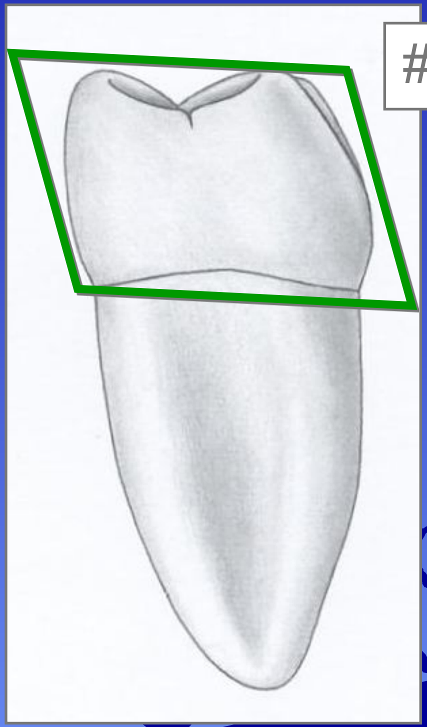


The longer of the parallel sides will be gingival, the shorter will be occlusal

Mandibular posterior teeth from the proximal aspect are Rhomboidal

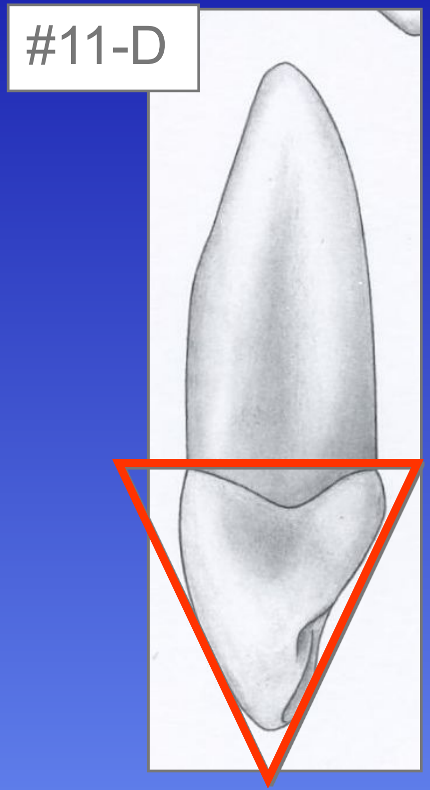
#28-D

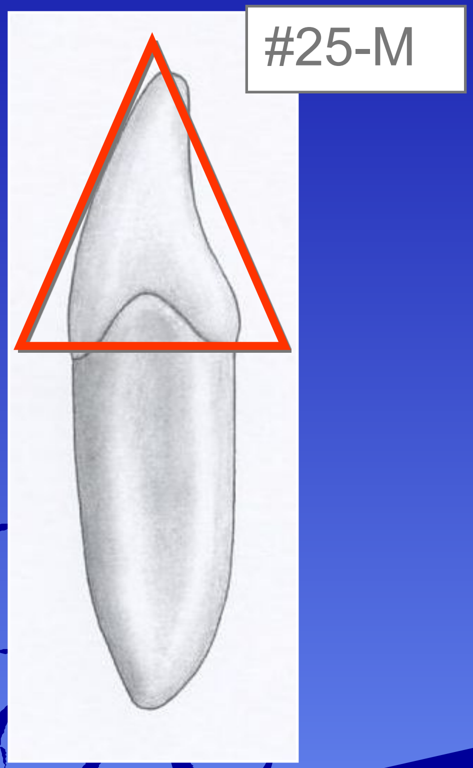




#19-M

Maxillary & Mandibular anterior teeth from the proximal aspect are Triangular

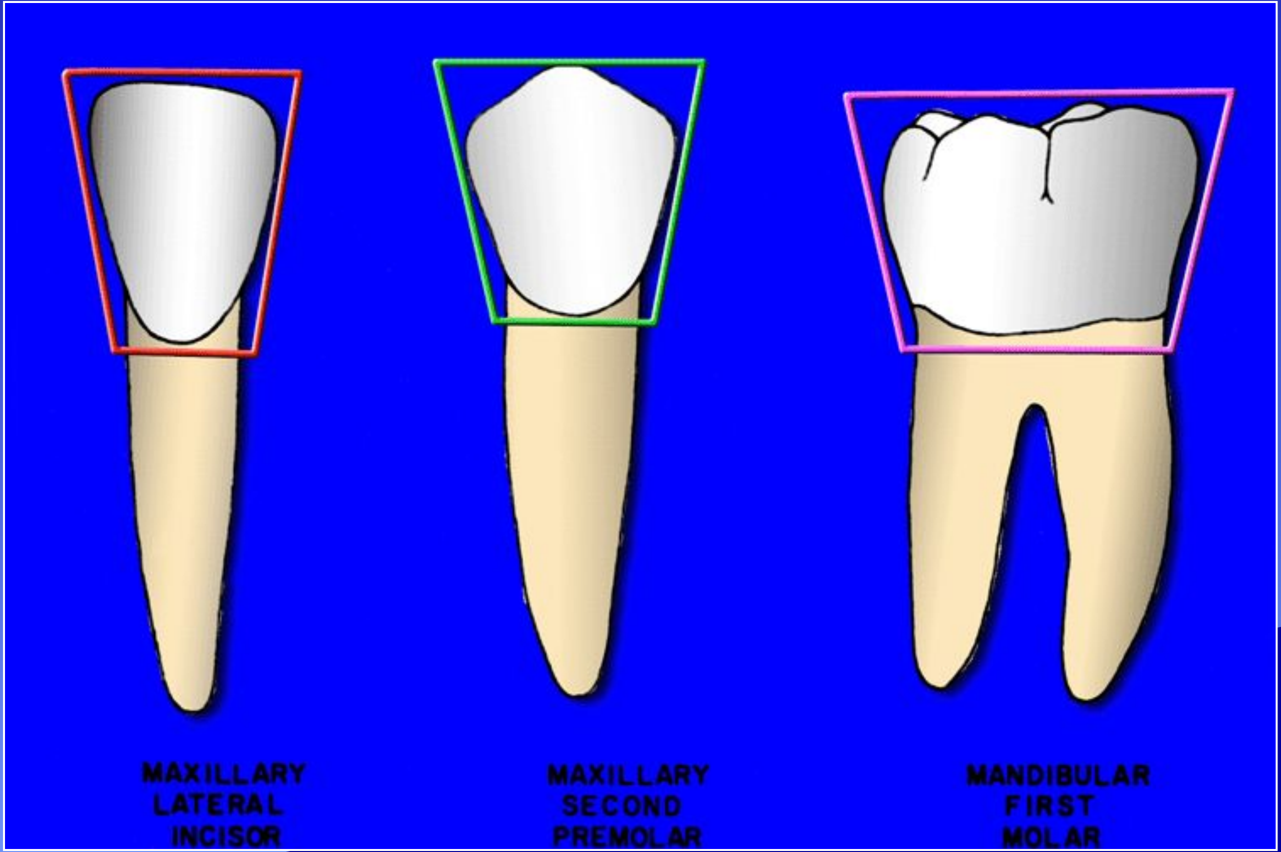




The base of the triangle is gingival, the apex is at the incisal

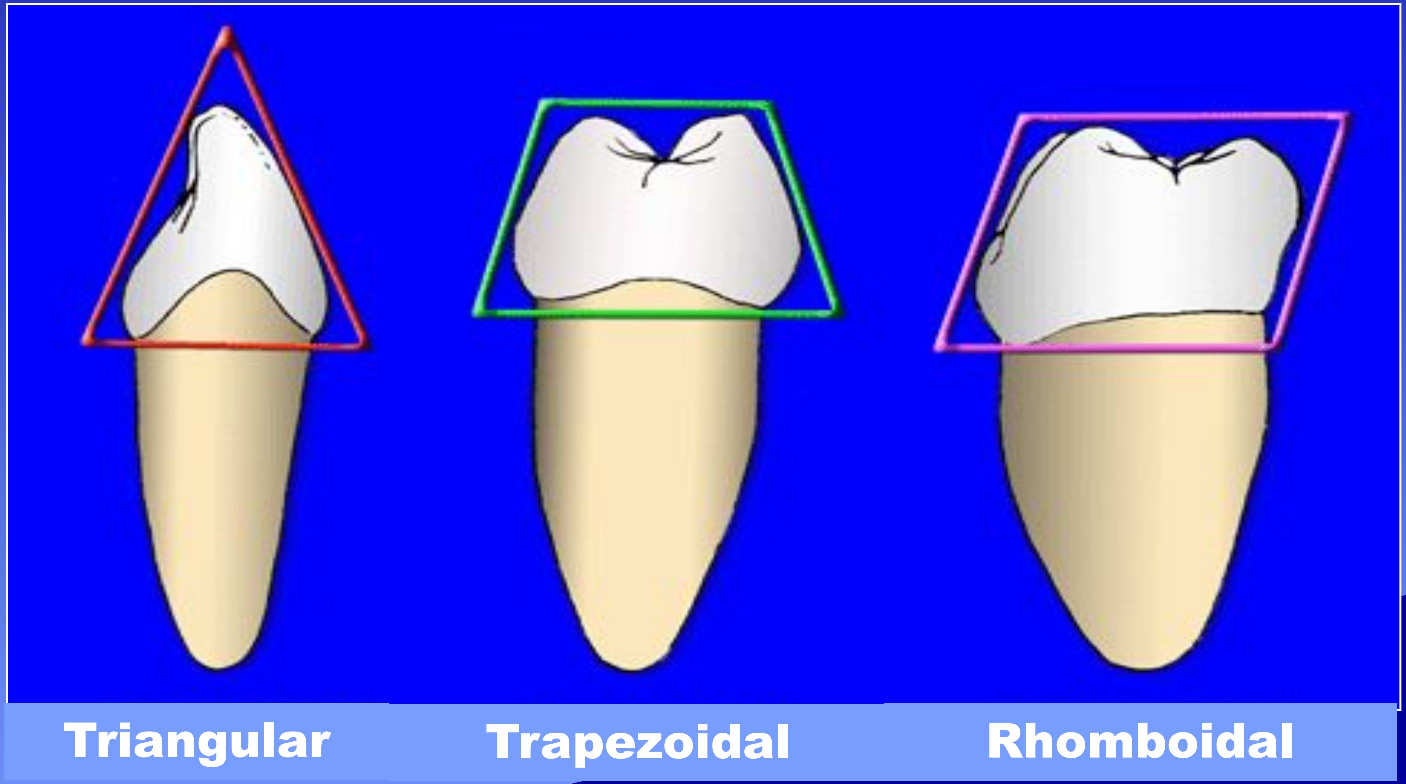
Geometric Concepts

From a Facial View:



Geometric Concepts

From a Proximal View:



Embrasures

When two teeth in the same arch are in contact with each other, four “V” shaped spaces are created around the contact area that form a spillway called *embrasures….*

Every contact area produces 4 embrasures

Embrasures

- Four embrasures surround each contact area
- Essentially, they are air spaces
  - Facial
  - Lingual
  - Occlusal or Incisal
  - Gingival

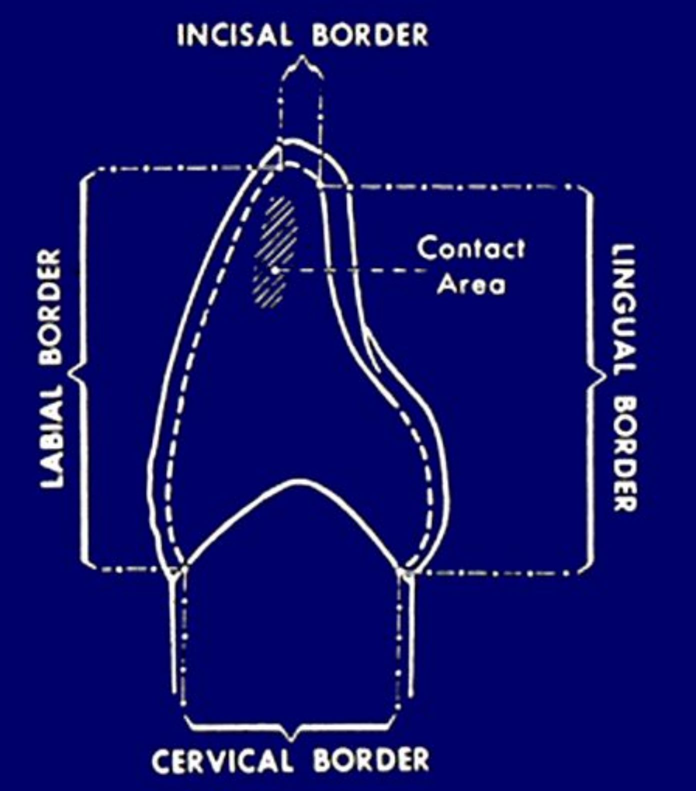
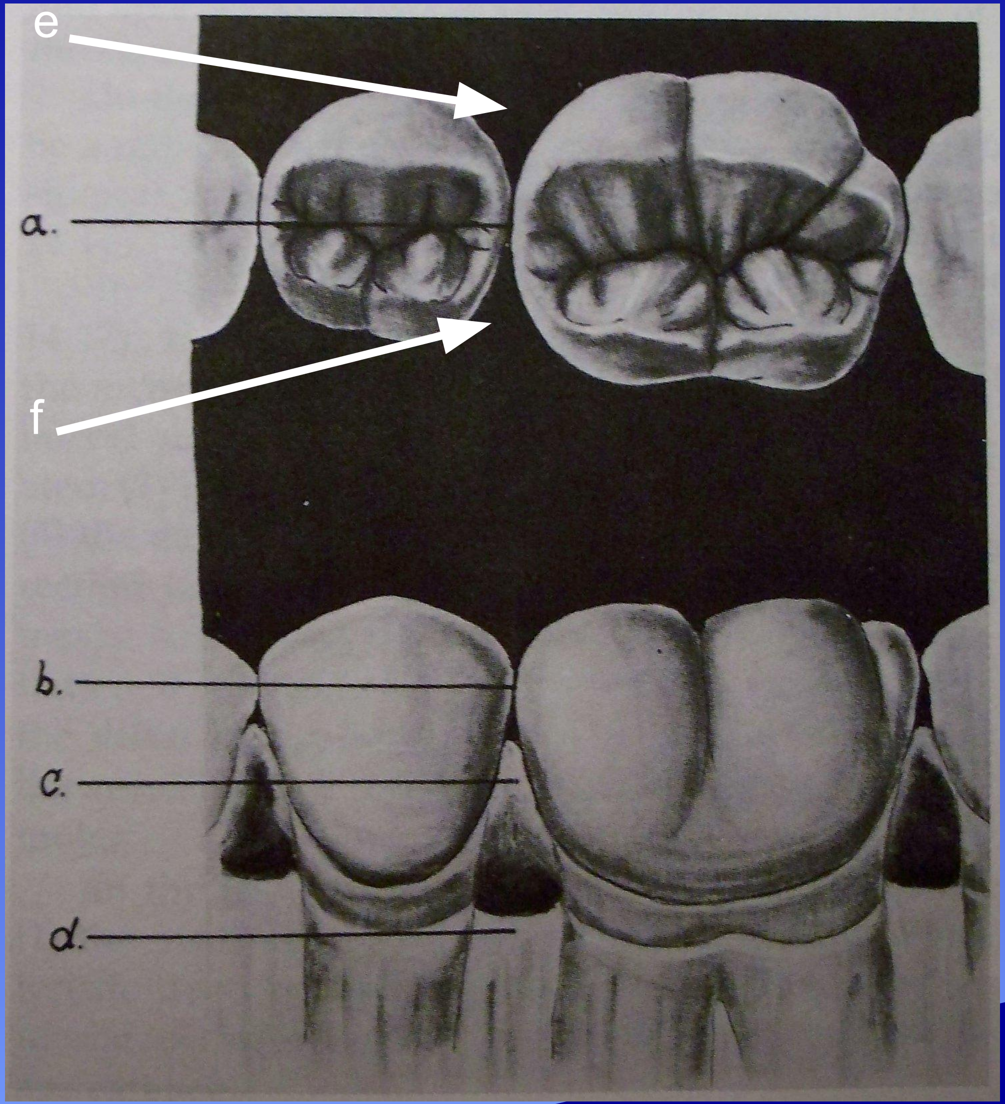


FIG. 6. DISTAL SURFACE

Embrasure Construct



Legend:

a = occlusal embrasure

b = contact area

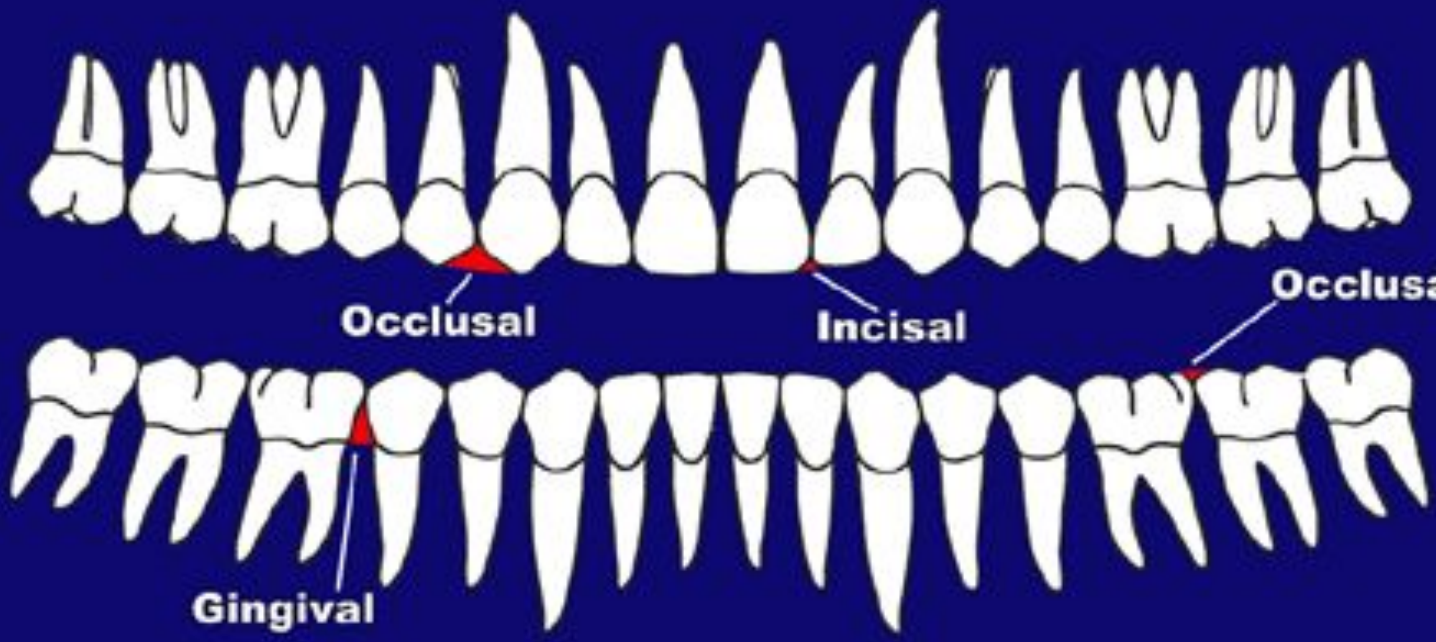
c = gingival or cervical embrasure

d = alveolar bone

e = facial embrasure

f = lingual embrasure

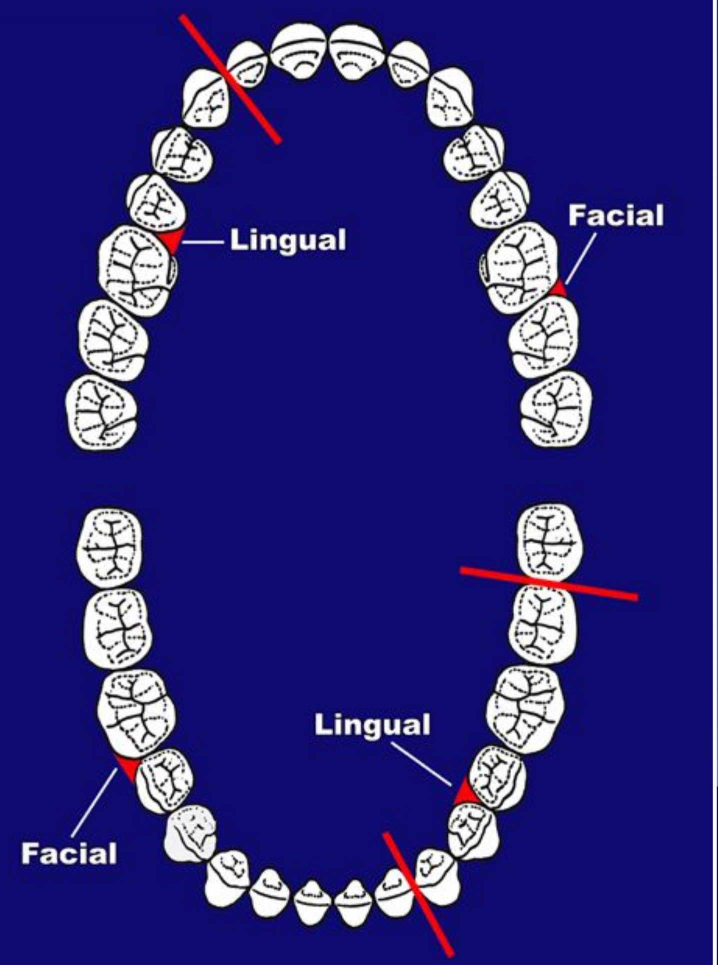
Embrasures- Facial Viewpoint



Gingival and incisal/occlusal embrasures are evident from a facial viewpoint

Embrasures-
Occlusal Viewpoint

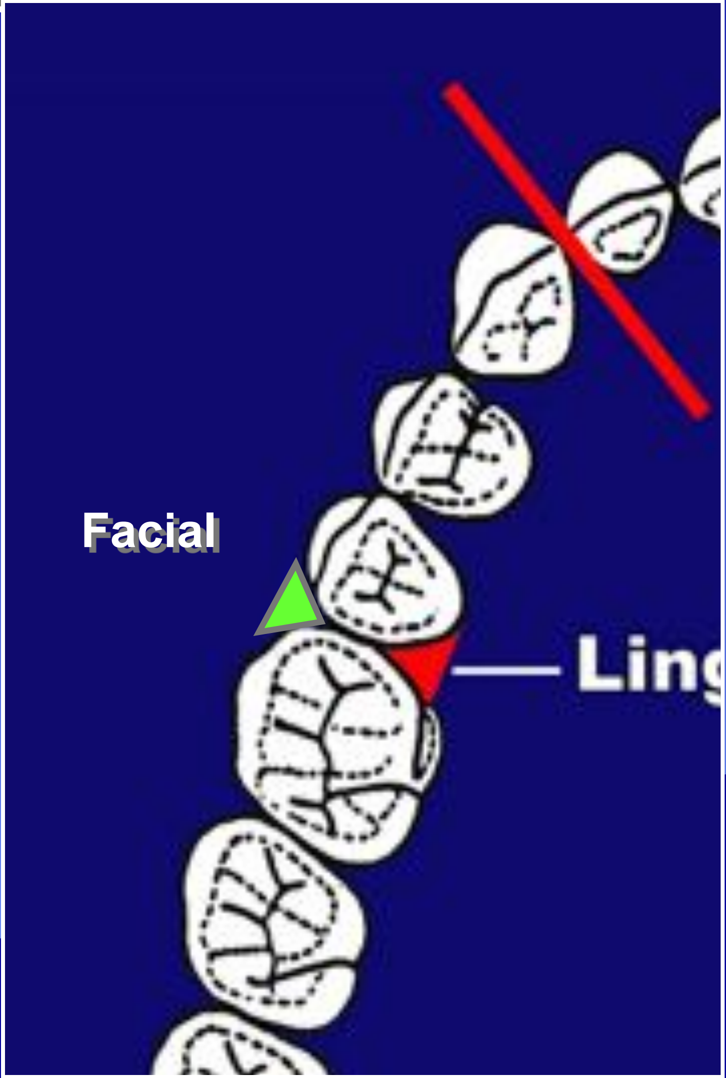
Facial and lingual embrasures are seen from the occlusal viewpoint.



Embrasures - Occlusal View

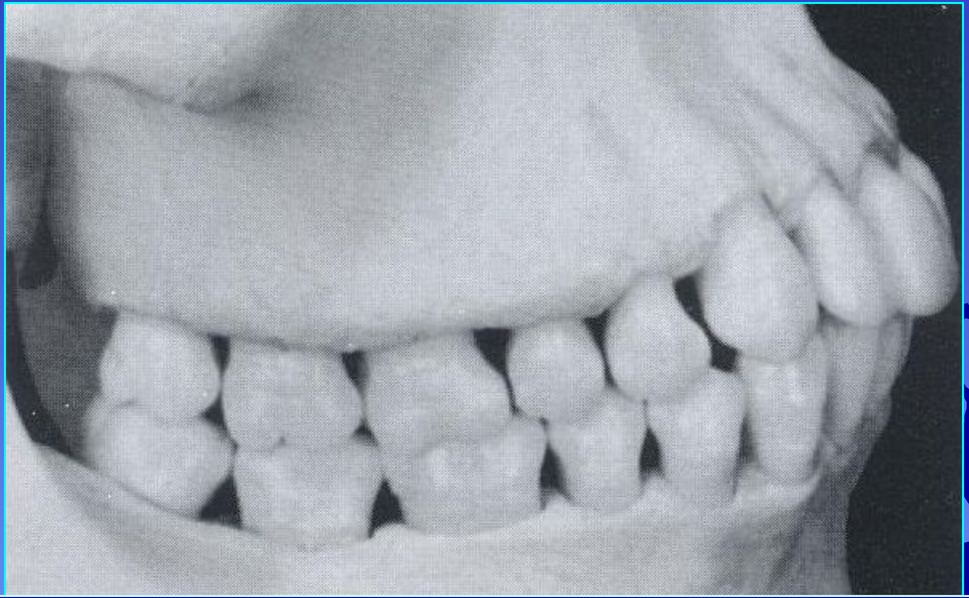
Embrasures- Occlusal Viewpoint

- Facial and lingual embrasures are seen from the occlusal viewpoint
- Embrasures are typically mirror images of each other
- Note the effect that underdeveloped or overdeveloped line angles have on the size of an embrasure
  - Facial vs. lingual



Gingival Embrasure

Triangular shaped space between the teeth normally filled with gingival tissue

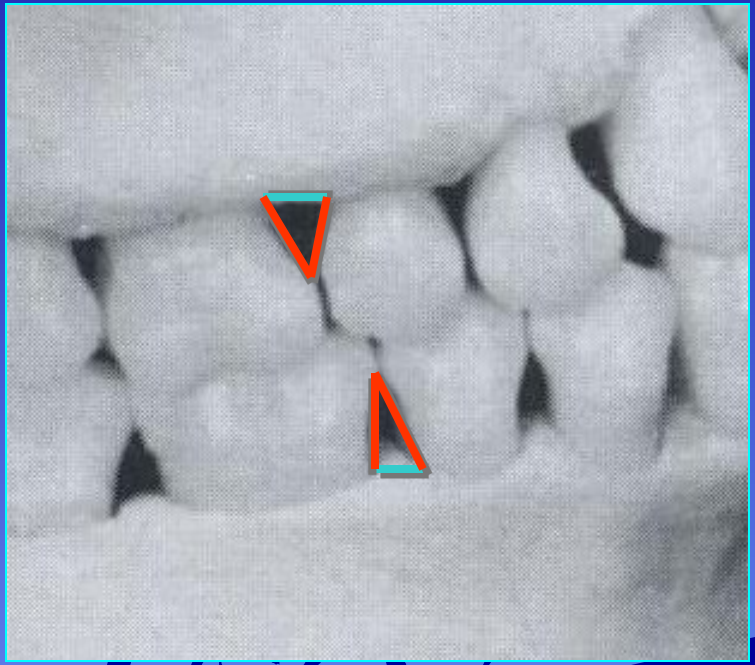


aka "interproximal space"

Gingival Embrasure

The base of the triangle is the alveolar process

The sides of the triangle are the proximal surfaces of the contacting teeth



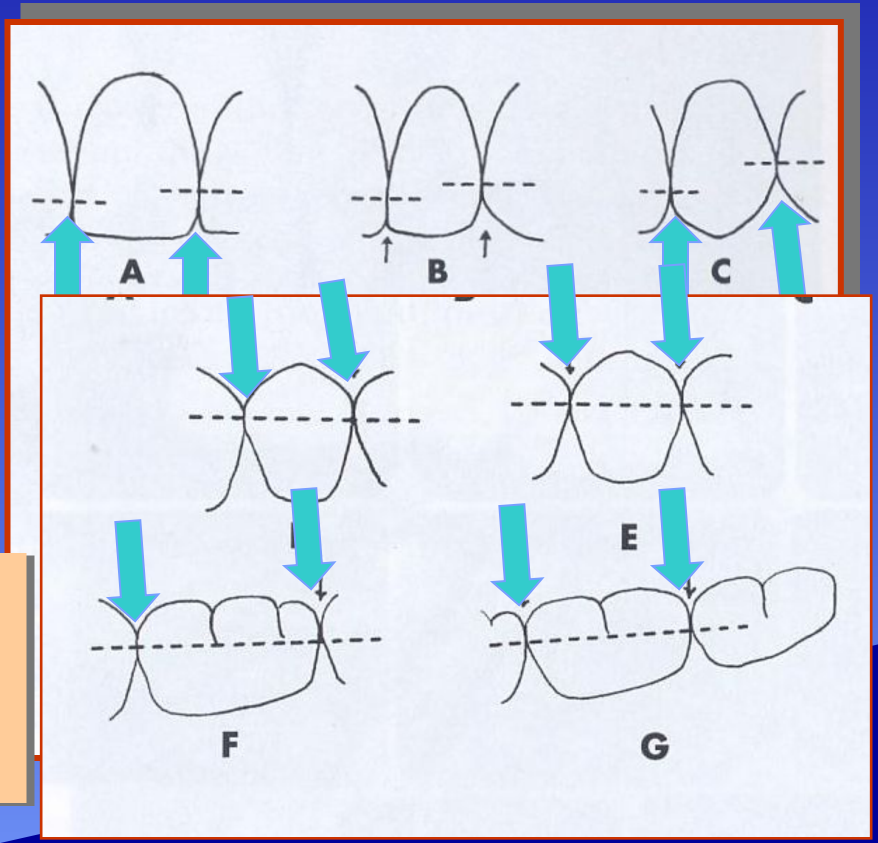
The apex of the triangle is the area of contact

Incisal or Occlusal Embrasures (facial view)

Above (occlusal to) the contact area:

Anterior Teeth:
Incisal Embrasure

Posterior Teeth:
Occlusal Embrasure



Largest = between Canine & 1st Premolar

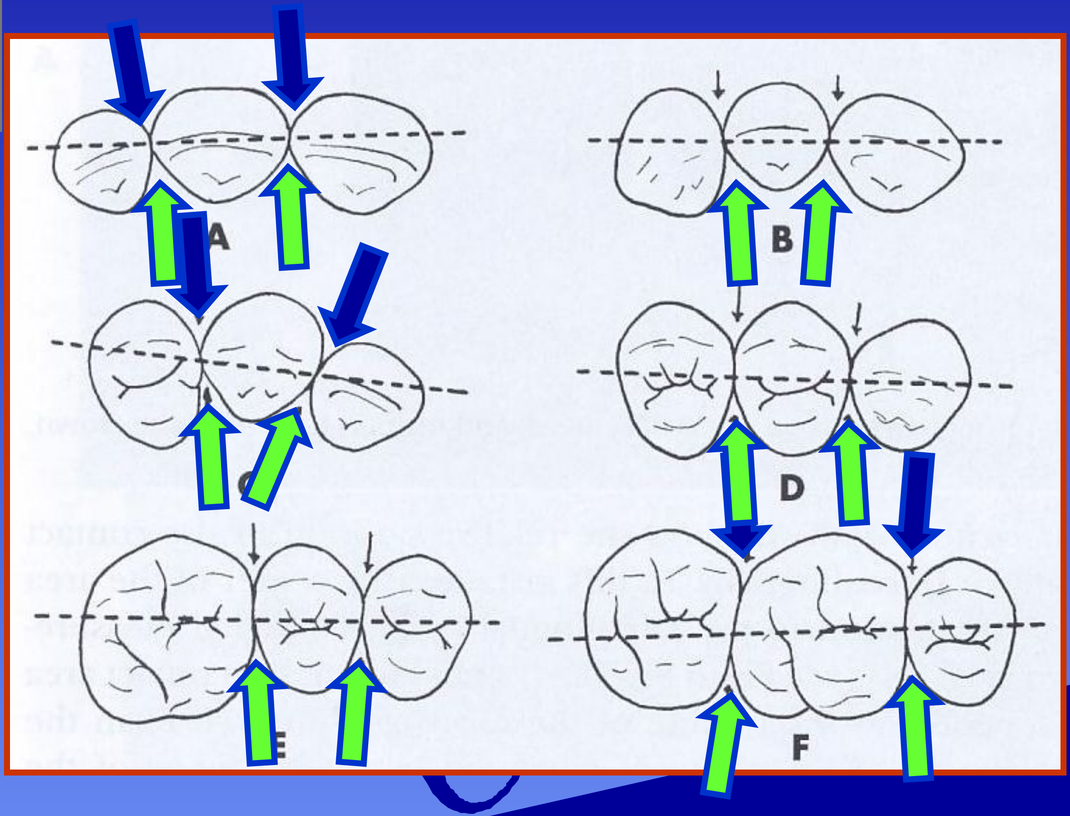
Smallest = at the midline

Facial & Lingual Embrasures (occlusal/incisal view)

Viewed from the occlusal/incisal:

Facial Embrasure:

Lingual Embrasure:



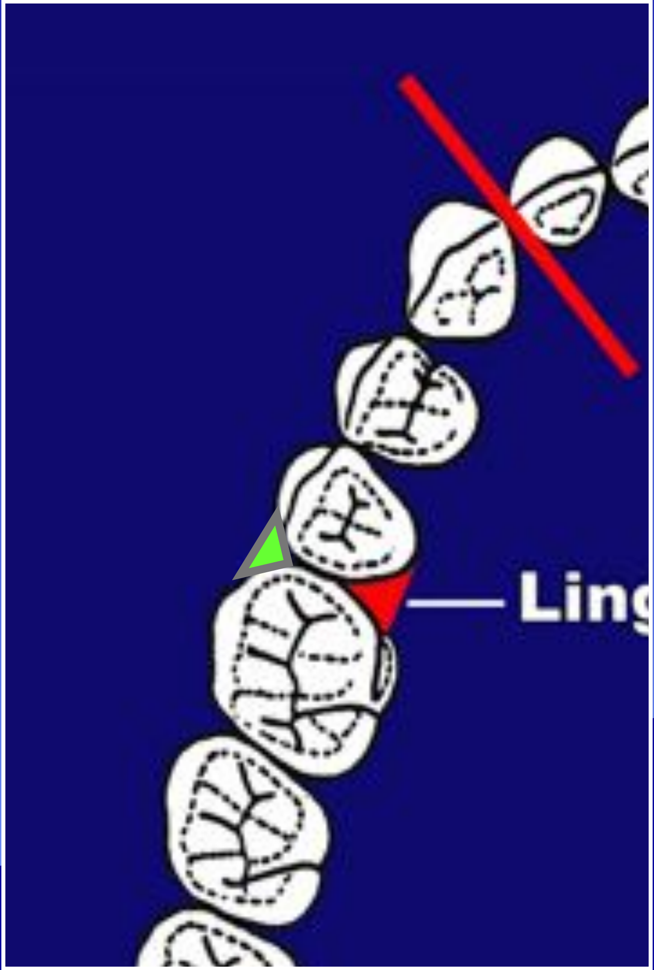
In the Maxillary arch, the lingual embrasures are larger than the respective facial, except surrounding the 1st Molar

Embrasures – Clinical Issues

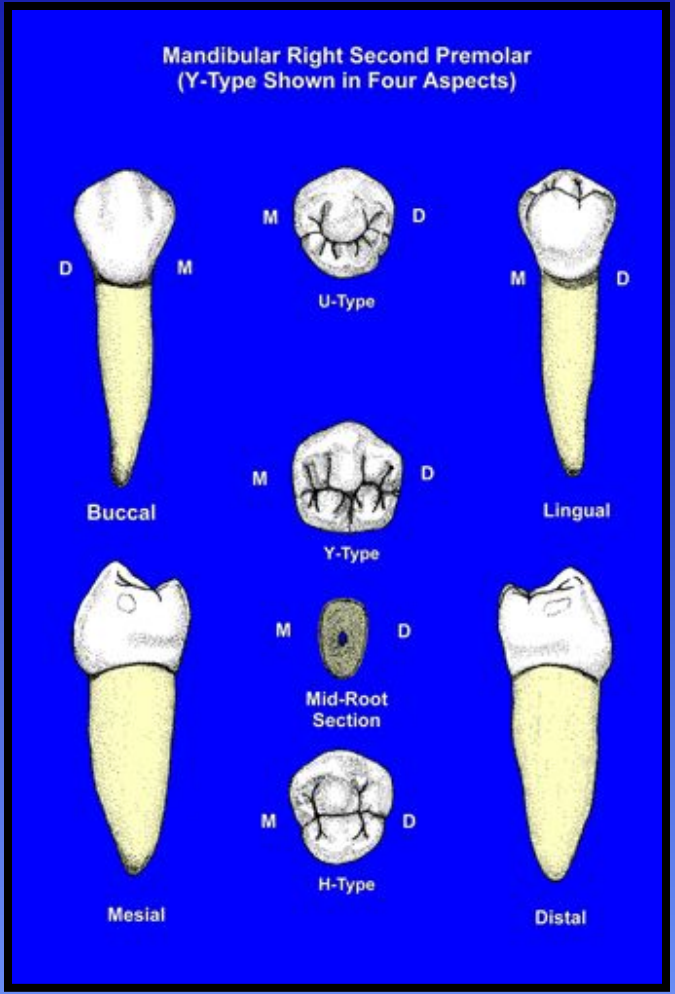
Ideal, open or closed and are determined by:
>malposition of teeth
>oversized proximal contacts
>evidence of caries

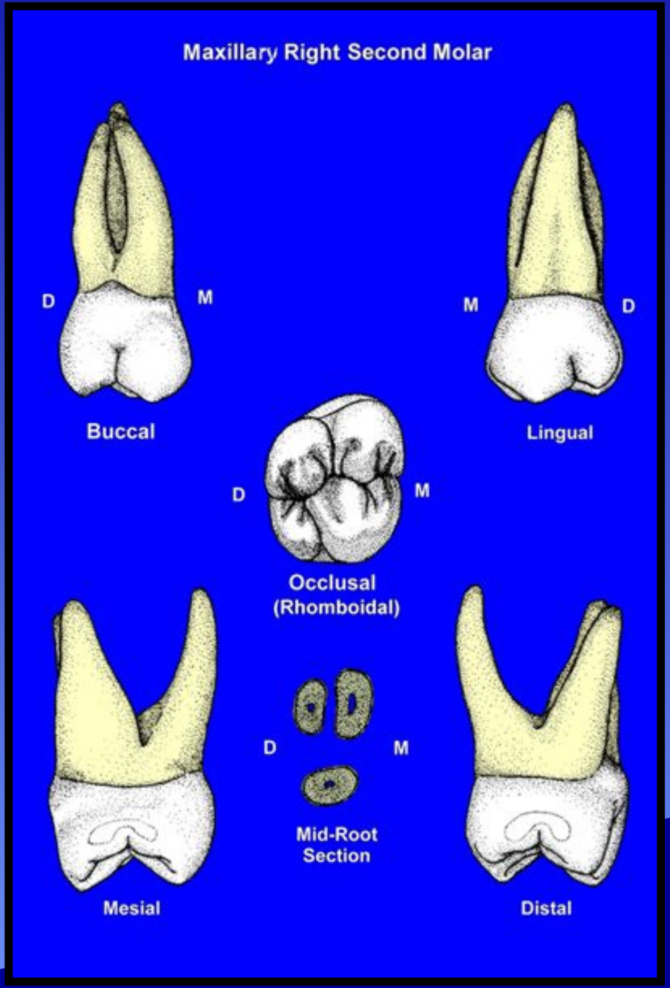
Embrasures & Line Angles Occlusal Viewpoint

- Facial and lingual embrasures are seen from the occlusal viewpoint
- Lingual embrasures generally larger than their respective facial embrasures
- Contacts on posterior teeth are slightly facially positioned
- Line angles
  - Facial more square
  - Lingual more round
- Mirror images



Tooth Form Generalizations: Cusp/ Marginal Ridge Measurement

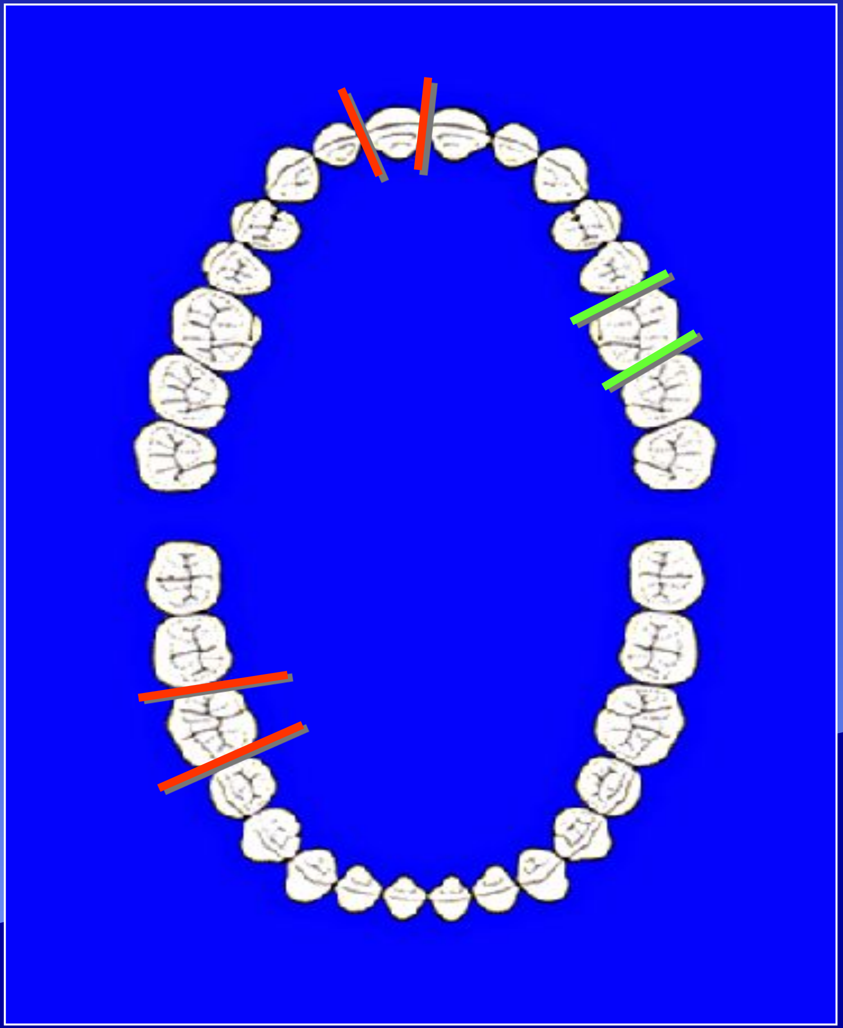




Mesial cusps are larger & Mesial marginal ridges are taller

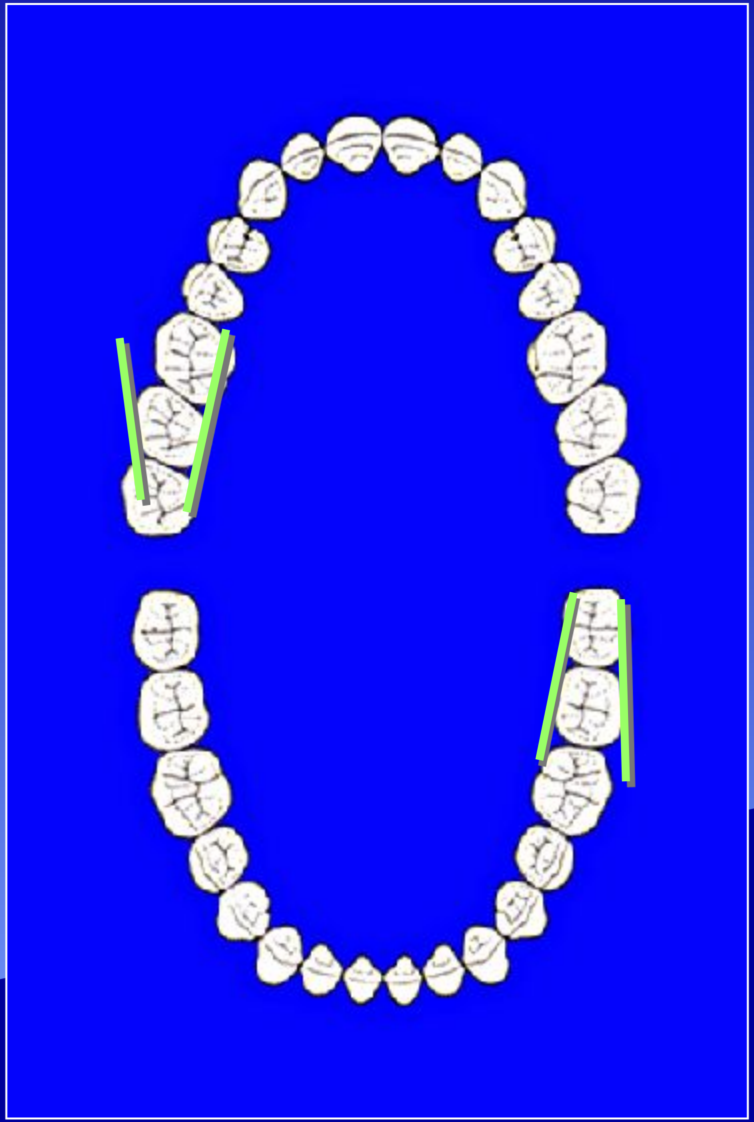
Lobe Forms: Occlusal View

- **All teeth, permanent and primary exhibit a *taper from facial to lingual***
  - 3 exceptions:
    1. Permanent Maxillary 1st Molar
    2. Mandibular 2nd premolar ("Y" type)
    3. Primary Maxillary 2nd Molar
- There is an observable affect on line angles



Lobe Forms: Occlusal View

- All molars, permanent or primary exhibit a *taper from mesial to distal*
- The *mesial* portion of a posterior tooth is *larger* than the distal portion
- Line angles are more *square* on the mesial, *rounder* on the distal



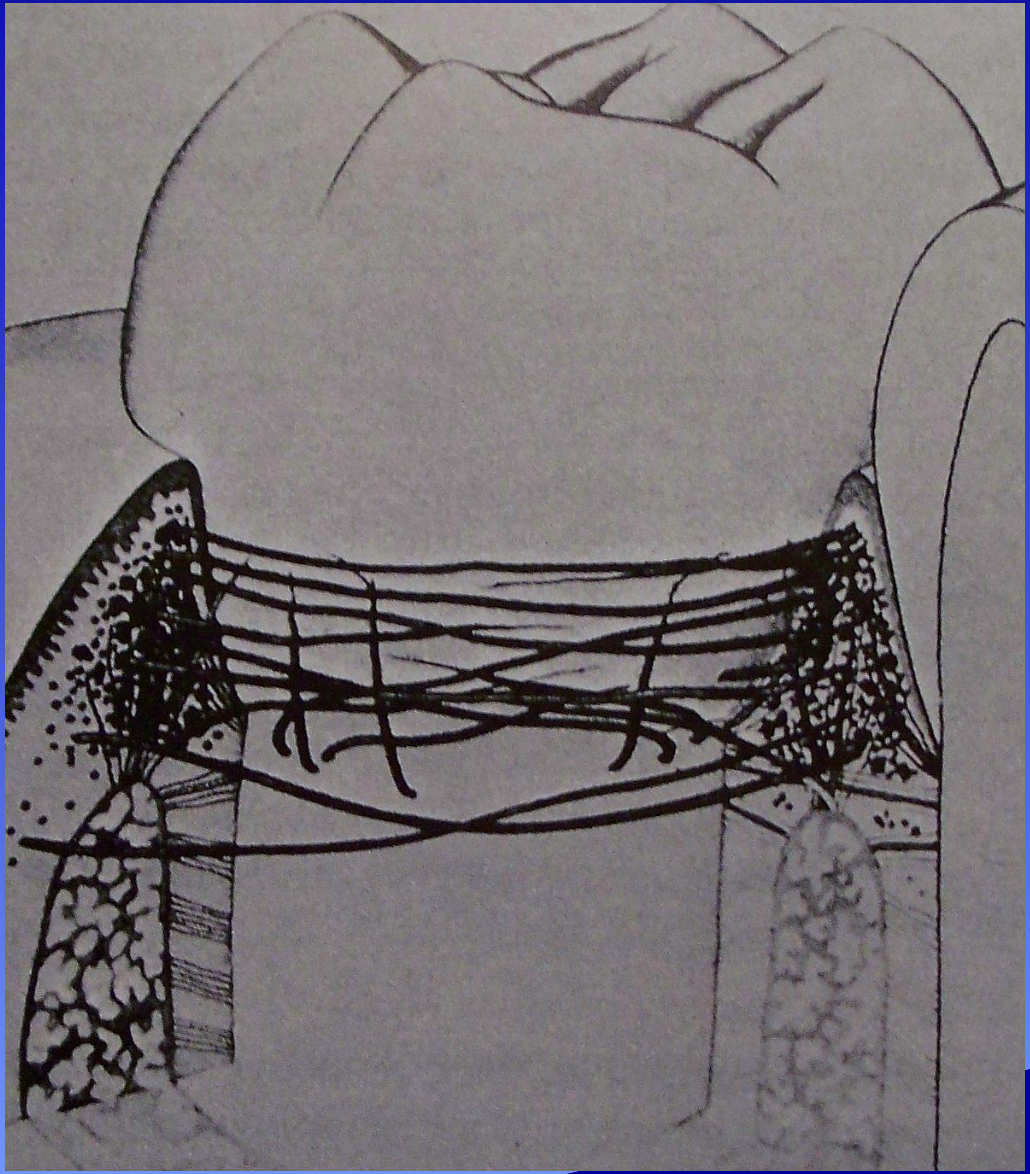
Tooth Form Generalizations:

- Mesial portions of crowns are generally more developed than the distal
- Mesial cusps are larger than distal cusps
- MMR usually taller than DMR (exception: mandibular 1st premolar)
- Molars taper M-D
- All teeth taper F-L except Maxillary 1st molar & Y-grooved Mand. 2nd premolar

The End

Lobe Forms & Geometric Concepts

Circular (Circumferential) Periodontal Fibers



Located at the cervical third of the crown and root of a tooth. Influences the rotation of a tooth within an alveolus

Review Topics

Arch Form

Tooth Alignment

Plane of Occlusion

Proximal Contact Locations

Height of Contour Measurements

Height of Curvature of the CEJ

Periodontal Tissues & Fibers